

IEEE Std 802®-2001
(Revision of IEEE Std 802-1990)

802®

IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture

IEEE Computer Society

Sponsored by the
LAN/MAN Standards Committee



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IEEE Std 802®2001
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802®

IEEE 局域网和城域网标准： 概述与体系结构

IEEE 计算机学会

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IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture

Sponsor
LAN/MAN Standards Committee
of the
IEEE Computer Society

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Abstract: IEEE Std 802-2001, IEEE Standards for Local and Metropolitan Area Networks: Overview and Architecture, provides an overview to the family of IEEE 802 Standards. It defines compliance with the family of IEEE 802 Standards; it describes the relationship of the IEEE 802 Standards to the Open Systems Interconnection Basic Reference Model [ISO/IEC 7498-1:1994] and explains the relationship of these standards to the higher layer protocols; it provides a standard for the structure of LAN MAC addresses; and it provides a standard for identification of public, private, and standard protocols.

Keywords: IEEE 802 standards compliance, Local Area Networks (LANs), LAN/MAN architecture, LAN/MAN reference model, Metropolitan Area Networks (MANs).

The Institute of Electrical and Electronics Engineers, Inc.
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IEEE 局域网和城域网标准：概述与体系结构

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的
IEEE 计算机学会

批准于2001年12月6日
IEEE-SA 标准委员会

摘要：IEEE 标准 802-2001《局域网和城域网 IEEE 802 标准：概述与体系结构》概述了 IEEE 802 标准系列。该标准定义了对 IEEE 802 标准系列的合规性；描述了 IEEE 802 标准与开放系统互连基本参考模型 [ISO/IEC 7498-1:1994]之间的关系，并说明这些标准与高层协议之间的关系；规定了局域网 MAC 地址的结构标准；并规定了用于识别公共、私有和标准协议的标准。

关键词：IEEE 802 标准合规性、局域网（LAN）、局域网/城域网体系结构、局域网/城域网参考模型、城域网（MAN）。

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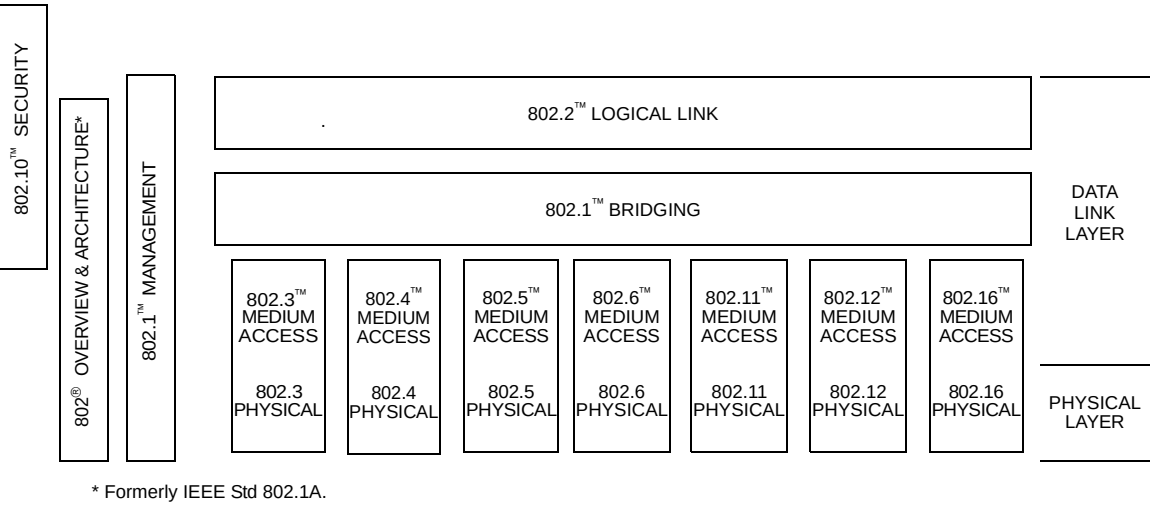
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Introduction

(This introduction is not part of IEEE Std 802-2001, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.)

IEEE Std 802-2001 provides an overview to the family of IEEE 802 Standards. It defines compliance with the family of IEEE 802 Standards; it describes the relationship of the IEEE 802 Standards to the Open Systems Interconnection Basic Reference Model [ISO/IEC 7498-1: 1994] and explains the relationship of these standards to higher layer protocols; it provides a standard for the structure of LAN MAC addresses; and it provides a standard for the identification of public, private, and standard protocols.

This standard is part of a family of standards for local and metropolitan area networks. The relationship between the standard and other members of the family is shown below. (The numbers in the figure refer to IEEE standard numbers.¹)



This family of standards deals with the Physical and Data Link Layers as defined by the International Organization for Standardization (ISO) Open Systems Interconnection Basic Reference Model (ISO/IEC 7498-1:1994). The access standards define several types of medium access technologies and associated physical media, each appropriate for particular applications or system objectives. Other types are under investigation.

The standards defining the technologies noted above are as follows:

- IEEE Std 802:² *Overview and Architecture.* This standard provides an overview to the family of IEEE 802 Standards. This document forms part of the IEEE Std 802.1 scope of work.
- IEEE Std 802.1B™ and 802.1K™ [ISO/IEC 15802-2]: *LAN/MAN Management.* Defines an Open Systems Interconnection (OSI) management-compatible architecture, and services and protocol elements for use in a LAN/MAN environment for performing remote management.

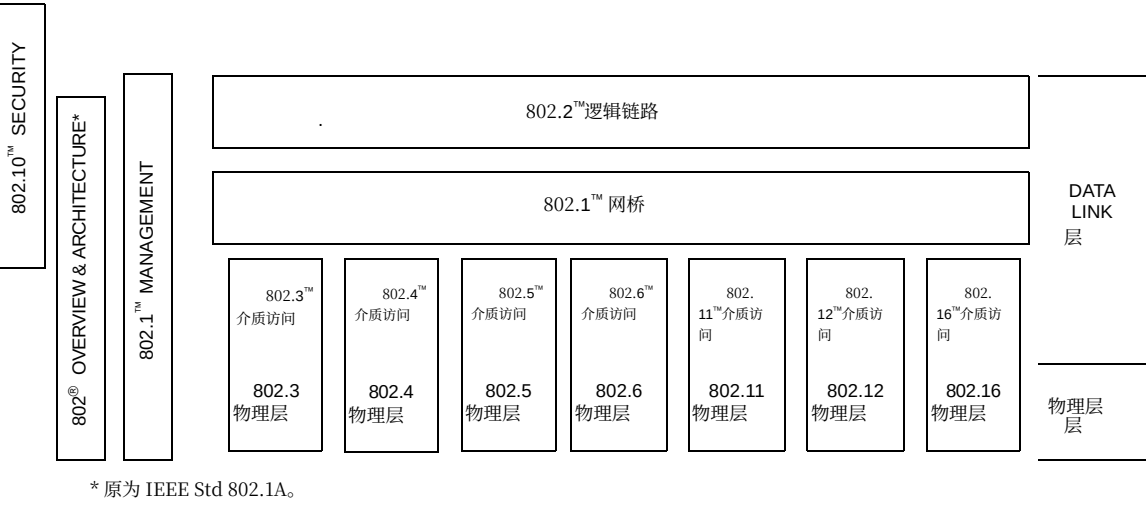
¹The IEEE Standards referred to in the above figure and list are trademarks owned by the Institute of Electrical and Electronics Engineers, Incorporated.
²The IEEE 802 Architecture and Overview Specification, originally known as IEEE Std 802.1A, has been renumbered as IEEE Std 802. This has been done to accommodate recognition of the base standard in a family of standards. References to IEEE Std 802.1A should be considered as references to IEEE Std 802.

引言

(本引言不属于 IEEE 标准 802-2001《IEEE 局域网和城域网标准：概述与体系结构》。)

IEEE Std 802-2001 概述了 IEEE 802 标准族。它规定了对 IEEE 802 标准族的合规性；描述了 IEEE 802 标准与开放系统互连基本参考模型 [ISO/IEC 7498-1: 1994]的关系，并说明了这些标准与高层协议之间的关系；为局域网 MAC 地址的结构制定了标准；并为公共、私有和标准协议的识别制定了标准。

本标准属于局域网和城域网标准族。下图展示了本标准与该标准族其他成员之间的关系。（图中的数字指 IEEE 标准编号。¹）



该标准族涉及国际标准化组织（ISO）开放系统互连基本参考模型（ISO/IEC 7498-1:1994）所定义的物理层和数据链路层。介质访问标准规定了若干类型的介质访问技术及其相关物理介质，每种技术和介质都适用于特定应用或系统目标。其他类型仍在研究中。

定义上述技术的标准如下：

- IEEE Std 802: ² *概述与体系结构。* 本标准概述了 IEEE 802 标准系列。本文件属于 IEEE Std 802.1 的工作范围。
- IEEE Std 802.1B ™和 802.1K™ [ISO/IEC 15802-2]: *局域网/城域网管理。* 定义了一种开放系统互连 (OSI) 兼容管理的体系结构，以及用于局域网/城域网环境中执行远程管理的服务和协议元素。

¹上述图表和列表中引用的 IEEE 标准是电气与电子工程师协会所有的商标。²IEEE 802 架构与概述规范最初称为 IEEE Std 802.1A，现已重新编号为 IEEE Std 802。此举是为了便于在标准族中识别基础标准。对 IEEE Std 802.1A 的引用应视为对 IEEE Std 802 的引用。

• IEEE Std 802.1D™	<i>Media Access Control (MAC) Bridges.</i> Specifies an architecture and protocol for the [ISO/IEC 15802-3]: interconnection of IEEE 802 LANs below the MAC service boundary.	•IEEE 标准 802.1D ™	<i>Media Access Control (MAC) Bridges.</i> Specifies an architecture and protocol for the [ISO/IEC 15802-3]: interconnection of IEEE 802 LANs below the MAC service boundary.
• IEEE Std 802.1E™ [ISO/IEC 15802-4]:	<i>System Load Protocol.</i> Specifies a set of services and protocol for those aspects of management concerned with the loading of systems on IEEE 802 LANs.	•IEEE 标准 802.1E ™ [ISO/IEC 15802-4]:	系统加载协议。规定一组服务和协议 适用于管理中涉及在 IEEE 802 局域网上加载系统的那些方面。
• IEEE Std 802.1F™	<i>Common Definitions and Procedures for IEEE 802 Management Information.</i>	•IEEE 标准 802.1F ™	<i>Common Definitions and Procedures for IEEE 802 Management Information.</i>
• IEEE Std 802.1G™ [ISO/IEC 15802-5]:	<i>Remote Media Access Control (MAC) Bridging.</i> Specifies extensions for the interconnection, using non-LAN systems communication technologies, of geographically separated IEEE 802 LANs below the level of the logical link control protocol.	•IEEE 标准 802.1G ™ [ISO/IEC 15802-5]:	远程媒体访问控制（MAC）桥接。规定 使用非局域网系统通信技术互连地理上分离的 IEEE 802 LANs below the level of the logical link control protocol.
• IEEE Std 802.1H™ [ISO/IEC TR 11802-5]	<i>Recommended Practice for Media Access Control (MAC) Bridging of Ethernet V2.0 in IEEE 802 Local Area Networks.</i>	•IEEE Std 802.1H ™ [ISO/IEC TR 11802-5]	媒体访问控制（MAC）推荐实践 <i>Bridging of Ethernet V2.0 in IEEE 802 Local Area Networks.</i>
• IEEE Std 802.1Q™	<i>Virtual Bridged Local Area Networks.</i> Defines an architecture for Virtual Bridged LANs, the services provided in Virtual Bridged LANs, and the protocols and algorithms involved in the provision of those services.	•IEEE 标准 802.1Q ™	虚拟桥接局域网。定义了虚拟桥接局域网的体系结构、提供的服务，以及提供这些服务所涉及的协议和算法。
• IEEE Std 802.2 [ISO/IEC 8802-2]:	<i>Logical Link Control.</i>	•IEEE Std 802.2 [ISO/IEC 8802-2]:	<i>Logical Link Control.</i>
• IEEE Std 802.3 [ISO/IEC 8802-3]:	<i>CSMA/CD Access Method and Physical Layer Specifications.</i>	•IEEE Std 802.3 [ISO/IEC 8802-3]:	载波侦听多路访问/冲突检测访问方法和物理层规范。
• IEEE Std 802.4 [ISO/IEC 8802-4]:	<i>Token Bus Access Method and Physical Layer Specifications.</i>	•IEEE Std 802.4 [ISO/IEC 8802-4]:	令牌总线访问方法和物理层规范。
• IEEE Std 802.5 [ISO/IEC 8802-5]:	<i>Token Ring Access Method and Physical Layer Specifications.</i>	•IEEE Std 802.5 [ISO/IEC 8802-5]:	令牌环访问方法和物理层规范。
• IEEE Std 802.6 [ISO/IEC 8802-6]:	<i>Distributed Queue Dual Bus Access Method and Physical Layer Specifications.</i>	•IEEE Std 802.6 [ISO/IEC 8802-6]:	分布式队列双总线访问方法和物理层规范。
• IEEE Std 802.10:	<i>Interoperable LAN/MAN Security.</i> Currently approved: Secure Data Exchange (SDE).	•IEEE 标准 802.10:	可互操作局域网/城域网安全。目前已获批准：安全数据交换（SDE）。
• IEEE Std 802.11: [ISO/IEC 8802-11]	<i>Wireless LAN Medium Access Control (MAC) Sublayer and Physical Layer Specifications.</i>	•IEEE Std 802.11: [ISO/IEC 8802-11]	无线局域网介质访问控制（MAC）子层和物理层规范。
• IEEE Std 802.12: [ISO/IEC 8802-12]	<i>Demand Priority Access Method, Physical Layer and Repeater Specification.</i>	•IEEE Std 802.12: [ISO/IEC 8802-12]	需求优先级访问方法、物理层和中继器规范。
• IEEE Std 802.15™:	<i>Wireless Medium Access Control (MAC) and Physical Layer (PHY) Specifications for: Wireless Personal Area Networks.</i>	•IEEE Std 802.15 ™:	无线介质访问控制（介质访问控制）和物理层 <i>(PHY) Specifications for: Wireless Personal Area Networks.</i>
• IEEE Std 802.16:	<i>Standard Air Interface for Fixed Broadband Wireless Access Systems.</i>	•IEEE Std 802.16:	固定宽带无线接入的标准空中接口系统。
• IEEE Std 802.17™:	<i>Resilient Packet Ring Access Method and Physical Layer Specifications.</i>	•IEEE Std 802.17 ™:	弹性分组环访问方法和物理层规范。

In addition to the family of standards, the following is a recommended practice for a common physical layer technology:

- IEEE Std 802.7™:
- IEEE Recommended Practice for Broadband Local Area Networks.

The reader of this standard is urged to become familiar with the complete family of standards.

Conformance test methodology

An additional standards series, identified by the number 1802™, has been established to identify the conformance test methodology documents for the IEEE 802 family of standards. Thus the conformance test documents for IEEE 802.3 are numbered 1802.3™, the conformance test documents for IEEE 802.5 will be 1802.5™, and so on. Similarly, ISO will use 18802 to number conformance test standards for 8802 standards.

Participants

At the time this standard was completed, the IEEE 802.1 Working Group had the following membership:

William P. Lidinsky, Chair Tony Jeffree, Vice Chair Alan Chambers, Tony Jeffree, Editors		
Steve Adams	Ariel Hendel	Anil Rijsinghani
Floyd Backes	Steve Horowitz	Ayman Sayed
John Bartlett	Altaf Hussain	Mick Seaman
Les Bell	Vipin K. Jain	Lee Sendelbach
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John Hart	John Pickens	Michael D. Wright
Richard Hausman	Steve Ramberg	Michele Wright
David Head	Shlomo Reches	Allen Yu
Deepak Hegde	James Richmond	Wayne Zakowski

除标准族外， 以下是针对一种通用物理层技术的推荐实践：

- IEEE Std 802.7 ™:
- IEEE 宽带局域网推荐实践。

建议本标准的读者熟悉完整的标准族。

一致性测试方法

已建立另一个标准系列， 以编号 1802™ 标识 IEEE 802 标准族的一致性测试方法文档。因此， IEEE 802.3 的一致性测试文档编号为 1802.3™， IEEE 802.5 的一致性测试文档将编号为 1802.5™， 依此类推。同样， ISO 将使用 18802 为 8802 标准的一致性测试标准编号。

参与者

本标准完成时， IEEE 802.1 工作组的成员如下：

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IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture

1. Scope

1.1 General

This document serves as the foundation for the family of IEEE 802^{®1} Standards published by IEEE for Local Area Networks (LANs) and Metropolitan Area Networks (MANs). It contains descriptions of the networks considered as well as a reference model (RM) for protocol standards. Compliance with the family of IEEE 802 Standards is defined, and a standard for the identification of public, private, and standard protocols is included.

1.2 Key concepts

The LANs described herein are distinguished from other types of data networks in that they are optimized for a moderate-sized geographic area, such as a single office building, a warehouse, or a campus. An IEEE 802 LAN is a peer-to-peer communication network that enables stations to communicate directly on a point-to-point, or point-to-multipoint, basis without requiring them to communicate with any intermediate switching nodes. LAN communication takes place at moderate-to-high data rates, and with short transit delays, on the order of a few milliseconds or less.

A LAN is generally owned, used, and operated by a single organization. This is in contrast to Wide Area Networks (WANs) that interconnect communication facilities in different parts of a country or are used as a public utility.

A MAN is optimized for a larger geographical area than is a LAN, ranging from several blocks of buildings to entire cities. As with local networks, MANs can also depend on communications channels of moderate-to-high data rates. A MAN might be owned and operated by a single organization, but it usually will be used by many individuals and organizations. MANs might also be owned and operated as public utilities. They will often provide means for internetworking of local networks.

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IEEE 局域网和城域网标准：概述和体系结构

1. 范围

1.1 总则

本文档为 IEEE 发布的 IEEE 802^{®1} 局域网（LAN）和城域网（MAN）标准系列奠定基础。文档介绍了所涉及的网络，并为协议标准提供了一个参考模型（RM）。文档规定了对 IEEE 802 标准系列的合规性，并包含一项用于识别公用、专用和标准协议的标准。

1.2 关键概念

本文所述的局域网与其他类型的数据网络不同之处在于，它们针对中等规模的地理区域进行了优化，例如单栋办公楼、仓库或校园。IEEE 802 局域网是一种对等通信网络，使站点能够以点对点或点对多点的方式直接通信，而无需通过任何中间交换节点进行通信。局域网通信以中到高速率进行，传输延迟很短，通常只有几毫秒或更短。

局域网通常由单个组织拥有、使用和运营。这与广域网（WAN）不同，广域网用于连接一个国家不同地区的通信设施，或作为公用事业供公众使用。

城域网面向比局域网更大的地理区域进行优化，覆盖范围从数个街区到整座城市不等。与局域网一样，城域网也可以依赖中高速率的数据通信信道。城域网可能由单个组织拥有并运营，但通常会供许多个人和组织使用。城域网也可能作为公用事业由相关机构拥有并运营。它们通常会提供局域网互连的方式。

¹IEEE 和 802 是电气与电子工程师协会拥有的注册商标，并已在美国专利商标局注册。

Although primarily aimed at deployment on the scale of a large building or a campus, LANs are also frequently applied in smaller areas, such as small offices or single laboratories, and increasingly in homes. At the small-scale application level, a LAN is different from the type of network, such as a data bus or backplane bus, that is optimized for the interconnection of devices on a desktop or of components within a single piece of equipment. However, desktop-scale applications of LANs are also possible, particularly where the nature of the application is more suited to peer-to-peer communication among autonomous components, as opposed to a system structure with more centralized control.

The original IEEE 802 LAN technologies used shared-medium communication, with information broadcast for all stations to receive. That approach has been varied and augmented subsequently, but in ways that preserve the appearance of simple peer-to-peer communications behavior for end stations. In particular, the use of bridges (see 6.3.2) for interconnecting LANs is now widespread. These devices allow the construction of networks with much larger numbers of LAN end stations, and much higher aggregate throughput, than would be achievable with a single shared-medium LAN. End stations attached to such a bridged LAN can communicate with each other just as though they were attached to a single shared-medium LAN (however, the ability to communicate with other stations can be limited by use of management facilities in the bridges, particularly where broadcast or multicast transmissions are involved). A further stage in this evolution has led to the use of point-to-point full duplex communication in LANs, either between an end station and a bridge or as a typically high-speed link between a pair of bridges.

The basic communications capabilities provided by all LANs and MANs are packet-based, as opposed to either cell-based or isochronous. That is, the basic unit of transmission is a sequence of data octets, which can be of any length within a range that is dependent on the type of LAN; for all LAN types, the maximum length is in excess of 1000 octets. (By contrast, cell-based communication transmits data in shorter, fixed-length units; isochronous communication transmits data as a steady stream of octets, or groups of octets, at equal time intervals.)

An optional function that may be offered by a LAN or a MAN is the provision of local networking of isochronous bearer services that are compatible with, or higher speed versions of, Integrated Services Digital Networks (ISDN) as defined by the ITU-T I-series Recommendations, to support voice, video, and data devices and terminals. These services are based on the use of end-user to end-user isochronous bearers that will span the supporting Integrated Services LAN (ISLAN) or MAN and an intervening ISDN-conformant WAN. Typically, the information streams for packet and isochronous services are multiplexed over the same physical media. In addition, capabilities are specified for a single integrated management of these various streams.

1.3 Application and support

The networks are intended to have wide applicability in many environments. The primary aim is to provide for moderate-cost devices and networks, suitable for commercial, educational, governmental, and industrial applications. Low-cost alternatives are possible for some networks, and application in other environments is not precluded. The following lists are intended to show some applications and devices and, as such, are not intended to be exhaustive, nor do they constitute a set of required items:

- File transfer
- Graphics
- Text processing
- Desktop publishing
- Electronic mail
- Database access
- Transaction processing

虽然局域网主要面向大型建筑或校园规模的部署，但也经常应用于较小的区域，例如小型办公室或单个实验室，并且越来越多地应用于家庭。在小规模应用层面，局域网不同于数据总线或背板总线等网络类型；后者针对桌面设备之间或单台设备内部组件之间的互连进行了优化。不过，局域网也可以用于桌面规模的应用，尤其是在应用性质更适合自主组件之间进行对等通信，而非采用集中式控制程度更高的系统结构时。

最初的 IEEE 802 局域网技术采用共享介质通信，将信息广播给所有站点接收。此后，这种方法不断变化并得到补充，但这些变化和补充都保留了端站所感知的简单对等通信行为。尤其是，使用网桥（参见 6.3.2）互连局域网的做法如今已十分普遍。这些设备使网络能够容纳数量更多的局域网端站，并实现远高于单个共享介质局域网所能达到的总吞吐量。连接到此类网桥连接的局域网的端站，彼此通信时就如同连接在同一个共享介质局域网中一样（不过，网桥中的管理设施可能会限制其与其他站点通信的能力，尤其是在涉及广播或多播传输时）。这一演进的下一阶段使局域网开始采用点对点全双工通信，既可用于端站与网桥之间，也可作为一对网桥之间通常具有较高速度的链路。

所有局域网和城域网提供的基本通信能力都是基于分组的，而非基于信元或等时的。也就是说，传输的基本单位是一串数据字节，其长度可在一定范围内变化，具体取决于局域网类型；对于所有局域网类型，最大长度都超过 1000 个八位组。（相比之下，基于信元的通信以较短的固定长度单位传输数据；等时通信则以相等的时间间隔，将数据作为连续的八位组流或八位组组传输。）

局域网或城域网可能提供的一项可选功能，是为等时承载业务提供本地联网。这些业务与 ITU-T I 系列建议所定义的综合业务数字网 (ISDN) 兼容，或是其更高速的版本，可支持语音、视频和数据设备及终端。这些服务基于端到端用户等时承载，承载范围覆盖提供支持的综合业务局域网 (ISLAN) 或城域网，以及中间的符合 ISDN 的广域网。通常，基于分组和等时服务的信息流通过同一物理介质进行复用。此外，还规定了对这些不同信息流进行统一集成管理的能力。

1.3 应用和支持

这些网络旨在适用于多种环境。其主要目标是提供成本适中的设备和网络，适用于商业、教育、政府和工业应用。某些网络也可以采用低成本替代方案，而且并不排除在其他环境中应用。以下列表旨在列举部分应用和设备，因此并非详尽无遗，也不构成必需项目清单：

- 文件传输
- 图形
- 文本处理
- 桌面出版
- 电子邮件
- 数据库访问
- 事务处理

- Multimedia
- Office automation
- Process control
- Robotics
- Integrated Services (voice, video and data) applications
- Client/server applications

The networks are intended to support various data devices, such as the following:

- Computers
- Terminals
- Mass storage devices
- Printers and plotters
- Photocopiers and facsimile machines
- Image and video monitors
- Wireless terminals
- Monitoring and control equipment
- Bridges, routers, and gateways
- Integrated Services devices, including ISDN terminals and end systems supporting combined voice, video, and data applications

1.4 An international family of standards

The terms LAN and MAN encompass a number of data communications technologies and applications of these technologies. So it is with the IEEE 802 Standards. In order to provide a balance between the proliferation of a very large number of different and incompatible local and metropolitan networks, on the one hand, and the need to accommodate rapidly changing technology and to satisfy certain applications or cost goals, on the other hand, several types of medium access technologies are currently defined in the family of IEEE 802 Standards. In turn, these medium access control (MAC) standards are defined for a variety of physical media. A logical link control (LLC) standard, a secure data exchange standard, and MAC bridging standards are intended to be used in conjunction with the MAC standards. In some ISLAN and MAN standards, provisions are made for optionally conveying isochronous bearer services in support of continuous voice, video, and synchronous data applications. An architecture and protocols for the management of IEEE 802 LANs are also defined.

The IEEE 802 Standards have been developed and applied in the context of an increasingly global data communications industry. This global context is recognized in that most IEEE 802 Standards are progressed to become also international standards, within ISO/IEC JTC 1 (Joint Technical Committee 1, Information Technology, of the International Organization for Standardization and the International Electrotechnical Commission); see Clause 2 and Annex A.

2. References

The following publications contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to

- 多媒体
- 办公自动化
- 过程控制
- 机器人技术
- 综合业务（语音、视频和数据）应用程序
- 客户端/服务器应用

这些网络旨在支持各种数据设备， 例如：

- 计算机
- 终端
- 大容量存储设备
- 打印机和绘图仪
- 复印机和传真机
- 图像和视频监视器
- 无线终端
- 监控和控制设备
- 网桥、路由器和网关
- 综合业务设备，包括 ISDN终端， 以及支持语音、 视频和数据应用组合的端系统

1.4 国际标准族

局域网和城域网这两个术语涵盖了多种数据通信技术及其应用。IEEE 802 标准也是如此。为了在 一方面避免大量不同且互不兼容的局域网和城域网不断涌现， 另一方面满足快速变化的技术需求以及某些应用或成本目标之间取得平衡， IEEE 802 标准族目前定义了几种媒介访问技术。相应地， 这些介质访问控制（MAC）标准适用于多种物理介质。逻辑链路控制（LLC）标准、安全数据交换标准和 MAC网桥标准旨在与 MAC标准配合使用。在一些 ISLAN 和城域网标准中， 还规定了可选地传送等时承载业务， 以支持连续语音、 视频和同步数据应用。标准还定义了用于管理 IEEE 802 局域网的体系结构和协议。

IEEE 802 标准是在数据通信行业日益全球化的背景下制定和应用的。大多数 IEEE 802 标准都会进一步发展成为国际标准， 这体现了这一全球化背景； 这些国际标准由 ISO/IEC JTC 1（国际标准化组织和国际电工委员会的信息技术联合技术委员会 1）制定； 参见第 2 条和附录 A。

2. 参考文献

以下出版物中的条款通过在本文本中的引用而构成本标准的条款。出版时， 所示版本均为有效版本。所有标准均可能修订，

revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards listed below.

ISO/IEC 7498-1:1994, Information technology—Open Systems Interconnection—Basic Reference Model: The basic model.²

ISO/IEC 8802-2:1998 (IEEE Std 802.2™-1998), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 2: Logical Link Control.³

ISO/IEC TR 11802-1:1997, Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Technical reports and guidelines—Part 1: The structure and coding of Logical Link Control addresses in Local Area Networks.

ISO/IEC TR 11802-2:1999, Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Technical reports and guidelines—Part 2: Standard Group MAC Addresses.

ISO/IEC 15802-1:1995, Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 1: Medium access control (MAC) service definition.

ISO/IEC 15802-3:1998 (IEEE Std 802.1D™-1998), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 3: Media Access Control (MAC) Bridges.

NOTE—Annex A lists, for information, the other LAN/MAN and related standards. Unlike those listed above, they do not contain detailed provisions that are used, by reference, by this standard.

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this standard, the following definitions apply. *The Authoritative Dictionary of IEEE Standards Terms*, Seventh Edition [B2],⁴ should be referenced for terms not defined in this clause.

3.1.1 access domain: A set of LAN or MAN stations together with interconnecting data transmission media and related equipment (e.g., connectors, repeaters), in which the LAN or MAN stations use the same MAC protocol to establish the sequence of stations that are in temporary control of the shared transmission media.

3.1.2 bit-reversed representation: The representation of a sequence of octet values in which the values of the individual octets are displayed in order from left to right, with each octet value represented as a two-digit hexadecimal numeral, and with the resulting pairs of hexadecimal digits separated by colons. The order of the hexadecimal digits in each pair, and the mapping between the hexadecimal digits and the bits of the octet value, are derived by reversing the order of the bits in the octet value and interpreting the resulting bit sequence as a binary numeral using the normal mathematical rules for digit significance.

²ISO/IEC publications are available from the ISO Central Secretariat, Case Postale 56, 1 rue de Varembé, CH-1211, Genève 20, Switzerland/Suisse (<http://www.iso.ch/>). ISO/IEC publications are also available in the United States from Global Engineering Documents, 15 Inverness Way East, Englewood, Colorado 80112, USA (<http://global.ihs.com/>). Electronic copies are available in the United States from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (<http://www.ansi.org/>).

³The IEEE standards referred to in Clause 2 are trademarks owned by the Institute of Electrical and Electronics Engineers, Incorporated.

⁴The numbers in brackets correspond to those of the bibliography in Annex A.

基于本标准达成协议的各方宜调查应用下列标准最新版本的可能性。

ISO/IEC 7498-1:1994, 信息技术—开放系统互连—基本参考模型：基本模型。²

ISO/IEC 8802-2:1998（IEEE Std 802.2™-1998）， 信息技术—系统间电信和信息交换—局域网和城域网—特殊要求—第2部分： 逻辑链路控制。³

ISO/IEC TR 11802-1:1997, 信息技术—系统间电信和信息交换—局域网和城域网—技术报告和指南—第1部分： 局域网中逻辑链路控制地址的结构和编码。

ISO/IEC TR 11802-2:1999, 信息技术——系统间电信和信息交换——局域网和城域网——技术报告和指南——第 2 部分： 标准组 MAC 地址。

ISO/IEC 15802-1:1995, 信息技术——系统间电信和信息交换——局域网和城域网——通用规范——第1部分： 介质访问控制（MAC）服务定义。

ISO/IEC 15802-3:1998（IEEE Std 802.1D™-1998）， 信息技术——系统间电信和信息交换——局域网和城域网——通用规范——第 3 部分： 媒体访问控制（MAC）网桥。

注——附录 A 列出了其他局域网/城域网及相关标准，供参考。与上述标准不同，这些标准不包含本标准通过引用而采用的详细规定。

3. 定义、首字母缩略词和缩略语

3.1 定义

本标准采用以下定义。对于本条款中未定义的术语，应参考《IEEE 标准术语权威词典》第七版 [B2],⁴。

3.1.1 接入域：由一组局域网或城域网站点、互连的数据传输介质以及相关设备（例如连接器、中继器）组成，其中局域网或城域网站点使用相同的介质访问控制协议，以确定暂时控制共享传输介质的站点序列。

3.1.2 位反转表示：一种用于表示八位组值序列的方式，其中各个八位组的值从左到右依次显示，每个八位组值表示为两位十六进制数，所得的十六进制数字对之间用冒号分隔。每对十六进制数字的顺序，以及十六进制数字与八位组值各位之间的映射，均通过反转八位组值中各位的顺序，再按照数字位权的常规数学规则将所得位序列解释为二进制数来确定。

²ISO/IEC 出版物可向 ISO 中央秘书处索取，地址为 Case Postale 56, 1 rue de Varembé, CH-1211, Genève 20, Switzerland/Suisse (<http://www.iso.ch/>)。在美国，也可向 Global Engineering Documents 索取 ISO/IEC 出版物，地址为 15 Inverness Way East, Englewood, Colorado 80112, USA (<http://global.ihs.com/>)。在美国，还可向美国国家标准学会索取电子版，地址为 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (<http://www.ansi.org/>)。

³第 2 条中提及的 IEEE 标准是电气与电子工程师协会拥有的商标。⁴方括号中的数字与附录 A 参考文献中的编号相对应。

OVERVIEW AND ARCHITECTURE	IEEE Std 802-2001	概述与体系结构	IEEE 802-2001 标准
NOTE—The bit-reversed representation is applicable to LAN MAC addresses for use in a Token Ring (IEEE 802.5™) or FDDI environment. See Figure 8 for a comparative example of bit-reversed and hexadecimal representation.		注——位反转表示适用于令牌环（IEEE 802.5™）或光纤分布式数据接口环境中的局域网 MAC 地址。有关位反转表示与十六进制表示的对比示例，请参见图 8。	
3.1.3 bridge, MAC bridge: A functional unit that interconnects two or more LANs or MANs that use the same Data Link layer protocols above the MAC sublayer, but can use different MAC protocols.		3.1.3 网桥，MAC 网桥：一种功能单元，用于互连两个或更多使用以下协议的局域网或城域网相同的数据链路层协议（位于 MAC 子层之上）， 但可以使用不同的 MAC 协议。	
3.1.4 canonical format: The format of a MAC data frame in which the octets of any MAC addresses conveyed in the MAC user data field have the same bit ordering as in the Hexadecimal Representation.		3.1.4 规范格式：MAC 数据帧的格式， 其中任何 MAC 地址的八位组在 MAC 用户数据字段中传输的内容， 其位顺序与十六进制表示中的相同。	
3.1.5 end station: A device attached to a LAN or MAN, which acts as a source of, and/or destination for, data traffic carried on the LAN or MAN.		3.1.5 终端站：连接到局域网或城域网的设备， 作为该局域网或城域网所承载数据流量的源和/或目的地。	
3.1.6 Ethernet frame: A MAC data frame structured in accordance with ISO/IEC 8802-3 and containing an Ethernet type value in the LENGTH / TYPE field.		3.1.6 以太网帧：按照 ISO/IEC 8802-3 结构化，并在 LENGTH / TYPE 字段中包含以太网类型值的 MAC 数据帧。	
3.1.7 fibre distributed data interface (FDDI) frame: A MAC data frame structured in accordance with ISO/IEC 9314-2.		3.1.7 光纤分布式数据接口（FDDI）帧：按照 ISO/IEC 9314-2 结构化的 MAC 数据帧。	
3.1.8 hexadecimal representation: The representation of a sequence of octet values in which the values of the individual octets are displayed in order from left to right, with each octet value represented as a two-digit hexadecimal numeral, and with the resulting pairs of hexadecimal digits separated by hyphens. The order of the hexadecimal digits in each pair, and the mapping between the hexadecimal digits and the bits of the octet value, are derived by interpreting the bits of the octet value as a binary numeral using the normal mathematical rules for digit significance.		3.1.8 十六进制表示：一种八位组值序列的表示方式， 其中各个八位组的值按从左到右的顺序显示，每个八位组值表示为一个两位十六进制数，所得的十六进制数字对之间用连字符分隔。 每对十六进制数字的顺序，以及十六进制数字与八位组值的各位之间的映射关系， 是通过按照数字位权的通常数学规则， 将八位组值的各位解释为二进制数而得出的数字有效位数的数学规则。	
NOTE—See Figure 8 for a comparative example of bit-reversed and hexadecimal representation.		注——位反转表示与十六进制表示的比较示例见图 8。	
3.1.9 IEEE 802.3™ frame, 802.3 frame: A MAC data frame structured in accordance with ISO/IEC 8802-3 and containing a length value in the LENGTH / TYPE field.		3.1.9 IEEE 802.3™ 帧，802.3 帧：按照 ISO/IEC 8802-3 规定的结构构成，且 LENGTH / TYPE 字段中包含长度值的介质访问控制数据帧。	
3.1.10 IEEE 802.5 frame, 802.5 frame: A MAC data frame structured in accordance with ISO/IEC 8802-5.		3.1.10 IEEE 802.5 帧，802.5 帧：按照 ISO/IEC 8802-5 结构化的介质访问控制数据帧。	
3.1.11 IEEE 802.n frame, 802.n frame: A MAC data frame structured in accordance with ISO/IEC 8802-n.		3.1.11 IEEE 802.n 帧，802.n 帧：按照 ISO/IEC 8802-n 结构化的介质访问控制数据帧。	
NOTES		注	
1—At the time of publication of this standard, relevant specifications, in addition to those cited explicitly in 3.1.9 and 3.1.10, are for n = 4, 6, 9, and 11.		1—在本标准出版时，除 3.1.9 和 . . 中明确引用的规范外，相关规范 3 1 10, 适用于 n = 4, 6, 9, 和 11 .	
2—ISO/IEC 8802-12 also defines a MAC protocol, but it does not specify its own MAC data frame format; instead, it uses the IEEE 802.3 and IEEE 802.5 frame formats.		2—ISO/IEC 8802-12 也定义了一种介质访问控制协议，但未规定自己的介质访问控制数据帧格式；而是采用 IEEE 802.3 和 IEEE 802.5 帧格式。	
3.1.12 IEEE 802 LAN, 802 LAN: A LAN consisting of an access domain using either a MAC protocol specified in one of the IEEE 802.n and ISO/IEC 8802-n Standards or the FDDI MAC protocol.		3.1.12 IEEE 802 局域网，802 局域网：由一个接入域构成的局域网，该接入域采用以下任一介质访问控制协议该协议在 IEEE 802.n 和 ISO/IEC 8802-n 标准之一中规定，或采用光纤分布式数据接口介质访问控制协议。	
3.1.13 IEEE 802 MAN, 802 MAN: A MAN consisting of one or more interconnected subnetworks each using a MAC protocol specified in an IEEE 802 or ISO/IEC 8802 MAN Standard.		3.1.13 IEEE 802 城域网，802 城域网：由一个或多个互连子网构成的城域网，每个子网均采用 IEEE 802 或 ISO/IEC 8802 城域网标准中规定的介质访问控制协议。	
NOTE—Part of the data communication capability of an IEEE 802 MAN is the provision of a data service equivalent to that provided by an IEEE 802 LAN, over the extended geographical area of the MAN.		NOTE—IEEE 802 城域网数据通信能力的一部分是提供一种等同于 IEEE 802 局域网所提供的数据服务，并覆盖城域网的扩展地理区域。	
3.1.14 interconnection: The provision of data communication paths between LAN or MAN stations.		3.1.14 互连：在局域网或城域网站点之间提供数据通信路径。	
3.1.15 interworking: The use of interconnected LAN or MAN stations for the exchange of data, by means of protocols operating over the underlying data transmission paths.		3.1.15 互通：利用互连的局域网或城域网站点，通过在底层数据传输路径上运行的协议进行数据交换。	
Copyright © 2002 IEEE. All rights reserved.	5	版权所有 © 2002 IEEE。保留所有权利。	5

3.1.16 LAN: A computer network, located on a user’s premises, within a limited geographical area.

3.1.17 MAC control frame: A data structure consisting of fields in accordance with a MAC protocol, for the communication of control information, only, in a LAN or MAN.

NOTE—ISO/IEC 8802-5 uses the term “MAC frame” in this sense.

3.1.18 MAC data frame: A data structure consisting of fields in accordance with a MAC protocol, for the communication of user data and control information in a LAN or MAN; one of the fields contains a sequence of octets of user data.

3.1.19 MAC protocol: The protocol that governs access to the transmission medium in a LAN or MAN, to enable the exchange of data between LAN or MAN stations.

3.1.20 MAN: A computer network, extending over a large geographical area such as an urban area and providing integrated communication services such as data, voice, and video.

3.1.21 noncanonical format: The format of a MAC data frame in which the octets of MAC addresses conveyed in the MAC user data field have the same bit ordering as in the Bit-reversed representation.

3.1.22 octet: A sequence of eight bits, the ends of the sequence being identified as the most significant bit (MSB) and the least significant bit (LSB).

NOTE—This identification of the ends of the sequence defines an unambiguous mapping from octet values, via binary numerals, to the integers 0–255, and hence a mapping also from octet values to the expressions of those integers as numerals in hexadecimal notation. See: Hexadecimal Representation.

3.1.23 station: An end station or bridge.

4. Abbreviations and acronyms

CMIP	common management information protocol (ISO/IEC 9596-1)
CSMA/CD	carrier sense multiple access with collision detection (ISO/IEC 8802-3)
DQDB	distributed queue dual bus
FDDI	fibre distributed data interface (ISO/IEC 9314)
IM	implementation model
I/G	individual/group
ISDN	integrated services digital network
ISLAN	integrated services LAN (ISO/IEC 8802-9)
LAN	local area network
LLC	logical link control (ISO/IEC 8802-2)
LSAP	link service access point (ISO/IEC 8802-2)
LSB	least significant bit
MAC	medium access control, media access control ⁵
MAN	metropolitan area network

⁵Both forms are used, with the same meaning. This standard uses “medium.”

3.1.16 局域网： 位于用户场所内、覆盖范围有限的地理区域中的计算机网络。

3.1.17 MAC 控制帧： 由符合 MAC 协议的字段构成的数据结构， 用于仅在局域网或城域网中传输控制信息。

注释——ISO/IEC 8802-5 在此意义上使用术语 “MAC 帧”。

3.1.18 MAC 数据帧： 一种数据结构， 由符合 MAC 协议的字段组成， 用于在局域网或城域网中传输用户数据和控制信息； 其中一个字段包含一串用户数据八位组。

3.1.19 MAC 协议： 一种用于管理局域网或城域网中传输介质访问的协议， 以便实现局域网或城域网站点之间的数据交换。

3.1.20 城域网： 一种覆盖较大地理区域（如城市区域）的计算机网络， 并且提供数据、 语音和视频等综合通信服务。

3.1.21 非规范格式： 一种 MAC 数据帧格式， 其中 MAC 地址的八位组在 MAC 用户数据字段中传输的内容具有与位反转表示相同的位顺序。

3.1.22 八位字节： 由八个位组成的序列， 其两端分别标识为最高有效位 (MSB) 和最低有效位 (LSB)。

注——对序列两端的这种标识定义了一种明确的映射： 通过二进制数， 将八位组值映射为整数 0–255， 并由此也将八位组值映射为这些整数的十六进制表示法。参见： 十六进制表示。

3.1.23 站点： 终端站或网桥。

4. 缩略语和首字母缩略词

CMIP	通用管理信息协议 (ISO/IEC 9596-1)
载波侦听多路访问/冲突检测	(ISO/IEC 8802-3)
DQDB	分布式队列双总线
FDDI	光纤分布式数据接口（ISO/IEC 9314）
IM	实现模型
I/G	个人/组
ISDN	综合业务数字网
ISLAN	综合业务局域网 (ISO/IEC 8802-9)
LAN	局域网
LLC	逻辑链路控制 (ISO/IEC 8802-2)
LSAP	链路服务访问点 (ISO/IEC 8802-2)
LSB	最低有效位
MAC	介质访问控制， 媒体访问控制 ⁵
MAN	城域网

⁵这两种形式都在使用， 含义相同。本标准使用 “介质”。

OVERVIEW AND ARCHITECTURE		IEEE Std 802-2001	概述与体系结构	IEEE 802-2001 标准
MIB	management information base		MIB	管理信息库
MOCS	managed object conformance statement		MOCS	受管对象一致性声明
MSAP	MAC service access point		MSAP	MAC 服务接入点
MSB	most significant bit		MSB	最高有效位
PDU	protocol data unit		PDU	协议数据单元
PhSAP	physical service access point		物理服务访问点	物理服务访问点
PICS	protocol implementation conformance statement		PICS	协议实现一致性声明
RM	reference model		RM	参考模型
OSI	open systems interconnection (ISO/IEC 7498-1)		OSI	开放系统互连 (ISO/IEC 7498-1)
OUI	organizationally unique identifier		OUI	组织唯一标识符
SNAP	subnetwork access protocol		SNAP	子网访问协议
SNMP	simple network management protocol (RFC 1157 ⁶)		SNMP	简单网络管理协议 (RFC 11576)
U/L	universally or locally administered		U/L	普遍管理或本地管理
WAN	wide area network		WAN	广域网

5. Compliance

NOTE—IEEE policy with respect to claims of compliance, conformance, or compatibility with IEEE standards can be found in the Trademark Policy statement in the front matter. This clause will be deleted in the next revision of this standard.

6. Reference and implementation models

6.1 Introduction

This clause defines the IEEE 802 LAN and MAN RM (LAN&MAN/RM) and implementation model (LAN&MAN/IM). The intent of presenting these models is as follows:

- a) To provide an overview of the standard
- b) To serve as a guide to reading other IEEE 802 Standards

The IEEE 802 LAN&MAN/RM is patterned after the Open Systems Interconnection (OSI) Basic Reference Model (OSI/RM), ISO/IEC 7498-1. It is assumed that the reader has some familiarity with the OSI/RM and its terminology. The IEEE 802 Standards encompass the functionality of the lowest two layers of the OSI/RM (i.e., Physical layer and Data Link layer) and the higher layers as they relate to LAN management. The LAN&MAN/RM is similar to the OSI/RM in terms of its layers and the placement of its service boundaries.

For the mandatory packet services supported by all LANs and MANs, the Data Link layer is structured as two sublayers, with the LLC sublayer operating over a MAC sublayer. In addition, some IEEE 802 LAN technologies provide direct support by the MAC sublayer for an alternative Ethernet sublayer operating at the same place in the architecture as does LLC; for the other IEEE 802 LAN technologies, the equivalent

⁶See A.1 for information on obtaining RFCs.

OVERVIEW AND ARCHITECTURE		IEEE Std 802-2001	概述与体系结构	IEEE 802-2001 标准
MIB	management information base		MIB	管理信息库
MOCS	managed object conformance statement		MOCS	受管对象一致性声明
MSAP	MAC service access point		MSAP	MAC 服务接入点
MSB	most significant bit		MSB	最高有效位
PDU	protocol data unit		PDU	协议数据单元
PhSAP	physical service access point		物理服务访问点	物理服务访问点
PICS	protocol implementation conformance statement		PICS	协议实现一致性声明
RM	reference model		RM	参考模型
OSI	open systems interconnection (ISO/IEC 7498-1)		OSI	开放系统互连 (ISO/IEC 7498-1)
OUI	organizationally unique identifier		OUI	组织唯一标识符
SNAP	subnetwork access protocol		SNAP	子网访问协议
SNMP	simple network management protocol (RFC 1157 ⁶)		SNMP	简单网络管理协议 (RFC 11576)
U/L	universally or locally administered		U/L	普遍管理或本地管理
WAN	wide area network		WAN	广域网

5. 合规性

注——关于 IEEE 标准合规性、符合性或兼容性声明的 IEEE 政策，请参阅前言部分的商标政策声明。本条款将在本标准的下次修订中删除。

6. 参考模型和实现模型

6.1 引言

本条定义了 IEEE 802 局域网和城域网参考模型（LAN&MAN/RM）以及实现模型（LAN&MAN/IM）。提出这些模型的目的如下：

- a) 概述该标准
- b) 为阅读其他 IEEE 802 标准提供指引

IEEE 802 LAN&MAN参考模型以开放系统互连（OSI）基本参考模型（开放系统互连参考模型）ISO/IEC 7498-1 为蓝本。假定读者对开放系统互连参考模型及其术语有所了解。IEEE 802 标准涵盖开放系统互连参考模型最低两层（即物理层和数据链路层）的功能，以及与 LAN 管理相关的高层功能。局域网和城域网参考模型在层次结构和服务边界的位置方面与开放系统互连参考模型相似。

对于所有局域网和城域网都支持的强制分组服务，数据链路层分为两个子层：LLC 子层运行在 MAC 子层之上。此外，某些 IEEE 802 局域网技术由 MAC 子层直接支持另一种以太网子层，该子层在架构中的位置与 LLC 相同；对于其他 IEEE 802 局域网技术，则通过封装以太网子层信息来提供等效功能。

⁶有关如何获取征求意见稿的信息，请参阅 A.1。

functionality is provided by encapsulation of the Ethernet sublayer information within LLC Protocol Data Units (PDUs), using the Subnetwork Access Protocol specified in Clause 10 of this standard.

Optionally, some Integrated Service LANs and MANs may also support ITU-T compatible isochronous bearer services at the Physical layer.

The OSI/RM is referred to by the IEEE 802 Standards because of the following:

- a)

The OSI/RM provides a common vehicle for understanding and communicating the various components and interrelationships of the standards.
- b)

The OSI/RM helps define terms.
- c)

The OSI/RM provides a convenient framework to aid in the development and enhancement of the standards.
- d)

The use of the OSI/RM facilitates a higher degree of interoperability than might otherwise be possible.

Figure 1 shows the architectural view of LAN&MAN/RM and its relation to the OSI/RM.

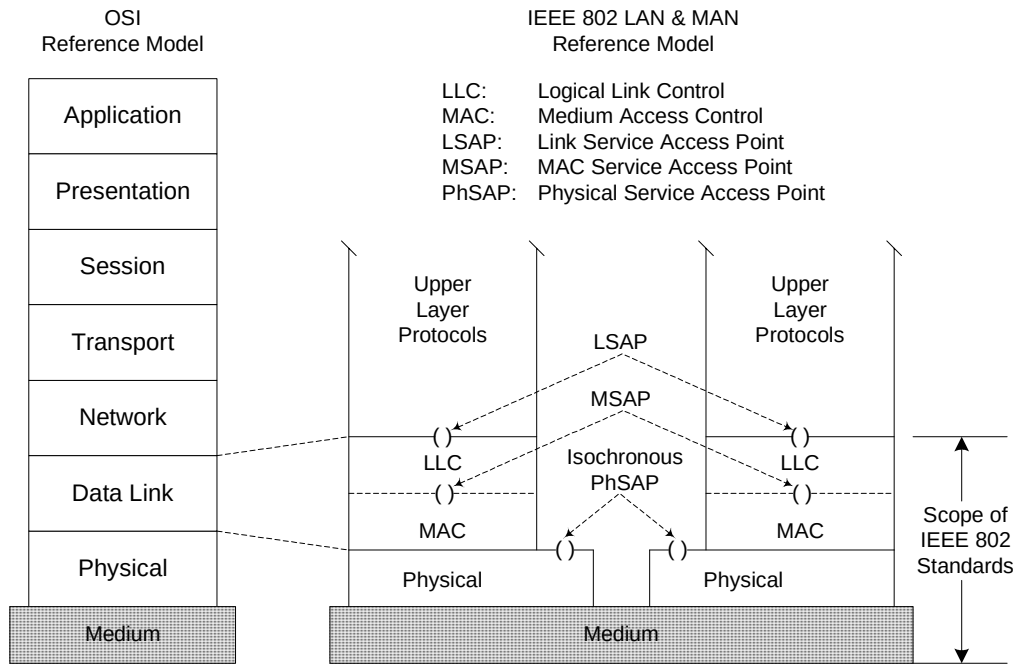


Figure 1—IEEE 802 RM for end stations (LAN&MAN/RM)

The LAN&MAN/IM is more specific than the LAN&MAN/RM, allowing differentiation between implementation approaches (e.g., of carrier sense multiple access with collision detection [CSMA/CD], and token passing). Figure 2 shows two implementation views of LAN&MAN/IMs and their relation to the LAN&MAN/RM.

Both Figure 1 and Figure 2 illustrate the application of the models in LAN end stations. A variation of the model applies within bridges (see 6.3.2). Also, Figure 1 and Figure 2 illustrate only the basic transfer of user data between end stations.

以太网子层信息被封装在 LLC 协议数据单元（PDU）中，所使用的子网访问协议由本标准第 10 条规定。

某些综合业务局域网和城域网也可以选择支持与 ITU-T 兼容的物理层等时承载业务。

IEEE 802 标准引用开放系统互连参考模型的原因如下：

- a)

开放系统互连参考模型为理解和交流各种标准的组件及其相互关系。
- b)

开放系统互连参考模型有助于定义术语。
- c)

开放系统互连参考模型提供了一个便捷的框架，有助于制定和改进标准。
- d)

使用开放系统互连参考模型有助于实现更程度的互操作性，否则可能无法达到这样的程度。

图 1 展示了局域网和城域网参考模型的架构视图及其与开放系统互连参考模型的关系。

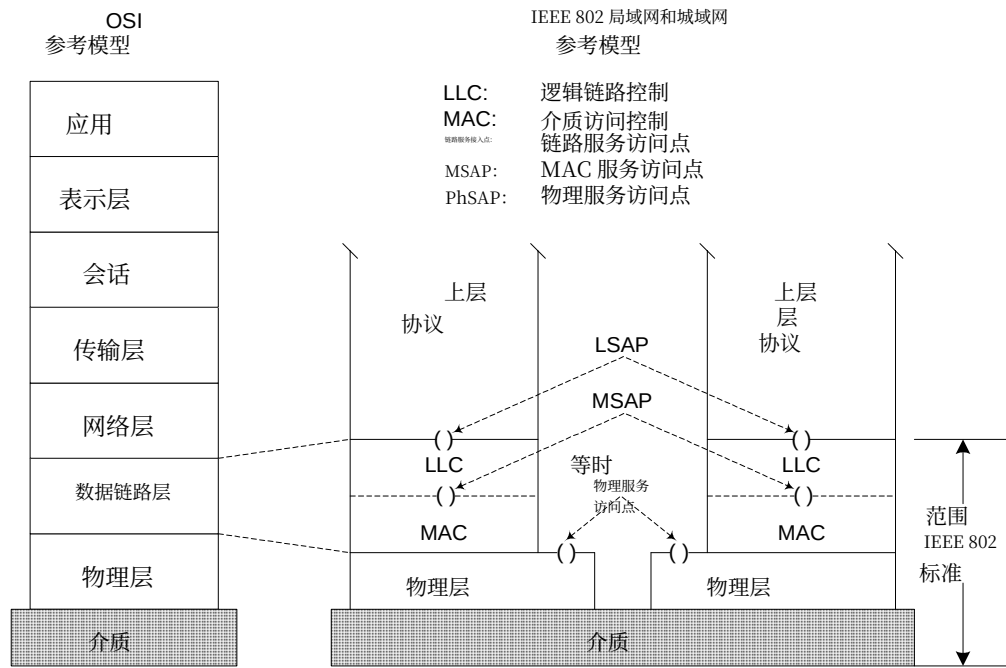


图 1—端站的 IEEE 802 参考模型（局域网和城域网参考模型）

LAN&MAN 实现模型比 LAN&MAN 参考模型更具体，可用于区分不同的实施方法（例如，载波侦听多路访问／冲突检测 [载波侦听多路访问/冲突检测], 和令牌传递）。图 2 展示了 LAN&MAN 实现模型的两种实施视图，以及它们与 LAN&MAN 参考模型的关系。

图 1 和图 2 均说明了这些模型在局域网端站中的应用。网桥内部则采用该模型的一种变体（见 6.3.2）。此外，图 1 和图 2 仅说明了端站之间用户数据的基本传输。

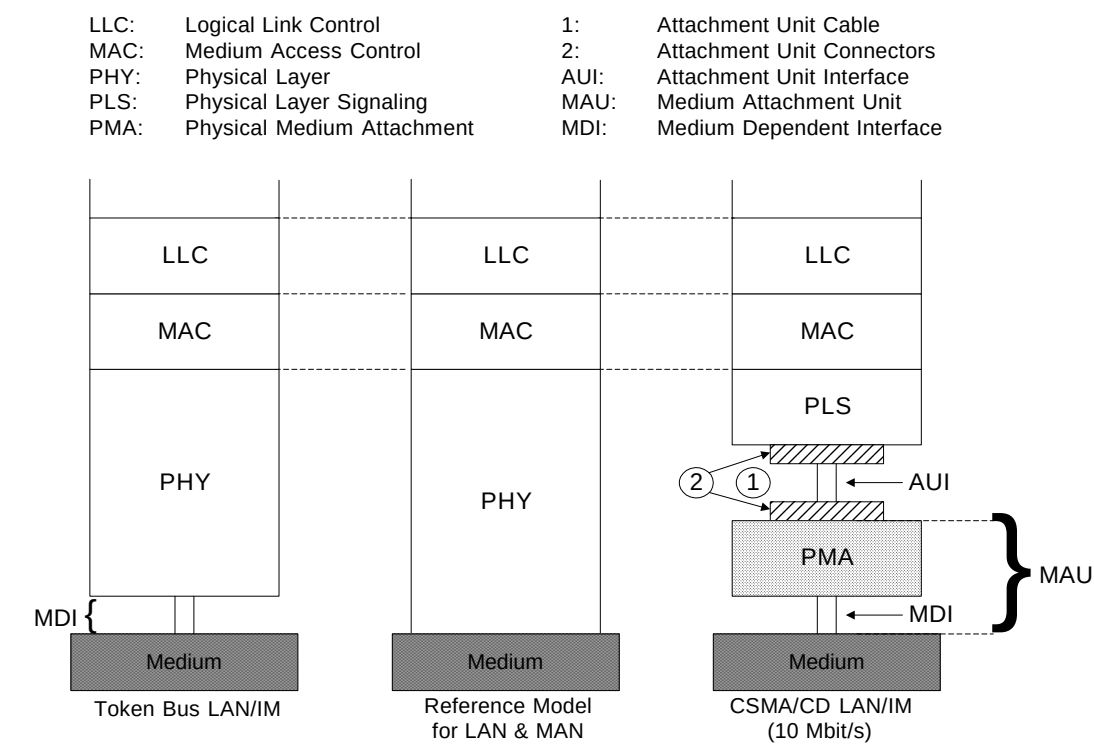


Figure 2—IEEE 802 RM and two examples of end-station implementation model

Considerations of management and security in LANs and MANs are also covered by IEEE 802 standards; these optional features lead to an elaboration of the RM (Figure 3). LAN/MAN management provides a Data Link layer management protocol for exchange of management information between LAN stations; managed objects are defined for all LAN/MAN protocol standards. The Secure Data Exchange (SDE) entity forms part of the LLC sublayer and provides a secure connectionless service immediately above the MAC sublayer.

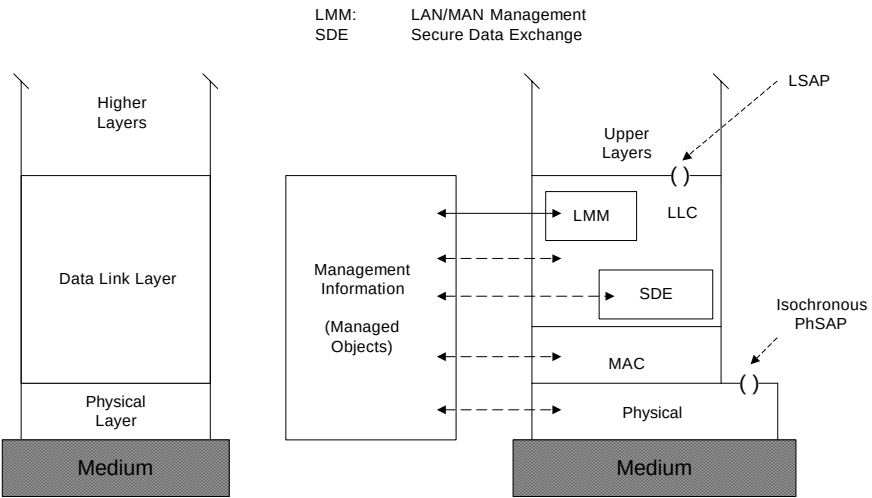


Figure 3—IEEE 802 RM with end-station management and security

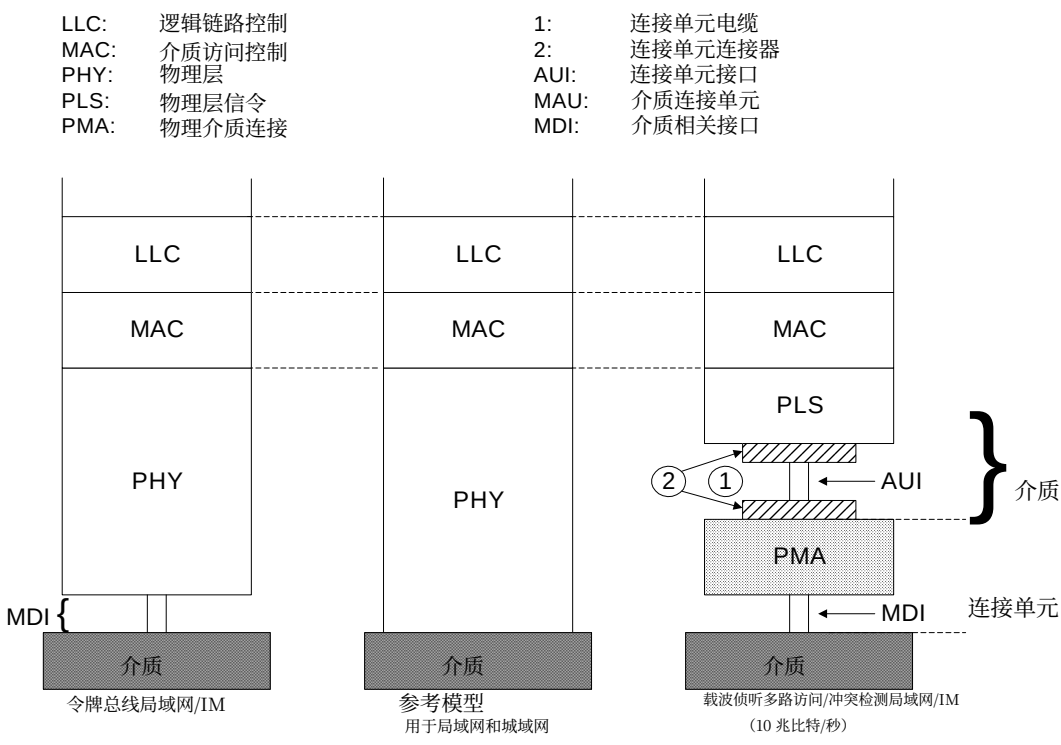


图 2—IEEE 802 参考模型和端站实现模型的两个示例

IEEE 802 标准也涵盖了局域网和城域网中的管理与安全相关事项；这些可选功能使参考模型得以扩展（图 3）。局域网/城域网管理提供数据链路层管理协议，用于在局域网站点之间交换管理信息；所有局域网/城域网协议标准均定义了受管对象。安全数据交换（SDE）实体是 LLC 子层的一部分，在 MAC 子层之上直接提供安全无连接服务。

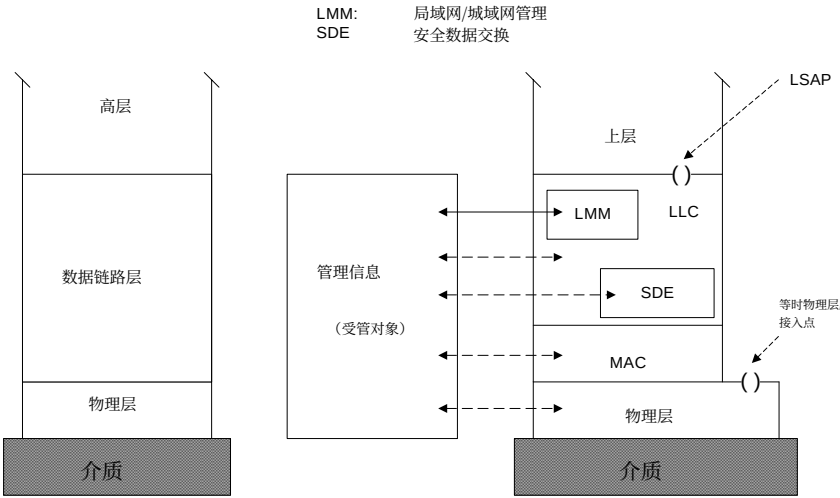


图 3—包含端站管理和安全的 IEEE 802 参考模型

6.2 RM description for end stations

The LAN&MAN/RM maps to the OSI/RM as shown in Figure 1. The applicable part of the OSI/RM consists of the lowest two layers: the Data Link layer and the Physical layer. These map onto the same two layers in the IEEE LAN&MAN/RM. The MAC sublayer of the LAN&MAN/RM exists between the Physical layer and the LLC sublayer to provide a common service for the LLC sublayer (certain MAC types provide additional MAC service features that can be used by LLC, in addition to the common core features). Service access points (SAPs) for addressing endpoints are shown.

6.2.1 Service access points (SAPs)

Multiple-link service access points (LSAPs) provide interface ports to support multiple higher layer users above the LLC sublayer.

The MAC sublayer provides a single MAC service access point (MSAP) as an interface port to the LLC sublayer in an end station. In general, the MSAP is identified (for transmission and reception) by a single individual MAC address and (for reception) by the LAN-wide broadcast MAC address; it can also be identified (for reception) by one or more group MAC addresses. Clause 9 provides details of how these MAC addresses are constructed and used; see also ISO/IEC 15802-1.

A user of LLC is identified by, at a minimum, the logical concatenation of the MAC address field and the LLC address field in a frame. See ISO/IEC 8802-2 and ISO/IEC TR 11802-1 for a description of LLC addresses.

The Physical layer provides an interface port to a single MAC station, and in the case of ISLANs and MANs, it may optionally offer isochronous bearer services at multiple Physical service access points (PhSAPs). See 11.3 for a more detailed description of ISLAN and MAN PhSAP addressing requirements.

6.2.2 LLC sublayer

The LLC sublayer standard, ISO/IEC 8802-2, describes three types of operation for data communication between service access points: unacknowledged connectionless-mode (type 1), connection-mode (type 2), and acknowledged connectionless-mode (type 3).

With type 1 operation, information frames are exchanged between LLC entities without the need for the prior establishment of a logical link between peers. The LLC sublayer does not provide any acknowledgments for these LLC frames, nor does it provide any flow control or error recovery procedures.

LLC type 1 also provides a TEST function and an Exchange Identification (XID) function. The capability to act as responder for each of these functions is mandatory: This allows a station that chooses to support initiation of these functions to check the functioning of the communication path between itself and any other station, to discover the existence of other stations, and to find out the LLC capabilities of other stations.

With type 2 operation, a logical link is established between pairs of LLC entities prior to any exchange of information frames. In the data transfer phase of operation, information frames are transmitted and delivered in sequence. Error recovery and flow control are provided, within the LLC sublayer.

With type 3 operation, information frames are exchanged between LLC entities without the need for the prior establishment of a logical link between peers. However, the frames are acknowledged to allow error recovery and proper ordering. Further, type 3 operation allows one station to poll another for data.

NOTE—ISO/IEC 8802-2 defines four classes of LLC, each of which groups together support for a different combination of LLC types. All classes include mandatory support of type 1.

6.2 终端站的参考模型描述

局域网和城域网参考模型映射到开放系统互连参考模型，如图 1 所示。适用的开放系统互连参考模型部分由最低的两层组成：数据链路层和物理层。这两层映射到 IEEE 局域网和城域网参考模型中的相同两层。局域网和城域网参考模型的介质访问控制子层位于物理层和 LLC 子层之间，为 LLC 子层提供通用服务（某些介质访问控制类型除通用核心功能外，还提供 LLC 可使用的其他介质访问控制服务功能）。图中还显示了用于寻址端点的服务访问点（SAP）。

6.2.1 服务访问点（SAP）

多链路服务访问点（LSAP）提供接口端口，以支持 LLC 子层之上的多个高层用户。

在终端站中，MAC 子层向 LLC 子层提供单个 MAC 服务接入点（MSAP）作为接口端口。通常，MSAP 通过单个单播 MAC 地址（用于发送和接收）以及局域网范围广播 MAC 地址（用于接收）来标识；也可以通过一个或多个组 MAC 地址（用于接收）来标识。第 9 条详细说明了这些 MAC 地址的构造和使用方法；另请参见 ISO/IEC 15802-1。

LLC 用户至少通过帧中的 MAC 地址字段与 LLC 地址字段的逻辑串接来标识。有关 LLC 地址的说明，请参见 ISO/IEC 8802-2 和 ISO/IEC TR 11802-1。

物理层向单个 MAC 站提供接口端口；对于综合业务局域网和城域网，它还可以选择在多个物理层服务访问点（PhSAP）提供等时承载业务。有关综合业务局域网和城域网 PhSAP 寻址要求的更详细说明，请参见 11.3。

6.2.2 逻辑链路控制子层

LLC 子层标准 ISO/IEC 8802-2 描述了服务访问点之间数据通信的三种操作类型：无确认无连接模式（类型 1）、连接模式（类型 2）和有确认无连接模式（类型 3）。

在类型 1 操作中，逻辑链路控制实体之间无需事先在对等方之间建立逻辑链路，即可交换信息帧。LLC 子层不为这些 LLC 帧提供任何确认，也不提供任何流量控制或差错恢复程序。

LLC 类型 1 还提供测试功能和交换标识（XID）功能。必须具备作为这两种功能响应方的能力：这样，选择支持发起这些功能的站点就可以检查自身与任何其他站点之间通信路径的运行情况、发现其他站点是否存在，并了解其他站点的 LLC 能力。

在类型 2 操作中，任何信息帧交换开始之前，都会在成对的 LLC 实体之间建立逻辑链路。在数据传输阶段，信息帧会按顺序传输和交付。LLC 子层提供差错恢复和流量控制。

在类型 3 操作中，LLC 实体之间无需事先在对等方之间建立逻辑链路，即可交换信息帧。不过，这些帧会得到确认，以便进行差错恢复并确保正确排序。此外，类型 3 操作允许一个站点轮询另一个站点以获取数据。

注——ISO/IEC 8802-2 定义了四种 LLC 类别，每种类别都组合支持不同的 LLC 类型。所有类别都必须支持类型 1。

The Secure Data Exchange (SDE) entity forms part of the LLC sublayer and provides a secure connectionless service immediately above the MAC sublayer. The operation of the SDE entity is described in IEEE Std 802.10™.

6.2.3 MAC sublayer

The MAC sublayer performs the functions necessary to provide packet-based, connectionless-mode (datagram style) data transfer between stations in support of the LLC sublayer or of the Ethernet sublayer (see 6.1) for LANs that support it. The term *MAC frame*, or simply *frame*, is used to describe the packets transferred within the MAC sublayer. In some MAC types (e.g., Token Ring), some MAC frames are used in support of the MAC sublayer functionality itself, rather than for transfer of LLC data.

The principal functions of the MAC sublayer comprise the following:

- Frame delimiting and recognition
- Addressing of destination stations (both as individual stations and as groups of stations)
- Conveyance of source-station addressing information
- Transparent data transfer of LLC PDUs, or of equivalent information in the Ethernet sublayer
- Protection against errors, generally by means of generating and checking frame check sequences
- Control of access to the physical transmission medium

Other functions of the MAC sublayer—applicable particularly when the supporting implementation includes interconnection devices such as hubs or bridges—include flow control between an end station and an interconnection device (see 6.3), and filtering of frames according to their destination addresses to reduce the extent of propagation of frames in parts of a LAN or MAN that do not contain communication paths leading to the intended destination end station(s).

The functions listed are those of the MAC sublayer as a whole. Responsibility for performing them is distributed across the transmitting and receiving end stations, and any interconnection devices such as bridges. Devices with different roles therefore can behave differently in support of a given function. For example, transmission of a MAC frame by a bridge is very similar to transmission by an end station, but not identical. Principally, the handling of source-station addressing is different.

The various MAC specifications all specify MAC frame formats in terms of a serial transmission model for the service provided by the supporting Physical layer. This model supports concepts such as “first bit (e.g., of a particular octet) to be transmitted,” and a strict order of octet transmission, in a uniform manner. However, the ways in which the model has been applied in different MAC specifications are not completely uniform with respect to bit-ordering within octets (see Clause 9, and particularly 9.5, for examples and explanation).

NOTE—The serial transmission model does not preclude current or future MAC specifications from using partly or wholly octet-oriented specifications of frame formats or of the interface to the Physical layer.

6.2.4 Physical layer

The Physical layer provides the capability of transmitting and receiving bits between Physical layer entities. A pair of Physical layer entities identifies the peer-to-peer unit exchange of bits between two MAC users, and in the case of ISLANs and MANs, it may optionally support the exchange of bits with end-to-end timing preserved between isochronous-service PhSAPs.

The Physical layer provides the capability of transmitting and receiving modulated signals assigned to specific frequency channels, in the case of broadband or wireless, or to a single-channel band, in the case of baseband.

安全数据交换（SDE）实体是 LLC 子层的一部分，在 MAC 子层正上方提供安全无连接服务。SDE 实体的操作在 IEEE Std 802.10™ 中进行了描述。

6.2.3 MAC 子层

MAC 子层执行必要的功能，以便在支持该功能的局域网中，为逻辑链路控制子层或以网子层（参见 6.1）提供站点之间基于分组、无连接模式（数据报方式）的数据传输。MAC 子层内传输的分组称为 MAC 帧，或简称帧。在某些 MAC 类型（例如令牌环）中，某些 MAC 帧用于支持 MAC 子层自身的功能，而非传输 LLC 数据。

MAC 子层的主要功能包括：

- 帧定界与识别
- 目的站寻址（既包括单个站点，也包括站点组）
- 传送源站寻址信息
- 对 LLC 协议数据单元，或以网子层中等效信息进行透明数据传输
- 错误保护，通常通过生成和校验帧校验序列来实现
- 控制对物理传输介质的访问

MAC 子层的其他功能——尤其适用于支持该子层的实施包含集线器或网桥等互连设备的情况——包括终端站与互连设备之间的流量控制（见 6.3），以及根据帧的目的地址对帧进行过滤，以减少帧在局域网或城域网中某些部分的传播范围；这些部分不包含通向预期目的地终端站的通信路径。

所列功能是 MAC 子层整体所具备的功能。执行这些功能的责任由发送端站、接收端站以及网桥等互连设备共同承担。因此，承担不同角色的设备在支持某项功能时可能会有不同的行为。例如，网桥发送 MAC 帧的方式与终端站发送 MAC 帧的方式非常相似，但并不完全相同。主要区别在于源站寻址的处理方式。

各种 MAC 规范都根据支持该服务的物理层所提供服务的串行传输模型，规定了 MAC 帧格式。该模型以统一的方式支持诸如“首先传输的比特（例如，某个特定八位字节中的比特）”以及严格的八位组传输顺序等概念。然而，不同 MAC 规范应用该模型的方式，在八位组内的位序方面并不完全一致（示例和说明见第 9 条，尤其是 9.5）。

注——串行传输模型并不排除当前或未来的 MAC 规范采用部分或完全以八位字节为导向的帧格式规范，或物理层接口规范。

6.2.4 物理层

物理层提供在物理层实体之间传输和接收位的能力。一对物理层实体定义了两个 MAC 用户之间的对等位单元交换；对于综合业务局域网和城域网，还可以选择支持在等时业务物理层服务接入点之间交换位，同时保留端到端定时。

物理层提供传输和接收调制信号的能力：对于宽带或无线通信，这些信号分配到特定频率信道；对于基带通信，则分配到单信道频带。

Note that, whereas the service offered to the MAC sublayer is expressed as the transfer of bits (in sequences representing MAC frames), the actual symbols that are encoded for transmission do not always represent individual bits. Particularly at speeds of 100 Mbit/s and above, or for wireless transmission, the Physical layer can map blocks of several bits (e.g., 4, 5, or 8 bits) to different multi-element symbols. In some Physical-layer encodings, these symbols are subject to further transformation before transmission, and in some cases, the transmission is spread over multiple physical data paths. The actual transmission on physical media can therefore be far removed from the simple bit-serial representation of a MAC frame (as was specified, for example, in the original ISO/IEC 8802-3 and ISO/IEC 8802-5 Physical layers).

6.2.5 Layer and sublayer management

The LLC sublayer, MAC sublayer, and Physical layer standards also include a management component that specifies managed objects and aspects of the protocol machine that provides the management view of these resources. See Clause 8 for further information.

6.3 Interconnection and interworking

In some cases, the end systems on a LAN or MAN have no need to communicate with end systems on other networks (other LANs, WANs, etc.). However, this is not expected to be the norm; there are many cases in which end stations on a LAN or MAN will need to communicate with end systems on other networks and so devices that interconnect the LAN or MAN with other kinds of networks are required. In addition, several standard methods have been developed that permit a variety of interconnection devices to operate transparently to end stations on a LAN or MAN in order to extend the LAN/MAN capabilities available to end stations, particularly in terms of the geographical extent and/or total number of end stations that can be supported.

Standard methods of interworking fall into the following three general categories, depending on the layer at which the corresponding interconnection devices operate:

- Physical-layer interconnection, using devices usually termed *repeaters* or *hubs* (6.3.1)
- MAC-sublayer interconnection, using devices termed *bridges* (6.3.2)
- Network-layer interconnection, using devices usually termed *routers* (6.3.3)

See also Clause 11 of this standard, for an outline of the optional methods by which ISLANs and MANs may support isochronous interworking with WANs and with remote ISLANs and MANs.

6.3.1 Physical-layer interconnection: repeaters and hubs

The original IEEE 802 LAN specifications were for end stations attached to a shared communication medium. This basic LAN configuration is referred to as a single *access domain*; the domain consists of the set of LAN stations such that at most one can be transmitting at a given time, with all other stations acting as (potential) receivers.

A repeater is a device used to interconnect segments of the physical communications media, for example, to extend the range of a LAN when the physical specifications of the technology would otherwise be exceeded, while providing a single access domain for the attached LAN stations. Repeaters used in support of multiple end stations attached by star-wired network topologies are frequently referred to as hubs.

请注意，虽然提供给 MAC 子层的服务表述为传输位（以表示 MAC 帧的序列形式），但经过编码以供传输的实际符号并不总是代表单个位。尤其是在速度达到 100 兆比特/秒及以上，或进行无线传输时，物理层可以将由多个位组成的数据块（例如 4、5 或 8 位）映射为不同的多元素符号。在某些物理层编码中，这些符号在传输前还要经过进一步转换；在某些情况下，传输还会分散到多条物理数据路径上。因此，物理介质上的实际传输可能与 MAC 帧的简单位串行表示大相径庭（例如，最初的 ISO/IEC 8802-3 和 ISO/IEC 8802-5 物理层就是如此规定的）。

6.2.5 层和子层管理

逻辑链路控制子层、介质访问控制子层和物理层标准还包含一个管理组件，用于指定受管对象以及协议机中提供这些资源管理视图的相关方面。更多信息请参见第 8 条。

6.3 互连和互通

在某些情况下，局域网或城域网中的端系统无需与其他网络（其他局域网、广域网等）中的端系统通信。不过，这并非通常情况；许多情况下，局域网或城域网中的端站需要与其他网络中的端系统通信，因此需要使用设备将局域网或城域网与其他类型的网络互连。此外，人们还开发了若干标准方法，使各种互连设备能够对局域网或城域网中的端站透明运行，从而扩展端站可用的局域网/城域网能力，尤其是扩大其地理覆盖范围和/或可支持的端站总数。

互通的标准方法根据相应互连设备运行所在的层，分为以下三个总则类别：

- 物理层互连，使用通常称为中继器或集线器的设备（6.3.1）
- 介质访问控制子层互连，使用称为网桥的设备（6.3.2）
- 网络层互连，使用通常称为路由器的设备（6.3.3）

另请参阅本标准第 11 条，其中概述了 ISLAN 和城域网可采用的可选方法，以支持与广域网以及远程 ISLAN 和城域网进行等时互通。

6.3.1 物理层互连：中继器和集线器

最初的 IEEE 802 局域网规范针对的是连接到共享通信介质的端站。这种基本的局域网配置称为单一接入域；该域由一组局域网站点组成，其中任意给定时间至多只有一个站点可以发送，其他所有站点则作为（潜在的）接收端。

中继器是一种用于互连物理通信介质网段的设备，例如，当技术的物理规范限制了局域网的传输范围时，可用它来扩展传输范围，同时为所连接的局域网站点提供单一接入域。用于支持通过星形布线网络拓扑连接的多个端站的中继器，通常称为集线器。

6.3.2 MAC-sublayer interconnection: bridges

6.3.2.1 Bridges and bridged LANs

Bridges (see 3.1.3) are devices that interconnect multiple access domains. ISO/IEC 15802-3 provides the basic specification for bridge interworking among IEEE 802 networks. A *bridged* LAN (see 3.1 of ISO/IEC 15802-3) consists of one or more bridges together with the complete set of access domains that they interconnect. A bridged LAN provides end stations belonging to any of its access domains with the appearance of a LAN that contains the whole set of attached end stations.

A bridged LAN can provide for the following:

- Communication between stations attached to LANs of different MAC types
- An increase in the total throughput of a LAN
- An increase in the physical extent of, or number of permissible attachments to, a LAN
- Partitioning of the physical LAN for administrative or maintenance reasons

The term *switch* is often used to refer to some classes of bridge. However, there is no consistent meaning applied to the distinction between the terms bridge and switch, and ISO/IEC 15802-3 does not make any such distinction. Hence, this standard only uses the term bridge.

6.3.2.2 Relaying and filtering by bridges

A bridge processes protocols in the MAC sublayer and is functionally transparent to LLC and higher layer protocols. MAC frames are forwarded between access domains, or filtered (i.e., not forwarded to certain access domains), on the basis primarily of MAC addressing information. Figure 4 shows the position of the bridging functions within the MAC sublayer; note particularly that the relaying and filtering functions are considered to belong entirely within the MAC sublayer.

Filtering by bridges tends to confine traffic to only those parts of the bridged LAN that lie between transmitting end stations and the intended receivers. This permits a bridged LAN to support several transmitting end stations at any given time (up to the total number of access domains present).

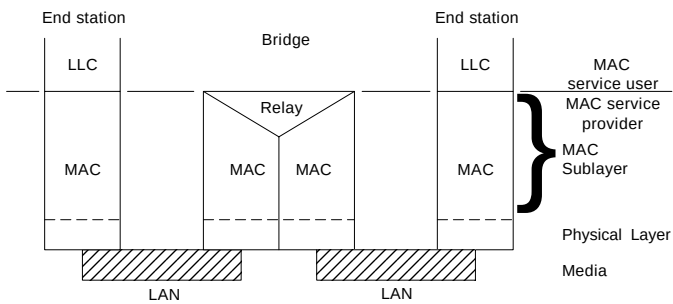


Figure 4—Internal organization of the MAC sublayer with bridging

6.3.2.3 The Spanning Tree Protocol

A key aspect of ISO/IEC 15802-3 is its specification of the Spanning Tree Protocol, which is used by bridges to configure their interconnections in order to prevent looping data paths in the bridged LAN. In the event that the basic interconnection topology of bridges and LANs contains multiple possible paths between certain points, use of the Spanning Tree Protocol blocks some paths in order to produce a simply connected

6.3.2 MAC 子层互连：网桥

6.3.2.1 网桥和网桥连接的局域网

网桥（见 3.1.3）是用于互连多个接入域的设备。ISO/IEC 15802-3 为 IEEE 802 网络之间的网桥互通提供了基本规范。网桥连接的局域网（见 ISO/IEC 15802-3 的 3.1）由一个或多个网桥以及它们互连的完整接入域集合组成。网桥连接的局域网使属于其任一接入域的端站看到一个包含所有已连接端站的局域网。

网桥连接的局域网可以提供以下功能：

- 不同 MAC 类型局域网所连接站点之间的通信
- 提高局域网的总吞吐量
- 扩大局域网的物理范围，或增加局域网允许的连接数量
- 出于管理或维护原因对物理局域网进行分区

“交换机”一词常用于指代某些类别的网桥。然而，“网桥”和“交换机”这两个术语之间的区别并没有统一的定义，ISO/IEC 15802-3 也未作此类区分。因此，本标准仅使用“网桥”一词。

6.3.2.2 网桥的中继与过滤

网桥在 MAC 子层中处理协议，在逻辑链路控制和高层协议看来，其功能是透明的。MAC 帧根据 MAC 寻址信息在接入域之间转发或过滤（即不转发到某些接入域）。图 4 展示了网桥功能在 MAC 子层中的位置；请特别注意，中继和过滤功能被认为完全属于 MAC 子层。

网桥的过滤功能通常会将流量限制在网桥连接的局域网中，只有位于发送端站与预期接收端之间的部分才能传递流量。这使网桥连接的局域网能够在任意时刻支持多个发送端站同时发送（最多可达接入域总数）。

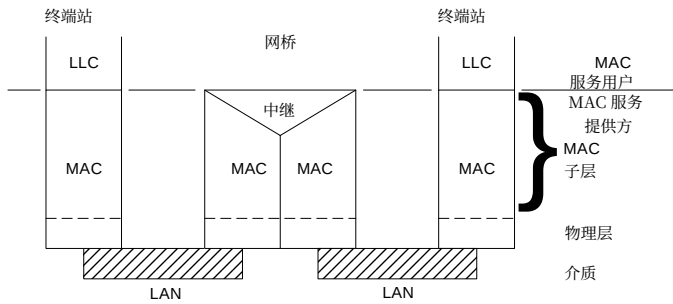


图 4——具有网桥功能的 MAC 子层内部组织结构

6.3.2.3 生成树协议

ISO/IEC 15802-3 的一个关键方面是对生成树协议的规范，该协议供网桥使用，用于配置网桥之间的互连，以防止网桥连接的局域网中出现数据环路。如果网桥和局域网的基本互连拓扑在某些点之间存在多条可能路径，则生成树协议会阻断其中一些路径，从而形成一个简单连通的

active topology for the flow of MAC user traffic between end stations. For each point of attachment of a bridge to a LAN, the Spanning Tree Protocol selects whether or not MAC user traffic is to be received and transmitted by the bridge at that point of attachment.

The Spanning Tree Protocol adapts to changes in the configuration of the bridged LAN, maintaining connectivity while avoiding data loops. Some configuration changes can cause temporary interruptions of connectivity between parts of the bridged LAN, typically lasting for a few tens of seconds at most.

6.3.2.4 Transparent bridging and source routing

ISO/IEC 15802-3 specifies *transparent bridging* operation, so called because the MAC bridging function does not require the MAC user frames transmitted and received to carry any additional information relating to the operation of the bridging functions; end-station operation is unchanged by the presence of bridges.

Also specified, in an annex to ISO/IEC 15802-3, is the operation of a Source Routing Transparent (SRT) bridge. This adds the ability for a *source-routing* function in a bridge to use explicit MAC-sublayer routing information contained in MAC user frames; this source-routing information is inserted by the transmitting end station. Source routing is specified only for ISO/IEC 8802-5 Token Ring and ISO/IEC 9314 FDDI LANs, and for IEEE 802.12™ Demand Priority LANs when operating with 8802-5 format frames.

6.3.2.5 Remote MAC bridging

ISO/IEC 15802-5 extends the specification in ISO/IEC 15802-3 to cover remote MAC bridging, where non-IEEE 802 communications technologies are used to interconnect bridges, for example, to allow a bridged LAN to include WAN links to provide greatly extended geographical range. Figure 5 shows the corresponding organization of the MAC sublayer.

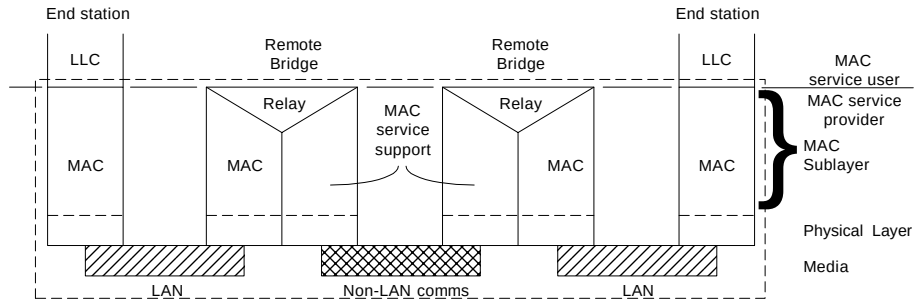


Figure 5—Internal organization of MAC sublayer with remote bridges

6.3.2.6 Bridging example

Some bridges are used to interconnect access domains each containing a very small number of end stations (often, a single end station). Others interconnect multiple access domains containing principally other bridges, thus, forming a backbone for the bridged LAN. (Hybrid configurations, with characteristics of both kinds of bridge, are of course possible.) Bridged LAN configurations involving these kinds of interconnection have become widespread as the IEEE 802 LAN technologies have developed. As noted in 1.2, they allow the construction of networks with much larger numbers of end stations and much higher aggregate throughput than was previously achievable.

Figure 6 illustrates the kind of bridged LAN that can be configured with bridge-style interconnection. The bridges A and B, and the CSMA/CD LAN configurations to which they attach, are typical of the older style of bridged LAN in which a bridge interconnects a small number of access domains each containing many end stations, as is similar with K, L, and M and their Token Ring LANs. On the other hand, the bridges S, T,

有源拓扑，以便端站之间的介质访问控制用户流量能够流通。对于网桥与局域网之间的每个连接点，生成树协议都会选择是否由网桥在该连接点接收和发送介质访问控制用户流量。

生成树协议能够适应网桥连接的局域网配置的变化，在避免数据环路的同时保持连通性。某些配置变化可能导致网桥连接的局域网各部分之间暂时中断连通，通常最长持续几十秒。

6.3.2.4 透明桥接和源路由

ISO/IEC 15802-3 规定了透明桥接操作之所以称为透明桥接，是因为 MAC 桥接功能不要求传输和接收的 MAC 用户帧携带任何与网桥功能操作有关的额外信息；网桥的存在不会改变终端端点的操作。

ISO/IEC 15802-3 的附录还规定了源路由透明（SRT）网桥的运行方式。此功能使网桥中的源路由功能能够使用 MAC 用户帧中包含的显式 MAC 子层路由信息；这些源路由信息由发送端终端站插入。源路由仅适用于 ISO/IEC 8802-5 令牌环和 ISO/IEC 9314 FDDI 局域网，以及采用 8802-5 格式帧运行的 IEEE802.12™ Demand Priority 局域网。

6.3.2.5 远程 MAC 桥接

ISO/IEC 15802-5 扩展了 ISO/IEC 15802-3 中的规范，涵盖远程 MAC 桥接，即使用非 IEEE 802 通信技术互连网桥，例如，使网桥连接的局域网能够包含广域网链路，从而大幅扩展地理覆盖范围。图 5 显示了相应的 MAC 子层组织结构。

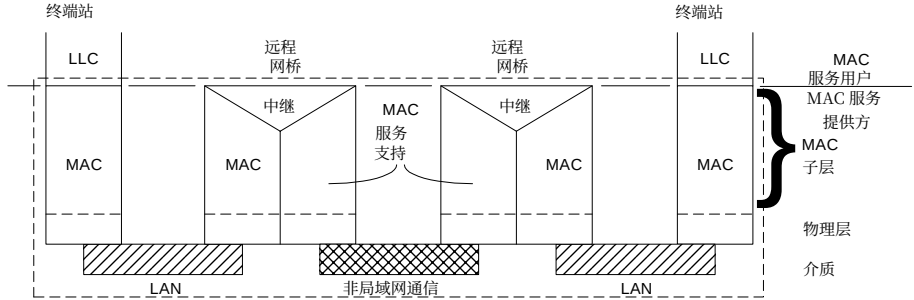


图 5—含远程网桥的 MAC 子层内部组织结构

6.3.2.6 网桥示例

有些网桥用于互连各个接入域，每个接入域只包含极少数端站（通常只有一个端站）。另一些网桥则用于互连多个接入域，这些接入域主要由其他网桥组成，从而为网桥连接的局域网构成骨干网。（当然，也可以采用兼具这两类网桥特征的混合配置。）随着 IEEE 802 局域网技术的发展，涉及这类互连的网桥连接的局域网配置已广泛应用。如 1.2 节所述，这些配置能够构建端站数量更多、总吞吐量远高于以往的网络。

图 6 展示了通过网桥式互连可以配置的网桥连接局域网类型。网桥 A 和 B，以及它们所连接的 CSMA/CD 局域网配置，是较早期网桥连接局域网的典型例子。在这种局域网中，一个网桥互连少量接入域，每个接入域都包含许多端站；K、L 和 M 及其令牌环局域网也具有类似的情况。另一方面，网桥 S、T，

and U function as bridges in a high-speed backbone that combines FDDI and 100-Mbit/s IEEE 802.3 LANs. S is a backbone bridge, handling a number of high-speed LAN attachments. T and U are bridges that support multiple end stations, with connection to the backbone. B and K also provide access to the backbone. The end station shown connected to S by a point-to-point link could be a server system, as might the end stations attached to the FDDI LAN.

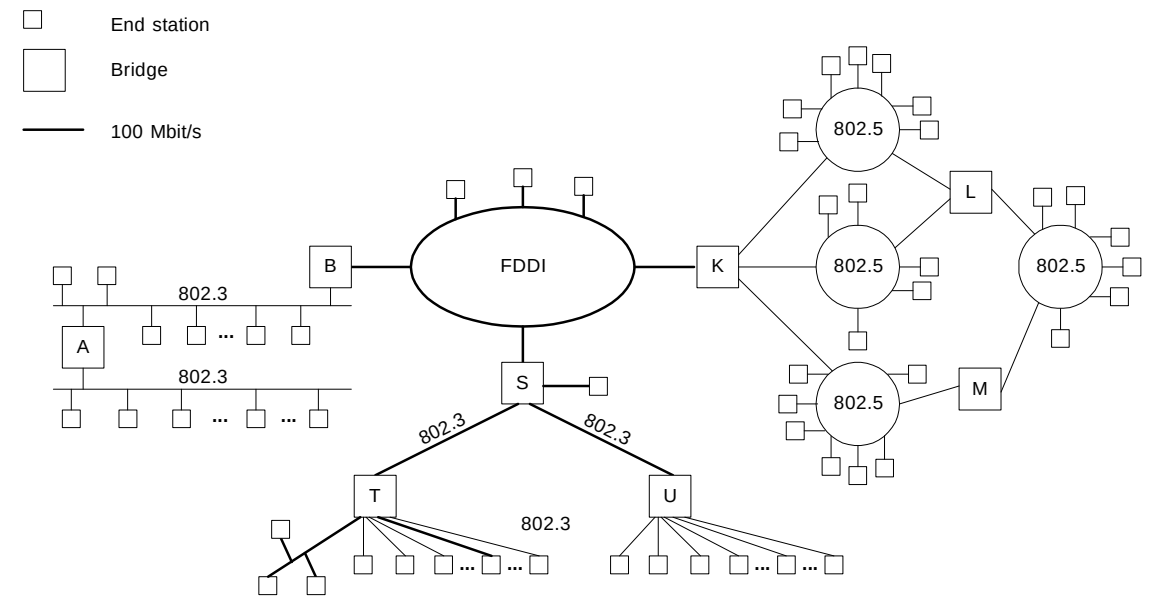


Figure 6—A bridged LAN

6.3.3 Network-layer interconnection: routers

The third category of interconnection uses Network-layer interconnection devices, generally known as routers, that operate as LAN/MAN end stations. These process Network layer protocols that operate directly above the LLC sublayer or equivalent, with forwarding decisions based on Network layer addresses. Details of this kind of interconnection lie outside of the scope of IEEE 802 LAN/MAN Standards, but the various standard and proprietary Network-layer protocols involved represent a very substantial part of the user traffic on many real IEEE 802 networks. In particular, IEEE 802 LANs and MANs are often interconnected by routers for the Internet Protocol (IP) and its related routing and management protocols, either directly to other LANs or by means of WAN links.

7. General requirements for an 802 LAN or MAN

7.1 Services supported

With the descriptions in Clause 6 as a basis, an IEEE 802 LAN or MAN can be characterized as a communication resource that provides sufficient capabilities to support the MAC service defined in ISO/IEC 15802-1, between two or more MSAPs. In particular, this requires the ability to convey LLC data from one MSAP to n other MSAPs, where n can be any number from 1 to all of the other MSAPs on the network. An IEEE 802 LAN is required, at a minimum, to support both LLC Type 1 and the MAC Internal Sublayer Service defined in ISO/IEC 15802-3. In addition, an ISLAN or MAN may optionally support isochronous bearer services compatible with ISDN services as defined in the ITU-T I-series Recommendations.

和 U 在一个高速骨干网中充当网桥，该骨干网结合了光纤分布式数据接口和 100 兆比特/秒 IEEE 802.3 局域网。S 是一台骨干网桥，负责处理多个高速局域网连接。T 和 U 是支持多个端站并连接到骨干网的网桥。B 和 K 也提供了对骨干网的接入。图中通过点对点链路连接到 S 的端站可能是一个服务器系统，连接到光纤分布式数据接口局域网的端站也可能是服务器系统。

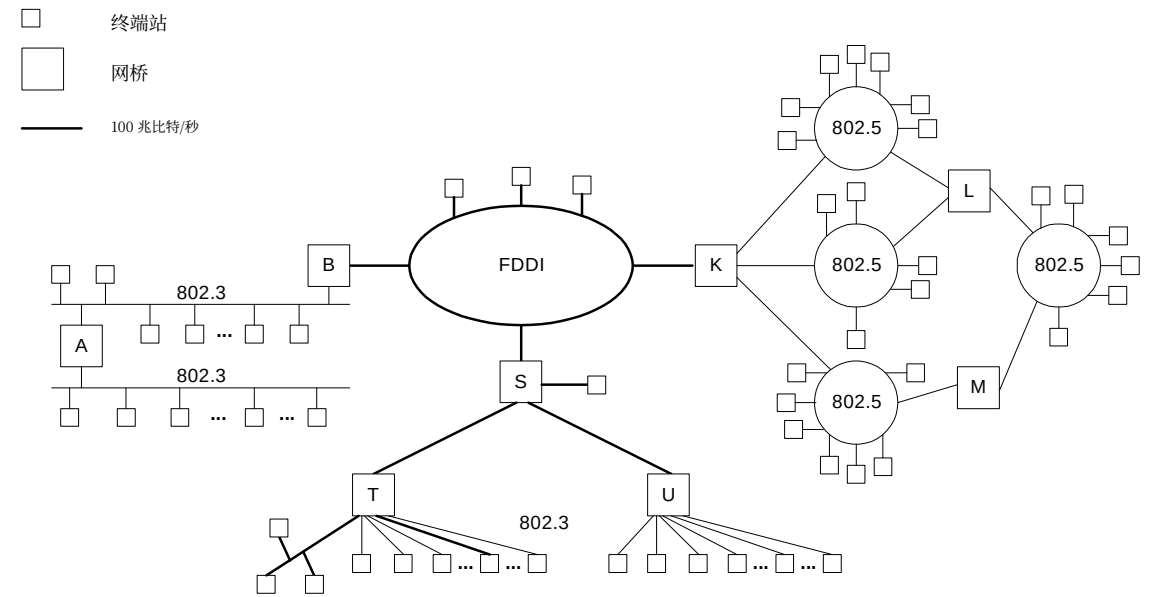


图 6—网桥连接的局域网

6.3.3 网络层互连：路由器

第三类互连采用网络层互连设备（通常称为路由器），这些设备作为局域网/城域网端站运行。它们处理直接运行在 LLC 子层或等效层之上的网络层协议，并根据网络层地址进行转发决策。这类互连的详细内容不属于 IEEE 802 局域网/城域网标准的范围，但涉及的各种标准和专有网络层协议，构成了许多实际 IEEE 802 网络上用户流量中非常重要的一部分。特别是，IEEE 802 局域网和城域网通常通过路由器互连，以支持互联网协议（IP）及其相关的路由和管理协议；互连方式可以是直接连接到其他局域网，也可以通过广域网链路实现。

7. 802 局域网或城域网的总则要求

7.1 支持的服务

以第 6 条中的说明为基础，可以将 IEEE 802 局域网或城域网描述为一种通信资源，它具备足够的的能力，可在两个或多个 MAC 服务访问点之间提供 ISO/IEC 15802-1 中定义的 MAC 服务。具体而言，这要求能够将 LLC 数据从一个 MAC 服务访问点传送到 n 个其他 MAC 服务访问点，其中 n 可以是 1 到网络上所有其他 MAC 服务访问点之间的任意数量。IEEE 802 局域网至少必须支持 LLC 类型 1 以及 ISO/IEC 15802-3 中定义的 MAC 内部子层服务。此外，ISLAN 或城域网还可以选择支持与 ITU-T I 系列建议中定义的综合业务数字网服务兼容的等时承载业务。

7.2 Size and extent

The original IEEE 802 LAN and MAN technologies were designed to be capable of supporting access domains containing at least 200 end stations and with geographical extent of at least 2 km for LANs (using Physical-layer repeaters if necessary) and 50 km for MANs. Subsequent developments in IEEE 802 LAN technology and performance have been accompanied by a reduction in the size and extent required in individual access domains, recognizing that these can readily and cost-effectively be interconnected in bridged LANs that are capable of offering at least the original minimum size and extent, with increased overall bandwidth and performance. Size and extent requirements for future IEEE 802 LAN technologies are, similarly, expected to be determined by application needs and opportunities.

7.3 Error rates

Error performance of IEEE 802 LANs and MANs is required to be such as follows:

- a) For wired or optical fiber physical media: Within a single access domain, the probability that a transmitted MAC frame (excluding any preamble) is not reported correctly at the Physical Service interface of an intended receiving peer MAC entity, due only to operation of the Physical layer, shall be less than 8×10^{-8} per octet of MAC frame length.
- b) For wireless physical media: Within a single access domain, the probability that a MAC Service Data Unit (MSDU) is not delivered correctly at an MSAP of an intended receiving MAC service user, due to the operation of the Physical layer and the MAC protocol, shall be less than 8×10^{-8} per octet of MSDU length.

NOTE—The performance measure stated in (a) defines a highly desirable characteristic of LAN performance, as it has a bearing on other aspects of the delivered service, such as frame loss and transmission delays caused by the need to retransmit. However, this measure is not realistic for all physical media; for example, wireless media may be unable to meet this level of physical layer performance due to the inherent transmission characteristics of the medium. In such cases, the operation of the MAC protocol must employ additional mechanisms, for example, error detection and correction mechanisms, in order to enable the MAC service provider to meet the performance levels implied by this condition in the service offered at the MAC service boundary.

- c) The probability that an MSDU delivered at an MSAP contains an undetected error, due to operation of the MAC service provider, shall be less than 5×10^{-14} per octet of MSDU length.

NOTE—For example, (a) the worst-case probability of losing a maximum-length IEEE 802.3 frame (1518 octets) through physical-layer damage is to be less than 1.21×10^{-4} , or approximately 1 in 8250; (c) the worst-case probability that a similar frame, which contains an MSDU of 1500 octets, is delivered with an undetected error is to be less than 7.5×10^{-11} , or approximately 1 in 13 300 000 000.

7.4 Service availability

Insertion of a device into, or removal of a device from, a LAN or MAN shall cause at most a transient loss of availability of the access domain(s) to which the device attaches, lasting not more than 1 s. Failure of a device, including loss of power, shall not cause more than a transient fault for the access domain(s) to which it attaches, with duration of order 1 s.

NOTE—In a bridged LAN, reconfiguration of the topology in response to logical insertion or removal of a bridge, or to changes in a bridge’s configuration parameters, can cause loss of communication between some access domains for longer periods, typically a few tens of seconds; ISO/IEC 15802-3 contains the full specification.

7.5 Safety, and lightning and galvanic protection

Equipment implementing IEEE 802 LAN and MAN standards is typically subject to guidance and requirements relating to safety and to protection of the equipment and its users from lightning and galvanic effects. Such guidance and requirements are outside of the scope of IEEE 802 standardization; they are

7.2 尺寸和范围

最初的 IEEE 802 局域网和城域网技术旨在支持至少包含 200 个端站的接入域；其地理范围至少为局域网 2 km（必要时使用物理层中继器），城域网 50 km。随着 IEEE 802 局域网技术和性能的发展，单个接入域所需的尺寸和范围也相应缩小，因为人们认识到，这些接入域可以轻松且经济高效地互连成桥接局域网，从而提供至少达到原始最低尺寸和范围的覆盖，同时提高整体带宽和性能。未来 IEEE 802 局域网技术的尺寸和范围要求预计也将同样由应用需求和机遇决定。

7.3 错误率

IEEE 802 局域网和城域网的错误性能应符合以下要求：

- a) 对于有线或光纤物理介质：在单一接入域内，仅因物理层运行而导致传输的 MAC 帧（不包括任何前导码）未能在预期接收方对等 MAC 实体的物理服务接口处被正确报告的概率，应小于每八位字节 MAC 帧长度 8×10^{-8} 。
- b) 对于无线物理介质：在单一接入域内，因物理层和 MAC 协议的运行而导致 MAC 服务数据单元 (MSDU) 未能在预期接收方 MAC 服务用户的 MSAP 处被正确交付的概率，应小于每八位字节 MSDU 长度 8×10^{-8} 。

注——(a) 中所述的性能指标定义了局域网一项极为理想的特性，因为它会影响交付服务的其他方面，例如帧丢失以及因需要重传而导致的传输时延。不过，这一指标并不适用于所有物理介质；例如，由于介质固有的传输特性，无线介质可能无法达到这一物理层性能水平。在这种情况下，MAC 协议必须采用其他机制，例如错误检测和

纠错机制，以使 MAC 服务提供者能够满足 MAC 服务边界所提供服务中该条件所隐含的性能级别。

- c) 由于 MAC 服务提供者的运行，在 MAC 服务访问点交付的介质访问控制服务数据单元包含未检测到错误的概率，按介质访问控制服务数据单元长度的每个八位组计算，应小于 5×10^{-14} 。

注——例如，(a) 由于物理层损坏而丢失最大长度 IEEE 802.3 帧（1518 个八位组）的最坏情况概率应小于 1.21×10^{-4} ，约为 8250 分之一；(c) 类似帧（包含 1500 个八位组的介质访问控制服务数据单元）以未检测到错误的状态交付的最坏情况概率应小于 7.5×10^{-11} ，约为 13 300 000 000 分之一。

7.4 服务可用性

将设备接入局域网或城域网，或从中移除设备，最多只能导致设备所接入的接入域暂时失去可用性，持续时间不得超过 1 秒。设备发生故障（包括断电）时，对其所接入的接入域造成的故障也不得超过暂时性故障，持续时间约为 1 秒。

注——在网桥连接的局域网中，为响应网桥的逻辑插入或移除，或网桥配置参数的变化而重新配置拓扑，可能导致某些接入域之间的通信中断较长时间，通常为几十秒；完整规范见 ISO/IEC 15802-3。

7.5 安全、雷电防护和电化学腐蚀防护

实现 IEEE 802 局域网和城域网标准的设备通常需要遵循有关安全以及保护设备及其用户免受雷击和电偶效应影响的指南和要求。这些指南和要求不属于 IEEE 802 标准化的范围；它们

typically specified by other organizations with different legal, geographical, and industrial scope. However, the general underlying concerns can have an influence on the Physical layer aspects of IEEE 802 LAN and MAN standards.

7.6 Regulatory requirements

Equipment implementing IEEE 802 LAN and MAN standards may be subject to regulations imposed within particular geographical and political domains. For example, the deployment of equipment implementing IEEE 802 wireless LAN standards may be subject to local regulations that pertain to the use of radio-frequency transmission. Such regulations are outside of the scope of IEEE 802 standardization; they are typically specified by other organizations with different legal, geographical, and industrial scope. However, the general underlying concerns can have an influence on the Physical layer aspects of IEEE 802 LAN and MAN standards.

8. LAN/MAN management

The provision of an adequate means of remote management is an important factor in the design of today's LAN equipment. Such management mechanisms fall into two broad categories: those that provide general-purpose management capability, allowing control and monitoring for a wide variety of purposes, and those that provide specific capabilities aimed at a particular aspect of management. These aspects of management are discussed in 8.1 and 8.2, respectively.

8.1 General-purpose LAN/MAN management

This subclause introduces the functions of management to assist in the identification of the requirements placed on IEEE 802 LAN/MAN equipment for support of management facilities, and it identifies general-purpose management standards that may be used as the basis of developing management specifications for such equipment.

8.1.1 Management functions

Management functions relate to users’ needs for facilities that support the planning, organization, supervision, control, protection and security of communications resources, and account for their use. These facilities may be categorized as supporting the functional areas of configuration, fault, performance, security, and accounting management. These can be summarized as follows.

- Configuration management provides for the identification of communications resources, initialization, reset and close-down, the supply of operational parameters, and the establishment and discovery of the relationships between resources.
- Fault management provides for fault prevention, detection, diagnosis, and correction.
- Performance management provides for evaluation of the behavior of communications resources and of the effectiveness of communication activities.
- Security management provides for the protection of resources.
- Accounting management provides for the identification and distribution of costs and the setting of charges.
- Management facilities in LAN/MAN equipment will address some or all of these areas, as appropriate to the needs of that equipment and the environment in which it is to be operated.

通常由法律、地理和行业范围各不相同的其他组织规定。不过，相关的基本问题可能会影响 IEEE 802 局域网和城域网标准的物理层方面。

7.6 监管要求

实施 IEEE 802 局域网和城域网标准的设备可能受特定地理和政治辖区内施行的法规约束。例如，实施 IEEE 802 无线局域网标准的设备部署可能受当地有关射频传输使用的法规约束。此类法规不属于 IEEE 802 标准化的范围；通常由其他组织制定，这些组织的法律、地理和行业范围各不相同。不过，相关的一般性根本问题可能会影响 IEEE 802 局域网和城域网标准的物理层方面。

8. 局域网/城域网管理

提供充分的远程管理手段是当今局域网设备设计中的一个重要因素。这些管理机制大致分为两类：一类提供通用管理能力，可用于各种用途的控制和监控；另一类则提供针对特定管理方面的专用能力。第 8.1 节和第 8.2 节分别讨论这些管理方面。

8.1 通用局域网/城域网管理

本小条款介绍管理功能，以帮助识别 IEEE 802 局域网/城域网设备在支持管理设施方面的要求，并指出可用于制定此类设备管理规范的通用管理标准。

8.1.1 管理功能

管理功能涉及用户对相关设施的需求，这些设施用于支持通信资源的规划、组织、监督、控制、保护和安全，并对资源的使用情况进行计费。这些设施可按其支持的功能领域分类，包括配置、故障、性能、安全和计费管理。概括如下。

- 配置管理提供了通信资源的识别、初始化、重置和关闭，运行参数的提供，以及建立和发现资源之间的关系。
- 故障管理提供故障预防、检测、诊断和纠正。
- 性能管理提供对通信资源行为的评估，以及对通信活动有效性的评估。
- 安全管理用于保护资源。
- 计费管理负责识别和分摊成本，并设置费用。
- LAN/MAN 设备中的管理设施将根据设备需求及其运行环境，处理上述部分或全部领域。

8.1.2 Management architecture

The management facilities defined in IEEE 802 LAN and MAN standards are based on the concept of managed objects, which model the semantics of management operations. Operations on a managed object supply information concerning, or facilitate control over, the process or entity associated with that object.

Operations on a managed object can be initiated by mechanisms local to the equipment being managed (e.g., via a control panel built into the equipment), or can be initiated from a remote management system by means of a general-purpose management protocol carried using the data services provided by the LAN or MAN to which the equipment being managed is connected.

There are two general-purpose management protocols of relevance to the management of LAN/MAN equipment, as follows:

- The Simple Network Management Protocol (SNMP), as described in RFC 1157
- The OSI common management information protocol (CMIP), as described in ISO/IEC 9595 and ISO/IEC 9596 and related standards

NOTE—In addition to operation of CMIP over a full OSI protocol stack, two standards define the use of CMIP over simpler protocol support. ISO/IEC 15802-2 (IEEE Std 802.1B™-1995) defines a means of using CMIP over a simple Data Link layer protocol stack in LANs; RFC 1095 describes the use of CMIP over a TCP/IP protocol stack.

Of the two protocols, SNMP is the more significant in terms of its wide application across the spectrum of LAN/MAN products in today’s marketplace; however, in some markets, and where it is desirable to integrate LAN management with management of wide-area networking and telephony equipment, use of the OSI management protocols may be important.

8.1.3 Managed object definitions

In order for an IEEE 802 standard to specify management facilities, it is necessary for them to define managed objects that model the operations that can be performed on the communications resources specified in the standard. There are essentially two components to a managed object definition, as follows:

- 1) A definition of the functionality provided by the managed object, and the relationship between this functionality and the resource to which it relates
- 2) A definition of the syntax that is used to convey management operations, and their arguments and results, in a management protocol

The functionality of a managed object can be described in a manner that is independent of the protocol that will be used; this abstract definition can then be used in conjunction with a definition of the syntactic elements required in order to produce a complete definition of the object for use with specific management protocols.

Each management protocol has its notation for defining managed objects, as follows:

- a) SNMP has standards for the structure of management information known as SMIV1 (RFC 1155, RFC 1212 and RFC 1215) and SMIV2 (RFC 1902, RFC 1903 and RFC 1904), which provides ASN.1-based macros for defining managed objects.
- b) CMIP has a standard language, known as GDMO (ISO/IEC 10165-4), which is used along with ASN.1 to define both the syntactic and semantic aspects of managed objects.

The choice of notational tools for defining managed objects will depend on which of the available management protocols the standard will support.

8.1.2 管理体系结构

IEEE 802 局域网和城域网标准中定义的管理设施基于受管对象的概念，这些对象对管理操作的语义进行建模。对受管对象执行的操作可提供有关与该对象关联的流程或实体的信息，或便于对其进行控制。

对受管理对象的操作可以通过受管理设备本地的机制发起（例如，通过设备内置的控制面板），也可以通过通用管理协议从远程管理系统发起。该协议通过受管理设备所连接的局域网或城域网提供的数据服务传输。

与 LAN/MAN 设备管理相关的通用管理协议有两种，如下所示：

- RFC 1157 中描述的简单网络管理协议（SNMP）
- ISO/IEC 9595 中描述的开放系统互连通用管理信息协议（CMIP），以及 ISO/IEC 9596 及相关标准

注——除了在完整的 OSI 协议栈上运行 CMIP 之外，还有两个标准规定了如何在较简单的协议支持上使用 CMIP。ISO/IEC 15802-2（IEEE Std 802.1B™-1995）规定了在局域网中通过简单的数据链路层协议栈使用 CMIP 的方法；RFC 1095 描述了通过 TCP/IP 协议栈使用 CMIP 的方法。

在这两种协议中，就其在当今市场各类局域网/城域网产品中的广泛应用而言，简单网络管理协议（SNMP）更为重要；不过，在某些市场中，以及在需要将 LAN 管理与广域网和电话设备的管理集成时，使用 OSI 管理协议可能很重要。

8.1.3 被管理对象定义

为了使 IEEE 802 标准能够规定管理设施，必须定义受管对象，以对可在该标准规定的通信资源上执行的操作进行建模。受管对象的定义主要包含以下两个组成部分：

- 1) 定义受管对象提供的功能，以及该功能与其所关联资源之间的关系
- 2) 定义用于在管理协议中传达管理操作及其参数的语法以及管理协议中的结果

受管对象的功能可以用独立于所用协议的方式描述；随后，可将此抽象定义与所需语法元素的定义结合起来，从而为该对象生成完整定义，以便在特定管理协议中使用。

每种管理协议都有用于定义受管对象的表示法，如下所示：

- a) 简单网络管理协议拥有称为 SMIV1（RFC 1155、RFC 1212 和 RFC 1215）和 SMIV2（RFC 1902、RFC 1903 和 RFC 1904）的管理信息结构标准，其中提供了基于 ASN.1、用于定义受管对象的宏。
- b) CMIP 有一种称为 GDMO（ISO/IEC 10165-4）的标准语言，该语言与 ASN.1 配合使用，用于定义受管对象的语法和语义方面。

用于定义受管对象的表示法工具应根据该标准将支持哪些可用管理协议来选择。

NOTES

- 1—IEEE Std 802.1F™ provides additional guidance for use of GDMO in LAN/MAN standards.
- 2—Some IEEE 802 standards have used GDMO as the notation for their managed object definitions, with SNMP management information base (MIB) definitions being developed subsequently within the IETF, using automatic tools for translating GDMO definitions into equivalent SNMP definitions.

8.2 Special-purpose LAN/MAN management standards

Special-purpose protocols relating to the management functionality of IEEE 802 stations can be developed where the use of a general-purpose management protocol is inappropriate. An example of a special-purpose management protocol is ISO/IEC 15802-4, which defines the services and protocols for remote station loading in a LAN/MAN environment. This protocol permits the simultaneous loading of multiple stations by use of the group-addressing capability in IEEE 802 technologies.

9. Universal addresses and protocol identifiers

The IEEE makes it possible for organizations to employ unique individual LAN MAC addresses, group addresses, and protocol identifiers. It does so by assigning Organizationally Unique Identifiers (OUIs), which are three octets (24 bits) in length. Because the assignment of the OUI in effect reserves a block of each derivative identifier (i.e., blocks of individual LAN MAC addresses, group addresses, and protocol identifiers), the address space of the OUI is chosen to be large. Although the OUIs are 24 bits in length, their true address space is 22 bits. The LSB of the first octet can be set to 1 or 0 depending on the application. The next-to-LSB of the first octet is 0, for all assignments. The remaining 22 bits, which shall not be changed by the assignee, result in 2²² (approximately 4 million) identifiers; see Figure 7.



Figure 7—OUI

The universal administration of LAN MAC addresses began with the Xerox Corporation administering Block Identifiers (Block IDs) for Ethernet addresses. Block IDs were assigned by the Ethernet Administration Office and were 24 bits in length (three octets). An organization developed addresses by assigning the remaining 24 bits. For example, the address as represented by the six octets P-Q-R-S-T-U comprises the Block ID, P-Q-R, and the locally assigned octets S-T-U.

The IEEE, because of the work in Project 802 on standardizing LAN technologies, has assumed the responsibility of defining and carrying out procedures for the universal administration of addresses for IEEE and ISO/IEC LANs (e.g., CSMA/CD, Token Bus, Token Ring, and FDDI). In carrying out the procedures, the IEEE acts as the Registration Authority for OUIs.⁷ The responsibility for defining the procedures is discharged by the IEEE Registration Authority Committee (RAC), which is chartered by the IEEE Standards Association Board of Governors.

The IEEE honors the Block ID assignments made by the predecessor administration office where those assignments fall—as the great majority of them do—within the space administered by the IEEE. The Block ID is referred to as the OUI by the IEEE.

⁷ Interested applicants should contact the IEEE Registration Authority, Institute of Electrical and Electronic Engineers Inc., 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, USA.

注释

- 1—IEEE 标准 802.1F™为在局域网/城域网标准中使用被管理对象定义指南提供了进一步指导。
- 2—一些 IEEE 802 标准曾使用被管理对象定义指南（GDMO）作为其被管理对象定义的表示法；随后，IETF 使用自动化工具将 GDMO 定义转换为等效的简单网络管理协议（SNMP）定义，并据此制定了 SNMP 管理信息库（MIB）定义。

8.2 专用局域网/城域网管理标准

如果不适合使用通用管理协议，可以开发与 IEEE 802 站点管理功能相关的专用协议。ISO/IEC 15802-4 就是专用管理协议的一个示例，它定义了 在局域网/城域网环境中进行远程站点加载的服务和协议。该协议利用 IEEE 802 技术的组寻址能力，允许同时加载多个站点。

9. 通用地址和协议标识符

IEEE 使组织能够使用唯一的单播局域网 MAC 地址、组地址和协议标识符。具体做法是分配组织唯一标识符（OUI），其长度为三个八位字节（24 位）。由于分配 OUI 实际上会预留每种衍生标识符的一段区块（即单播局域网 MAC 地址、组地址和协议标识符的区块），因此 OUI 的地址空间被设定得较大。虽然 OUI 的长度为 24 位，但其实际地址空间为 22 位。根据应用情况，第一个字节的最低有效位可以设为 1 或 0。对于所有分配，第一个字节的次低有效位均为 0。剩余的 22 位不得由受让方更改，因此可产生 2²²（约 400 万）个标识符；见图 7。

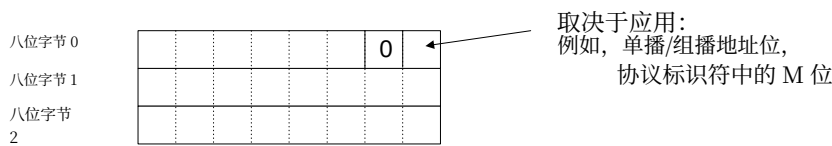


图 7—组织唯一标识符

局域网 MAC 地址的统一管理始于施乐公司为以太网地址管理块标识符（块 ID）。块标识符由以太网管理办公室分配，长度为 24 位（三个八位字节）。组织通过分配其余 24 位来生成地址。例如，由六个八位组 P-Q-R-S-T-U 表示的地址包含块标识符 P-Q-R，以及本地分配的八位组 S-T-U。

IEEE 因 802 项目在制定局域网技术标准方面所开展的工作，承担了为 IEEE 和 ISO/IEC 局域网（例如载波侦听多路访问/冲突检测、令牌总线、令牌环和光纤分布式数据接口）的地址制定并执行统一管理程序的责任。在执行这些程序时，IEEE 作为组织唯一标识符（OUI）的注册机构。⁷ 制定这些程序的责任由 IEEE 注册机构委员会（RAC）履行，该委员会由 IEEE 标准协会理事会授权成立。

IEEE 承认前任管理办公室所作的块标识符分配，只要这些分配（绝大多数情况下确实如此）属于 IEEE 管理的地址空间。IEEE 将块标识符称为 OUI。

⁷ 有意申请者应联系 IEEE 注册管理机构，电气与电子工程师协会，Hoes Lane 445 号，邮政信箱 1331，皮斯卡塔韦，新泽西州 08855-1331，美国。

9.1 OUI

OUIs allow a general means of assuring unique identifiers for a number of purposes. Currently, the IEEE assigns OUIs to be used for generating LAN MAC addresses and protocol identifiers. Assuming correct administration by the IEEE Registration Authority and the assignee, the LAN MAC addresses and protocol identifiers will be universally unique.

OUIs are assigned as three-octet values. Both values (0, 1) are assigned to the LSB of the first octet. The next-to-LSB of the first octet is set to 0; this bit of the OUI being set to 0 indicates that the assignment is universal. Three-octet values occupying the same fields as OUIs can occupy, but with the next-to-LSB of the first octet set to 1, are locally assigned and have no relationship to the IEEE-assigned values (as described herein).

The standard representation of the OUI is as a string of three octets, using the hexadecimal representation (see 3.1.8).

9.2 48-bit universal LAN MAC addresses

9.2.1 Concept

The concept of universal addressing is based on the idea that all potential members of a network need to have a unique identifier (if they are going to coexist in the network). The advantage of a universal address is that a station with such an address can be attached to any LAN in the world with an assurance that the address is unique.

A 48-bit universal address consists of two parts. The first 24 bits correspond to the OUI as assigned by the IEEE, except that the assignee may set the LSB of the first octet to 1 for group addresses or set it to 0 for individual addresses. The second part, comprising the remaining 24 bits, is administered by the assignee. In the 48-bit LAN MAC address, an example of which is shown in Figure 8, the OUI is contained in octets 0, 1, 2, and the value assigned by the assignee is contained in octets 3, 4, 5. This address, including its OUI, is used throughout this document as the basis for examples of LAN MAC addresses and protocol identifiers.

NOTE—The requirement for the use of 64-bit addresses in new applications is under consideration by the IEEE Registration Authority (RAC).

The standard representation of a 48-bit LAN MAC address is as a string of six octets, using the hexadecimal representation (3.1.8). In certain contexts associated with use of IEEE 802.5 frame formats, LAN MAC addresses may be represented using the alternative bit-reversed representation (3.1.2). See 9.5 for further specification relating to use of the bit-reversed representation.

NOTE—The upper, bit-stream representation of the universal address in Figure 8 shows the LSB of each octet first; this corresponds to the data-communications convention for representing bit-serial transmission in left-to-right order, applied to the model for transmission of LAN MAC address fields (see 6.2.3). See also 9.5 for further discussion of bit-ordering issues. The lower, octet-sequence representation shows the bits within each octet in the usual order for binary numerals; the order of octet transmission is from the top downward.

9.1 组织唯一标识符（OUI）

OUI 为确保多种用途所需的唯一标识符提供了一种通用方式。目前，IEEE 分配 OUI，用于生成局域网 MAC 地址和协议标识符。在 IEEE 注册管理机构和受让方正确管理的前提下，局域网 MAC 地址和协议标识符将具有全球唯一性。

组织唯一标识符（OUI）以三字节值的形式分配。两个值（0、1）分配给第一个字节的最低有效位。第一个字节中紧邻最低有效位的下一位设为 0；组织唯一标识符（OUI）的这一位设为 0，表示该分配是通用的。占据组织唯一标识符（OUI）可占据的相同字段、但第一个字节中紧邻最低有效位的下一位设为 1 的三字节值，是本地分配的，与 IEEE 分配的值（如本文所述）无关。

组织唯一标识符（OUI）的标准表示法是由三个八位组组成的字符串，采用十六进制表示（见 3.1.8）。

9.2 48 位通用局域网 MAC 地址

9.2.1 概念

通用寻址的概念基于这样一种理念：网络的所有潜在成员都需要拥有一个唯一标识符（如果它们要在网络中共存）。通用地址的优点是，具有此类地址的站点可以连接到世界上任何局域网，并确保该地址是唯一的。

48 位通用地址由两部分组成。前 24 位对应 IEEE 分配的 OUI，但受让方可以将第一个字节的最低有效位置为 1，以表示组地址；也可以将其置为 0，以表示单播地址。其余 24 位构成第二部分，由受让方管理。在 48 位局域网 MAC 地址中（示例如图 8 所示），OUI 位于八位组 0、1、2 中，受让方分配的值位于八位组 3、4、5 中。本文档始终以此地址（包括其 OUI）为基础，举例说明局域网 MAC 地址和协议标识符。

注——IEEE 注册管理机构（RAC）正在考虑是否要求在新应用中使用 64 位地址。

48 位局域网 MAC 地址的标准表示法是由六字节组成的字符串，采用十六进制表示法（3.1.8）。在某些与 IEEE 802.5 帧格式相关的情境中，局域网 MAC 地址可采用另一种位反转表示（3.1.2）。有关使用位反转表示的进一步规范，请参见 9.5。

注——图 8 中通用地址的上方位流表示首先显示每个八位字节的最低有效位；这符合数据通信惯例，即从左到右表示位串行传输，并应用于局域网 MAC 地址字段的传输模型（见 6.2.3）。有关位排序问题的进一步讨论，另见 9.5。下方的八位组序列表示按照二进制数的常用顺序显示每个八位字节中的各位；八位字节的传输顺序是从上到下。

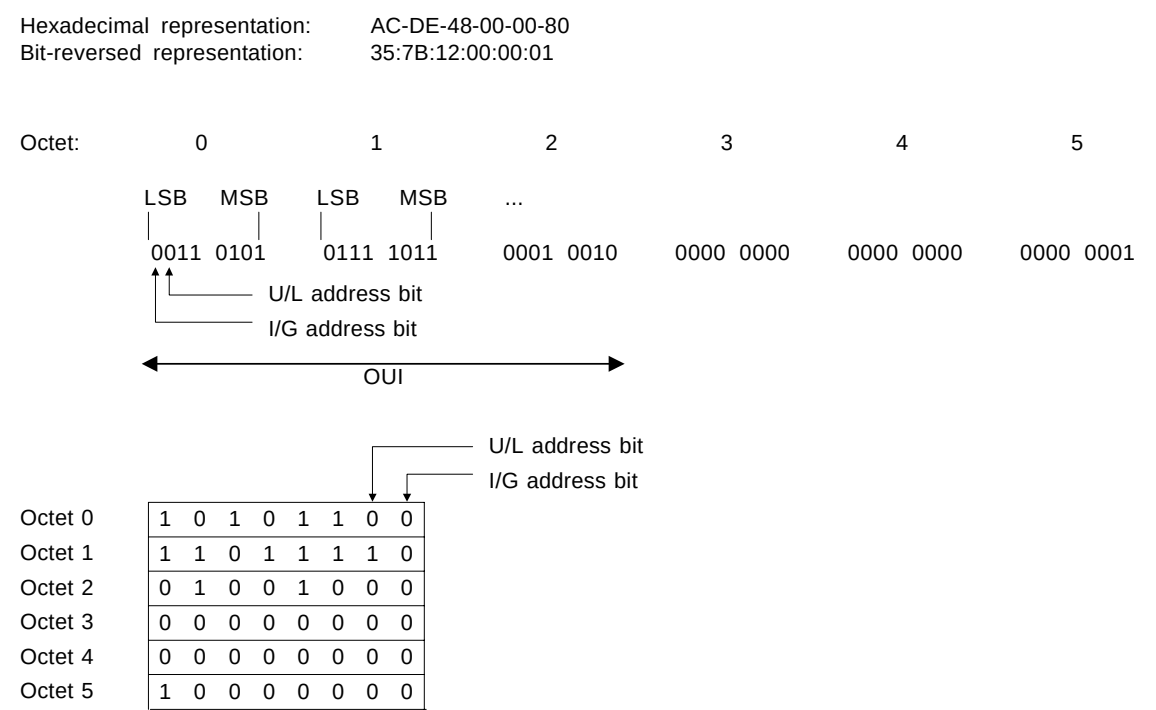


Figure 8—Universal address

The Individual/Group (I/G) address bit (LSB of octet 0) is used to identify the destination address as an individual address or a group address. If the I/G address bit is 0, it indicates that the address field contains an individual address. If this bit is 1, the address field contains a group address that identifies one or more (or all) stations connected to the LAN. The all-stations broadcast address is a special, predefined group address of all 1's.

The Universally or Locally administered (U/L) address bit is the bit of octet 0 adjacent to the I/G address bit. This bit indicates whether the address has been assigned by a local or universal administrator. Universally administered addresses have this bit set to 0. If this bit is set to 1, the entire address (i.e., 48 bits) has been locally administered.

9.2.2 Assignment by organizations

Varying the last 24 bits in the block of MAC addresses for a given OUI allows the OUI assignee approximately 16 million unique individual addresses and 16 million unique group addresses that no other organization may assign (i.e., universally unique). The IEEE intends not to assign additional OUIs to any organization unless the organization has exhausted this address block. Therefore, it is important for the IEEE to maintain a single point of contact with each assignee to avoid complicating the assignment process. It is important to note that in no way should these addresses be used for purposes that would lead to skipping large numbers of them (for example, as product identifiers for the purpose of aiding company inventory procedures). The IEEE asks that organizations not misuse the assignments of the last 24 bits and thereby unnecessarily exhaust the block. There are sufficient identifiers to satisfy most needs for a long time, even in volume production; however, no address space is infinite.

The method that an assignee uses to ensure that no two of its devices carry the same address will, of course, depend on the assignment or manufacturing process, the nature of the organization, and the organization's philosophy. However, the users of networks worldwide expect to have unique addresses. The ultimate responsibility for assuring that user expectations and requirements are met, therefore, lies with the organization offering such devices.

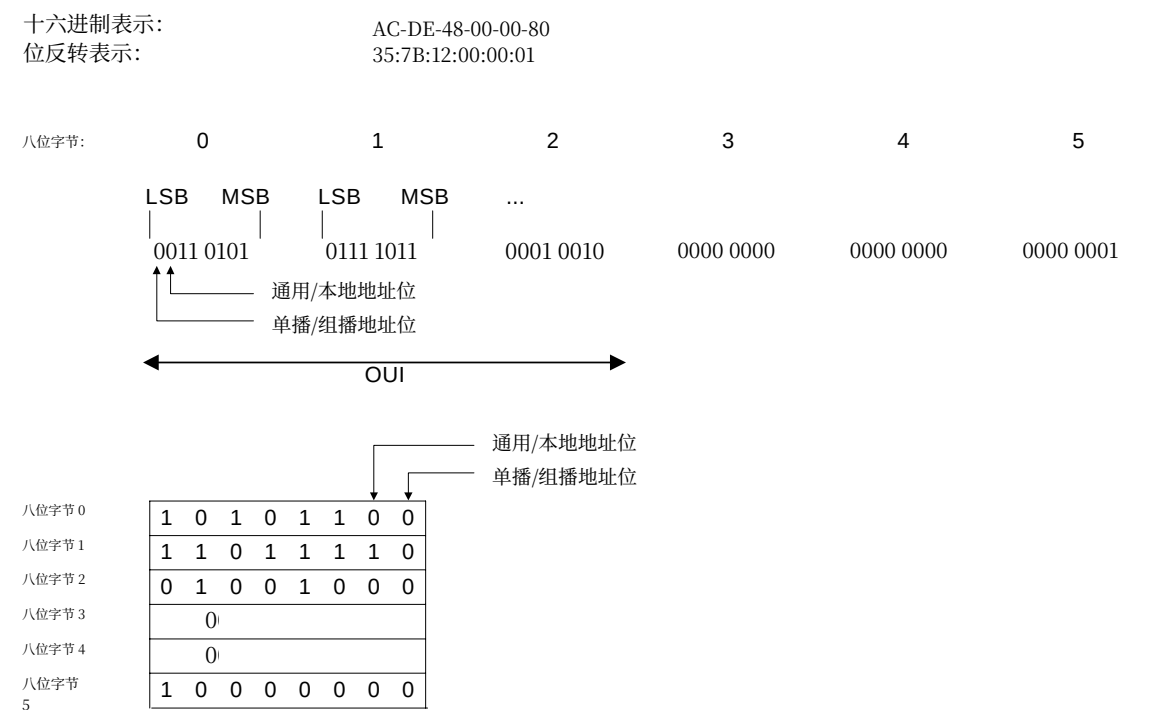


图 8—通用地址

单播/组播地址位（八位字节 0 的最低有效位）用于将目的地址识别为单播地址或组播地址。如果单播/组播地址位为 0，则表示地址字段包含一个单播地址。如果该位为 1，则地址字段包含一个组播地址，用于标识连接到局域网的一个或多个（或全部）站点。全站广播地址是一种特殊的预定义组播地址，其各位均为 1。

通用或本地管理地址位是八位字节 0 中紧邻单播/组播地址位的比特。此位表示该地址是由本地管理员还是通用管理员分配的。通用管理员分配的地址，此位设为 0。如果此位设为 1，则整个地址（即 48 位）均由本地管理员分配。

9.2.2 组织分配

对于给定组织唯一标识符（OUI）的 MAC 地址块，更改其最后 24 位，可让该 OUI 受让方获得约 1600 万个唯一单播地址和 1600 万个唯一组地址，且其他组织均不得分配这些地址（即这些地址具有全局唯一性）。IEEE 打算不再向任何组织分配额外的组织唯一标识符，除非该组织已用尽此地址块。因此，IEEE 必须与每个受让方保持单一联系点，以免使分配流程复杂化。需要注意的是，绝不应将这些地址用于会导致大量地址被跳过的用途（例如，将其用作产品标识符以协助公司库存流程）。IEEE 要求各组织不要滥用最后 24 位的分配，从而避免不必要地耗尽该地址块。即使在批量生产中，现有标识符也足以满足大多数需求很长一段时间；然而，地址空间并非无限。

受让方采用的方法，以确保其设备不会出现地址重复的情况，当然取决于分配或制造流程、组织的性质以及组织的理念。然而，全球网络用户都期望拥有唯一地址。因此，确保满足用户期望和要求的最终责任在于提供此类设备的组织。

The standard representation of a protocol identifier is as a string of five octets, using the hexadecimal representation (3.1.8).

The LSB of the first octet of a protocol identifier is referred to as the M bit. All identifiers derived from OUIs assigned by the IEEE shall have the M bit set to 0. Values with the M bit set to 1 are reserved.

Protocol identifiers may be assigned universally or locally. The X bit of a protocol identifier is the bit of the first octet adjacent to the M bit. All identifiers derived from OUIs assigned by the IEEE will have the X bit set to 0 and are universally assigned. Values with the X bit set to 1 are locally assigned and have no relationship to the IEEE-assigned values. They may be used, but there is no assurance of uniqueness.

9.4 Standard group MAC addresses

The previous subclauses described the assignment of individual and group MAC addresses, and protocol identifiers for public or private use by private organizations. There is also a need for standard group MAC addresses to be used with standard protocols. The administration of these addresses, including the procedure for application and a list of currently assigned values, is defined in ISO/IEC TR 11802-2. These standard group MAC addresses come from a block of universally administered LAN MAC addresses derived from an OUI that has been assigned by the IEEE for this purpose.

ISO/IEC 8802-5 also defines *functional addresses*, for use in Token Ring LAN environments to identify well-known functional entities. These addresses are a subset of the locally administered group MAC addresses, identified by having the 15 address bits that follow the I/G and U/L address bits set to zero. Certain values are used consistently throughout Token Ring LAN environments and, therefore, play a very similar role to standard group MAC addresses; these functional address values are also recorded in ISO/IEC TR 11802-2.

9.5 Bit-ordering and different MACs

NOTE—Throughout this subclause, considerations relating to the order of bit and/or octet transmission refer to the basic bit-serial model of transmission that applies to the representation of MAC frames at the boundary between the MAC sublayer and the Physical layer; see 6.2.3.

9.5.1 General considerations

The transmission of data on IEEE 802.3 and 802.4™ LAN types is represented (6.2.3) as occurring LSB first within each octet. This is true for the entire frame: LAN MAC address fields (source and destination), MAC-specific fields (e.g., length field in IEEE 802.3 LANs), and the MAC Information field. (The MAC Information field is defined to be that part of the frame starting directly after the MAC header and ending immediately before the frame check sequence: e.g., LLC information, such as the Protocol Identification field, is contained in the MAC Information field.)

On some other LAN types, for which IEEE 802.5 is here used as the typical example, each octet of the MAC Information field is represented as being transmitted MSB first. The LAN MAC address fields (source and destination), however, are represented as being transmitted with the LSB of each octet first. Thus, the first bit transmitted is the I/G Address Bit, as on IEEE 802.3 and IEEE 802.4 LANs. For frames that originate within the MAC (e.g., MAC-embedded management frames), the ordering of bits within the MAC Information field is defined by the MAC specification—ISO/IEC 8802-5, etc.

For most purposes, the difference in the bit-orderings used to represent transmission of the octets of the MAC Information field is of no consequence, whether considered within a given MAC type, or across different MAC types. Each octet of user data is mapped to and from the appropriate ordering, symmetrically by the transmitting and receiving MAC entities. An unfortunate exception has occurred, however, where the octets concerned are those of a MAC address that is embedded, as user data, in the MAC Information field.

协议标识符的标准表示法是由五个八位组构成的字符串，采用十六进制表示（3.1.8）。

协议标识符第一个字节的最低有效位称为 M 位。所有源自 IEEE 分配的组织唯一标识符（OUI）的标识符，其 M 位均应设为 0。M 位设为 1 的值为保留值。

协议标识符可以进行全球分配，也可以进行本地分配。协议标识符的 X 位是第一个字节中与 M 位相邻的比特。所有源自 IEEE 分配的组织唯一标识符（OUI）的标识符，其 X 位均设为 0，属于全球分配。X 位设为 1 的值为本地分配，与 IEEE 分配的值无关。这些值可以使用，但无法保证其唯一性。

9.4 标准组 MAC 地址

前面的子条款介绍了如何分配供私人组织公开或私下使用的单播和组 MAC 地址，以及协议标识符。此外，还需要用于标准协议的标准组 MAC 地址。这些地址的管理，包括申请流程和当前已分配值的列表，均在 ISO/IEC TR 11802-2 中定义。这些标准组 MAC 地址来自一个由 IEEE 为此目的分配的 OUI 所派生的通用管理局域网 MAC 地址块。

ISO/IEC 8802-5 还定义了功能地址，用于令牌环局域网环境中标识众所周知的功能实体。这些地址是本地管理的组 MAC 地址的子集，其特征是 I/G 和 U/L 地址位之后的 15 个地址位均置为零。某些值在令牌环局域网环境中始终一致地使用，因此其作用与标准组 MAC 地址非常相似；这些功能地址值也记录在 ISO/IEC TR 11802-2 中。

9.5 位序与不同的 MAC 地址

注——在本小节中，凡是涉及比特和/或八位字节传输顺序的内容，均指适用于 MAC 子层与物理层边界处 MAC 帧表示的基本位串行传输模型；参见 6.2.3。

9.5.1 总则

IEEE 802.3 和 802.4™ 局域网类型上的数据传输表示为（6.2.3）每个八位字节内均先传输最低有效位。整个帧均如此：局域网 MAC 地址字段（源和目的地）、MAC 专用字段（例如 IEEE 802.3 局域网中的长度字段）以及 MAC 信息字段。（MAC 信息字段定义为帧中紧接 MAC 首部之后开始、在帧校验序列之前立即结束的部分；例如，逻辑链路控制信息（如协议标识字段）包含在 MAC 信息字段中。）

对于某些其他局域网类型，本文以 IEEE 802.5 为典型示例，MAC 信息字段中的每个八位字节都表示为先传输最高有效位。然而，局域网 MAC 地址字段（源和目的地）则表示为先传输每个八位字节的最低有效位。因此，与 IEEE 802.3 和 IEEE 802.4 局域网一样，首先传输的是单播/组播地址位。对于源自介质访问控制（MAC）的帧（例如，MAC 嵌入式管理帧），MAC 信息字段中的位排序由 MAC 规范（ISO/IEC 8802-5 等）定义。

在大多数情况下，用于表示 MAC 信息字段中八位组传输的位序差异并无影响，无论是在某一种 MAC 类型内考察，还是在不同 MAC 类型之间考察。每个用户数据八位字节都会由发送和接收 MAC 实体以对称方式映射为相应的位序，并从相应位序映射回来。然而，当相关八位组属于嵌入在 MAC 信息字段中、作为用户数据的 MAC 地址时，却出现了一个不幸的例外。

The consequences particularly affect the use of MAC addresses in mixed environments containing Token Ring LANs and non-Token-Ring LANs.

The following subclauses describe the problem and some of the issues arising from it.

9.5.2 Illustrative examples

This subclause illustrates the various bit- and octet-transmission scenarios that can occur, and it is intended as a basis for clarifying the issue of bit ordering for LAN MAC addresses across different MACs. Throughout, the examples make use of the OUI value AC-DE-48, introduced in 9.2.1. This three-octet value is considered in its two possible roles: as the first part of a five-octet protocol identifier, and as the first part of a six-octet LAN MAC address. The consistent representations of the OUI in its role as part of a protocol identifier are contrasted with the sometimes variable representations that apply to its role as part of a MAC address.

NOTE—Protocol identifiers always form part of the normal user data in a MAC Information field; hence, there is nothing special about OUI octets in their protocol identifier role.

For the examples, the bit significance of an OUI in general is defined to be as in Figure 10.

	MSB						LSB
Octet 0	h	g	f	e	d	c	b a
Octet 1	p	o	n	m	l	k	j i
Octet 2	x	w	v	u	t	s	r q

When used in LAN MAC addresses:
Bit “a” of the OUI = I/G address bit.

Bit “b” of the OUI = U/L address bit.

When used in protocol identifiers:
Bit “a” of the OUI = M bit.

Bit “b” of the OUI (always zero) = X bit.

Figure 10—Bit significance of an OUI

When transmitted on an IEEE 802.3 or IEEE 802.4 LAN (all data octets transmitted LSB first), the OUI portions of a protocol identifier and of a LAN MAC address appear as in Figure 11. When transmitted on an IEEE 802.5 LAN (data octets in the MAC Information field transmitted MSB first), the OUI portions of a protocol identifier, and of a LAN MAC address contained in a MAC Address field, appear as in Figure 12

In some circumstances, it is necessary to convey LAN MAC addresses as data within MAC Information fields, e.g., as part of a management protocol or a Network layer routing protocol.

For LAN types in which Figure 11 applies, such as IEEE 802.3, the bit ordering within the octets of a LAN MAC address conveyed as data is the same as both the ordering when the address appears in a MAC address field and the ordering for octets of nonaddress information (see Figure 13).

For LAN types in which Figure 12 applies, such as IEEE 802.5 and FDDI, there appears to be a choice of representations for MAC addresses conveyed as data, as follows:

- a) The octets of the MAC address can be treated like any other data octets and transmitted with the bit-ordering of Figure 12 (A)
- b) The bit-ordering of Figure 12 (B) can be treated as a property of the MAC address rather than of the MAC address field as transmitted in MAC frames, and the MAC address octets can be transmitted with the bit-ordering reversed compared with normal data octets

其影响尤其体现在包含令牌环局域网和非令牌环局域网的混合环境中 MAC 地址的使用上。

以下各小条款将描述该问题以及由此产生的一些问题。

9.5.2 示例说明

本小条款通过示例说明可能出现的各种比特和八位字节传输情形，旨在作为澄清不同介质访问控制（MAC）设备之间局域网 MAC 地址位顺序问题的依据。文中示例均使用 9.2.1 中引入的组织唯一标识符（OUI）值 AC-DE-48。这个三八位字节值有两种可能的用途：作为五八位字节协议标识符的首部分，以及作为六八位字节局域网 MAC 地址的首部分。文中对比了该 OUI 用作协议标识符一部分时的一致表示方式，以及用作 MAC 地址一部分时有时会变化的表示方式。

注——协议标识符始终是 MAC 信息字段中常规用户数据的一部分；因此，OUI 八位组作为协议标识符时并无特殊之处。

对于这些示例，OUI 的位重要性通常定义为图 10 所示。

	MSB						LSB
八位字节 0	h	g	f	e	d	c	b a
八位字节 1	p	o	n	m	l	k	j i
八位字节 2	x	w	v	u	t	s	r q

用于局域网 MAC 地址时：OUI 的 “a” 位 =
单播/组播地址位。

组织唯一标识符中的 “b” 比特 = 通用/本地地址位。

在协议标识符中使用时：组织唯一标识符
= M 位的 “a” 比特。

组织唯一标识符中的 “b” 比特（始终为零）= X 位。

图 10—组织唯一标识符的位重要性

在 IEEE 802.3 或 IEEE 802.4 局域网上传输时（所有数据字节均按最低有效位优先的顺序传输），协议标识符和局域网 MAC 地址中的 OUI 部分如图 11 所示。在 IEEE 802.5 局域网上传输时（MAC 信息字段中的数据字节按最高有效位优先的顺序传输），协议标识符中的 OUI 部分，以及 MAC 地址字段中所含局域网 MAC 地址中的 OUI 部分，如图 12 所示

在某些情况下，需要将局域网 MAC 地址作为数据置于 MAC 信息字段中，例如，作为管理协议或网络层路由协议的一部分。

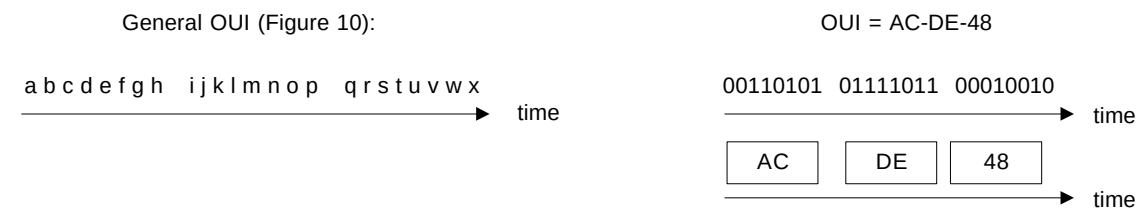
对于适用图 11 的局域网类型（如 IEEE 802.3），作为数据传送的局域网 MAC 地址，其八位组内的位排序与地址出现在 MAC 地址字段中时的排序以及非地址信息八位组的排序相同（见图 13）。

对于适用图 12 的局域网类型（例如 IEEE 802.5 和光纤分布式数据接口），作为数据传递的 MAC 地址似乎有以下几种表示方式可供选择：

- a) MAC 地址的八位组可以像其他任何数据字节一样处理，并与图 12 (A) 中的位序
- b) 图 12 (B) 中的位序可以视为 MAC 地址的属性，而非在 MAC 帧中传输的 MAC 地址字段的属性；与常规数据字节相比，MAC 地址八位字节可以采用相反的位序进行传输

.

(A) OUI within a protocol identifier (in MAC Information field, as normal data):



(B) OUI within a MAC Address field:

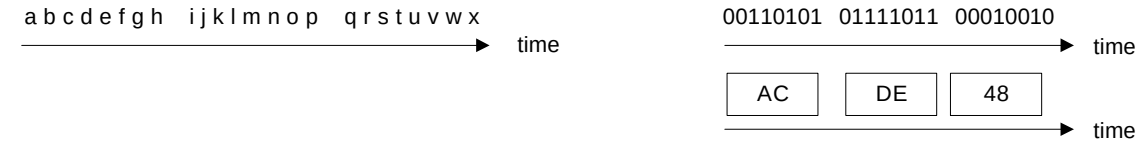
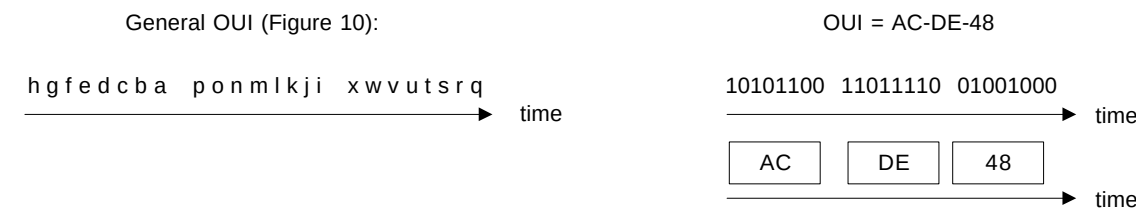


Figure 11—IEEE 802.3, etc., frame: Order of bit and octet transmission for an OUI

(A) OUI within a protocol identifier (in MAC Information field, as normal data):



(B) OUI within a MAC Address field:

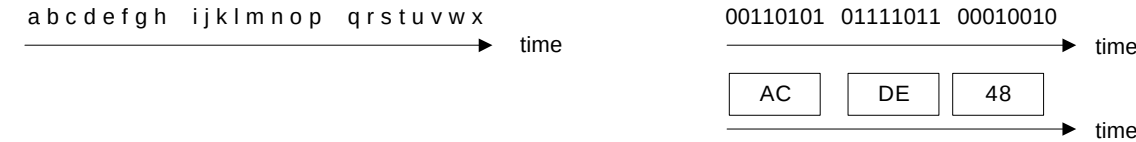


Figure 12—IEEE 802.5, etc., frame: Order of bit and octet transmission for an OUI

OUI within a MAC address contained in the MAC Information field:

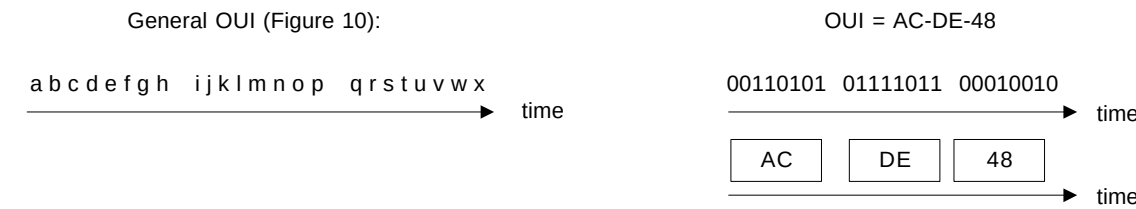


Figure 13—IEEE 802.3, etc., frame: Order of bit and octet transmission for an OUI in a MAC address contained in a MAC Information field

In the case of IEEE 802.5 Token Ring LANs, approach b) was adopted early in the development and deployment of Token Ring technology. This has the unfortunate consequence that applications operating in environments containing a mixture of LAN types have to handle different representations of MAC addresses, according to the environment in which the MAC address is to be used; see Figure 14.

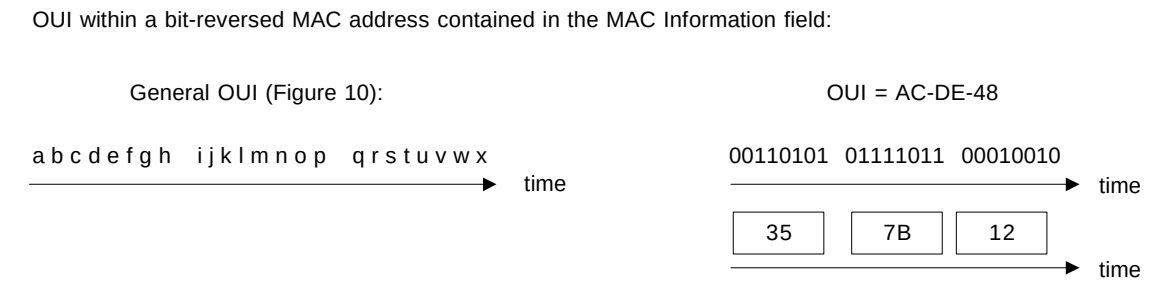


Figure 14—IEEE 802.5, etc., frame: Order of bit and octet transmission for an OUI in a MAC address contained in a MAC Information field (noncanonical format)

For other LAN types in which Figure 12 applies, including, in particular, FDDI, approach a) was adopted (Figure 15), at least in environments involving interconnection with IEEE 802.3, and so on, LANs. However, where FDDI LANs are used in an IEEE 802.5 Token Ring environment, approach b) is used for consistency with the interconnected IEEE 802.5 LANs. In a mixed environment of FDDI, IEEE 802.3 and IEEE 802.5 LANs, frames constructed according to both approaches can occur on the FDDI LANs, at least; some care is needed in managing such an installation to avoid confusion between the formats.

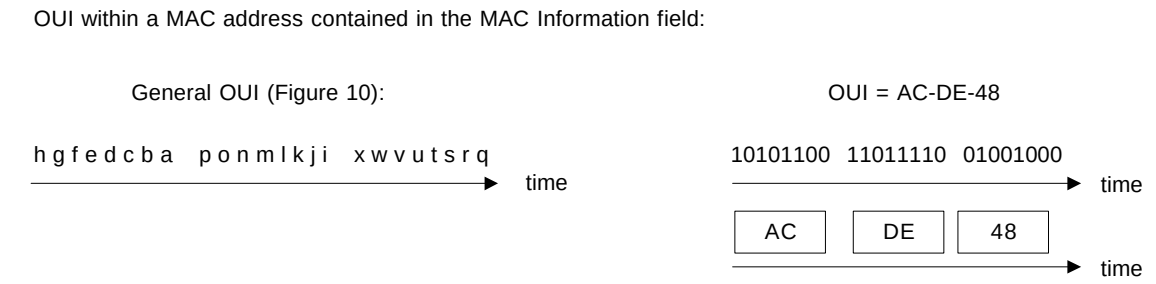


Figure 15—FDDI frame: Order of bit and octet transmission for an OUI in a MAC address contained in a MAC Information field (canonical format)

In Figure 11 through Figure 15, it can be seen that the interpretation of OUI bits as octet values is consistent in every case except Figure 14, in which the octet values are bit reversed. This reversal of the bit order applies only to all six octets (not just the OUI) of a MAC address placed in the MAC Information field of a frame by a protocol that uses the Token Ring Bit-reversed view of MAC addresses derived from Figure 12 (B). Frames containing, or possibly containing, such MAC addresses are described as having noncanonical format. Frames (on any LAN or LAN type) that cannot contain such MAC addresses are described as having canonical format.

Note that there is no way of knowing, from MAC layer information only, whether a particular frame is in canonical or noncanonical format. In general, this depends on which higher layer protocols are present in the frame.

对于 IEEE 802.5 令牌环局域网，在令牌环技术的早期开发和部署阶段就采用了方法 b)。遗憾的是，这导致运行于包含多种局域网类型的环境中的应用程序，必须根据 MAC 地址的使用环境来处理 MAC 地址的不同表示形式；见图 14。

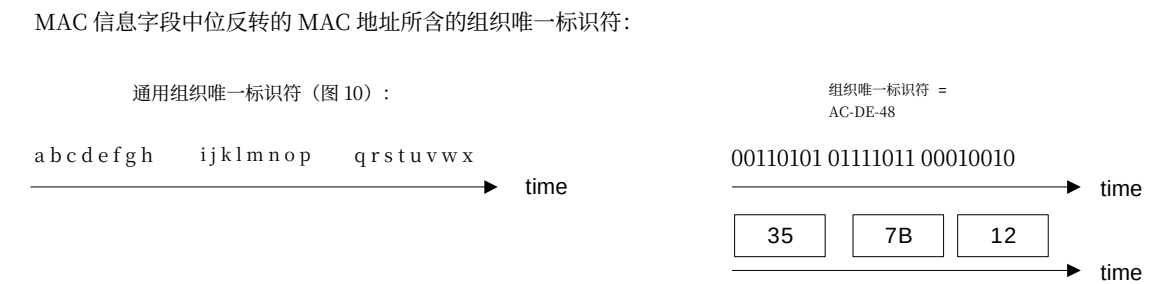


图 14—IEEE 802.5 等帧：MAC 信息字段中 MAC 地址内组织唯一标识符的比特和八位组传输顺序（非规范格式）

对于适用图 12 的其他局域网类型，尤其包括光纤分布式数据接口，采用了方法 a)（图 15），至少在涉及与 IEEE 802.3 等局域网互连的环境中是如此。但是，在 IEEE 802.5 令牌环环境中使用光纤分布式数据接口局域网时，为了与互连的 IEEE 802.5 局域网保持一致，采用方法 b)。在光纤分布式数据接口、IEEE 802.3 和 IEEE 802.5 局域网混合的环境中，光纤分布式数据接口局域网上至少可能出现按照这两种方法构造的帧；管理此类安装时需要谨慎，以免混淆这些格式。

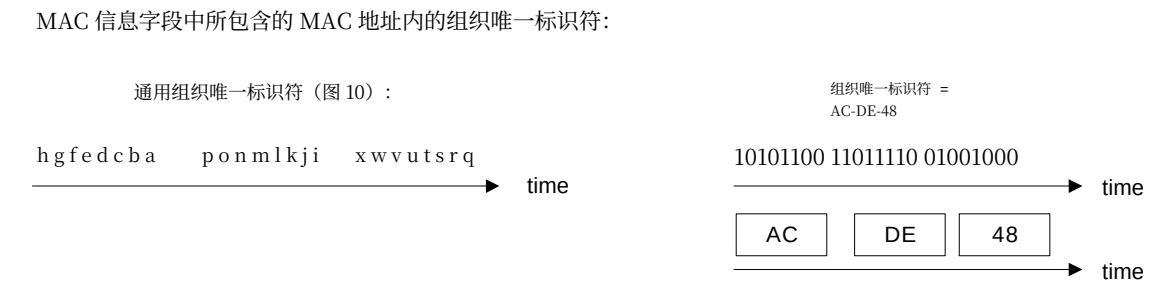


图 15—FDDI 帧：MAC 信息字段中 MAC 地址内的组织唯一标识符的比特和八位组传输顺序（规范格式）

从图 11 到图 15 可以看出，除图 14 外，在所有情况下，将 OUI 位解释为八位组值都是一致的；图 14 中的八位组值经过了位反转。这种位顺序反转仅适用于由使用 MAC 地址令牌环位反转视图的协议放入帧的 MAC 信息字段中的 MAC 地址的全部六个八位组（而不仅仅是 OUI），这些地址源自图 12 (B)。包含或可能包含此类 MAC 地址的帧称为非规范格式。无法包含此类 MAC 地址的帧（无论属于何种局域网或局域网类型）称为规范格式。

请注意，仅凭 MAC 层信息，无法判断特定帧采用的是规范格式还是非规范格式。一般来说，这取决于帧中包含哪些高层协议。

9.5.3 Recommendation

Designers of protocols that operate above the Data Link layer are strongly recommended to avoid specifying new protocols that result in frames of noncanonical format, except where such a protocol is clearly an extension of existing practice in a strongly Token Ring environment.

10. Protocol discrimination above the MAC sublayer: Subnetwork Access Protocol (SNAP) and Ethernet types

10.1 Introduction

This clause outlines the mechanisms for the coexistence of multiple standard, public, and private network layer protocols within a single 802 station (10.2). It then describes the functions, features, and protocol format conventions for public and private protocols sharing a single LSAP (10.3). All public and private protocols using the IEEE 802 reserved LLC address assigned for public and private protocol use shall conform to this standard.

This clause further describes Ethernet types used to identify different protocols operating over the alternative Ethernet sublayer (10.4), and it describes the standard encapsulation specified for conveyance of such Ethernet-supported protocols on IEEE 802 LANs that do not intrinsically support an Ethernet sublayer (10.5).

A standard protocol is defined to be a protocol whose specification is published and known to the public but controlled by a standards body. A public protocol is defined to be a protocol whose specification is published and known to the public, but controlled by an organization other than a formal standards body. A private protocol is defined to be a protocol whose use and specification are controlled by a private organization.

By providing for the coexistence of multiple Network layer protocols, the migration of existing LANs to future standard protocols is facilitated, and multiple higher layer protocols are more easily accommodated.

10.2 Basic concepts

10.2.1 Coexistence of multiple protocols

Within a given layer, entities can exchange data by a mutually agreed upon protocol mechanism. A pair of entities that do not support a common protocol cannot communicate with each other. For multiple protocols to coexist within a layer, it is necessary to determine which protocol is to be invoked to process a service data unit delivered by the lower layer.

The following subclauses specify mechanisms for use when the LLC sublayer is present above the MAC layer, and when the alternative Ethernet sublayer is present above the MAC sublayer.

10.2.2 Multiple protocols above the LLC sublayer

Standard Network layer protocols have been assigned reserved LLC addresses, as recorded in ISO/IEC TR 11802-1. These addresses permit multiple standard network layer protocols to coexist at a single MAC station. One half of the LLC address space is reserved for such assignment.

9.5.3 建议

强烈建议设计运行在数据链路层之上的协议的人员避免规定会产生非规范格式帧的新协议，除非此类协议显然是对令牌环环境中既有做法的扩展。

10. MAC 子层之上的协议区分：子网访问协议 (SNAP) 和以太网类型

10.1 引言

本条概述了在单个 802 站点（10.2）中共存多种标准、公共和专用网络层协议的机制。随后介绍了共享单个链路服务接入点（LSAP）的公共和专用协议的功能、特性和协议格式约定（10.3）。所有使用分配给公共和专用协议的 IEEE 802 保留 LLC 地址的公共和专用协议，均应符合本标准。

本条进一步介绍了用于标识在替代以太网子层上运行的不同协议的以太网类型（10.4），并介绍了为在本身不支持以太网子层的 IEEE 802 局域网上传送此类以太网支持的协议而规定的标准封装（10.5）。

标准协议是指其规范已发布且为公众所知，但由标准组织控制的协议。公共协议是指其规范已发布且为公众所知，但由正式标准组织以外的组织控制的协议。专用协议是指其使用和规范由私人组织控制的协议。

通过支持多个网络层协议共存，可促进现有局域网向未来标准协议迁移，也能更轻松地支持多个高层协议。

10.2 基本概念

10.2.1 多种协议共存

在给定层内，实体可以通过双方共同认可的协议机制交换数据。不支持共同协议的一对实体无法相互通信。要使多个协议在一层内共存，必须确定调用哪种协议来处理由下层传递的服务数据单元。

以下子条款规定了在 MAC 层之上存在 LLC 子层，以及在 MAC 子层之上存在另一种以太网子层时所使用的机制。

10.2.2 LLC 子层之上的多种协议

标准网络层协议已分配保留的逻辑链路控制地址，如 ISO/IEC TR 11802-1 中所记录。这些地址允许多个标准网络层协议共存于单个介质访问控制站。逻辑链路控制地址空间的一半保留用于此类分配。

Other protocols are accommodated in two ways. One way is by local assignment of LSAPs, for which the other half of the LLC address space is available. Thus, users can agree to use locally assigned LSAPs for either an instance of communication or a type of communication.

The second way is through the use of a particular reserved LLC address value that has been assigned for use in conjunction with the Subnetwork Access Protocol (SNAP, specified in 10.3), which provides for multiplexing and demultiplexing of public and private protocols among multiple users of a data link.

10.2.3 Multiple protocols above the Ethernet sublayer

The Ethernet MAC frame format includes a 16-bit type value, whose function is to identify the particular protocol pertaining to the user data contained in the frame. See 10.4 for further details.

10.3 Subnetwork Access Protocol

10.3.1 SNAP address

The reserved LLC address for use with SNAP is called the SNAP address. It is defined to be the bit pattern (starting with the LSB) Z1010101, in which the symbol Z indicates that either value 0 or 1 can occur, depending on the context in which the address appears (as specified in ISO/IEC 8802-2). The two possible values have Hexadecimal Representation AA and AB.

The SNAP address identifies, at each MSAP, a single LSAP for standard, public, and private protocol usage. To permit multiple public and private network layer protocols to coexist at one MSAP, each public or private protocol using SNAP must employ a protocol identifier that enables SNAP to discriminate among these protocols.

10.3.2 SNAP PDU format

Each SNAP PDU shall conform to the format shown in Figure 16 and shall form the entire content of the LLC information field.

In Figure 16, the Protocol Identification field is a five-octet field containing a protocol identifier whose format and administration are as described in 9.3. The Protocol Data field is a field whose length, format, and content are defined by a public or private protocol specification. Each public or private protocol begins its PDU format with the Protocol Identification field, which shall contain the protocol identifier assigned to the protocol.

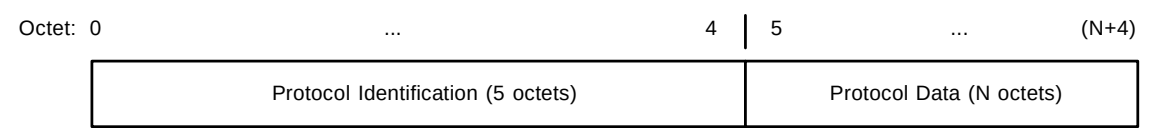


Figure 16—SNAP PDU format

Figure 17 illustrates how a SNAP PDU appears in a complete MAC frame (the IEEE 802.3 MAC format is used for the example). The LLC control field (CTL) is shown for PDU type UI, Unnumbered Information, which is the most commonly used PDU type in this context; however, other information-carrying LLC PDU types may also be used with SNAP.

其他协议可通过两种方式得到支持。一种方式是本地分配链路服务接入点，逻辑链路控制地址空间的另一半可用于此目的。因此，用户可以商定使用本地分配的链路服务接入点来表示某个通信实例或某种通信类型。

第二种方式是使用一个特定的保留的 LLC 地址值，该值已指定用于配合子网访问协议（SNAP，详见 10.3）使用，从而在一个数据链路的多个用户之间复用和解复用公用协议与专用协议。

以太网子层之上的多种协议

以太网 MAC 帧格式包含一个 16 比特的类型值，其作用是标识与帧中所含用户数据相关的特定协议。更多详情请参见 10.4。

10.3 子网访问协议

10.3.1 SNAP 地址

与 SNAP 配合使用的保留 LLC 地址称为 SNAP 地址。它被定义为位模式（从最低有效位开始）Z1010101，其中符号 Z 表示根据地址出现的上下文（如 ISO/IEC 8802-2 中所规定），其值可以是 0 或 1。这两个可能的值的十六进制表示为 AA 和 AB。

在每个 MAC 服务访问点，SNAP 地址都标识一个用于标准、公共和专用协议的单一链路服务接入点。为使多个公共和专用网络层协议能够在一个 MAC 服务访问点共存，每个使用 SNAP 的公共或专用协议都必须采用一个协议标识符，以便 SNAP 区分这些协议。

10.3.2 SNAP 协议数据单元格式

每个 SNAP 协议数据单元均应符合图 16 所示的格式，并构成整个 LLC 信息字段的内容。

在图 16 中，协议标识字段是一个五个八位组的字段，其中包含一个协议标识符，其格式和管理方式如 9.3 所述。协议数据字段的长度、格式和内容由公开或私有协议规范定义。每种公开或私有协议都以协议标识字段作为其 PDU 格式的开头，该字段应包含分配给该协议的协议标识符。

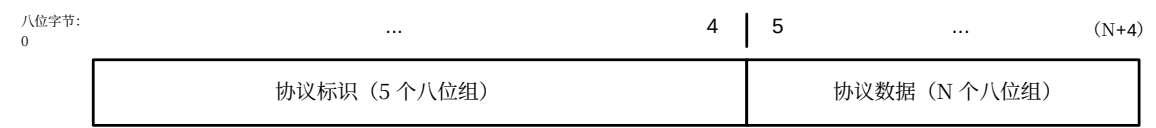


图 16—SNAP PDU 格式

图 17 展示了 SNAP 协议数据单元在完整 MAC 帧中的形式（示例采用 IEEE 802.3 MAC 格式）。图中显示的是 UI 类型 PDU（无编号信息）的 LLC 控制字段（控制字段，CTL）。这是此上下文中最常用的 PDU 类型；不过，SNAP 也可以使用其他承载信息的 LLC PDU 类型。

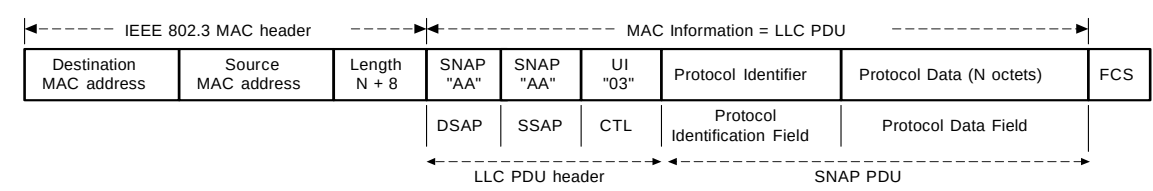


Figure 17—SNAP PDU in IEEE 802.3 MAC frame

10.4 Ethernet types: Format, function, and administration

The IEEE 802.3 MAC frame format is compatible with the alternative Ethernet MAC frame format, in the sense that frames of both formats can be freely intermixed on a given LAN and at a given LAN station. The service provided by use of the Ethernet frame format differs from the ISO/IEC 15802-1 MAC service in that there is a 16-bit type value associated with each frame transferred, and that the minimum amount of MAC-sublayer user data transferred is 64 octets.

An Ethernet type value is a sequence of two octets, interpreted as a 16-bit numeric value with the first octet containing the most significant 8 bits and the second octet containing the least significant 8 bits. Values in the range 0–1535 are not available for use.

The function of the Ethernet type value is to identify the protocol that is to be invoked to process the user data in the frame.

As with OUIs, administration of Ethernet types was originally undertaken by the Xerox Corporation, and it is now the responsibility of the IEEE using procedures defined by the IEEE RAC (see Clause 9). All assignments of Ethernet types made by the predecessor administration remain in effect under the IEEE’s administration.

10.5 Encapsulation of Ethernet frames over LLC

This subclause specifies the standard method for conveying Ethernet frames across IEEE 802 LANs that offer only the LLC sublayer, and not the Ethernet sublayer, directly above the MAC sublayer.

An Ethernet frame conveyed on an LLC-only LAN shall be encapsulated in a SNAP PDU contained in an LLC PDU of type UI, as follows (see Figure 18):

- a) The Protocol Identification field of the SNAP PDU shall contain a protocol identifier in which
 - 1) The three OUI octets each take the value zero.
 - 2) The two remaining octets take the values, in the same order, of the two octets of the Ethernet frame’s Ethernet type.
- b) The Protocol Data field of the SNAP PDU shall contain the user data octets, in order, of the Ethernet frame.
- c) The values of the Destination MAC Address and Source MAC Address fields of the Ethernet frame shall be used in the Destination MAC Address and Source MAC Address fields, respectively, of the MAC frame in which the SNAP PDU is conveyed.

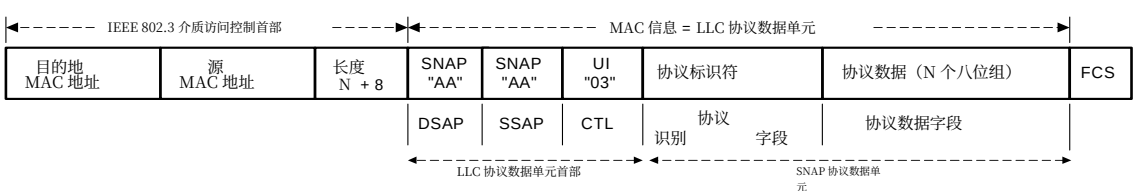


图 17—IEEE 802.3 MAC 帧中的 SNAP 协议数据单元

10.4 以太网类型：格式、功能和管理

IEEE 802.3 MAC 帧格式与另一种以太网 MAC 帧格式兼容，也就是说，在给定的局域网及其站点上，两种格式的帧可以自由混用。使用以太网帧格式所提供的服务与 ISO/IEC 15802-1 MAC 服务有所不同：每个传输的帧都关联一个 16 位类型值，且传输的 MAC 子层用户数据最少为 64 个八位组。

以太网类型值由两个八位字节组成，是一个 16 位数值，其中第一个字节包含最高 8 位，第二个字节包含最低 8 位。0–1535 范围内的值不可使用。

以太网类型值的功能是标识将被调用以处理帧中用户数据的协议。

与组织唯一标识符（OUI）一样，Ethernet 类型最初由施乐公司管理；现在则由 IEEE 按照 IEEE 注册管理委员会规定的程序负责管理（见第 9 条）。前任管理机构分配的所有 Ethernet 类型在 IEEE 管理下仍然有效。

10.5 通过 LLC 封装以太网帧

本条规定了在仅提供 LLC 子层、而不在 MAC 子层之上直接提供以太网子层的 IEEE 802 局域网上传送以太网帧的标准方法。

在仅支持 LLC 的局域网上传送的以太网帧，应封装在 SNAP 协议数据单元中，而该 SNAP 协议数据单元包含在类型为无编号信息的 LLC 协议数据单元中，如下所示（见图 18）：

- a) SNAP 协议数据单元的协议标识字段应包含一个协议标识符，其中
 - 1) 三个 OUI 八位组的值均为零。
 - 2) 剩余的两个八位组依次取以太网中两个八位组的值
帧的以太网类型。
- b) SNAP 协议数据单元的协议数据字段应按顺序包含以太网帧
- c) 以太网帧的目的 MAC 地址字段和源 MAC 地址字段的值，应分别用于承载 SNAP 协议数据单元的 MAC 帧中的目的 MAC 地址字段和源 MAC 地址字段。

NOTES

- 1—This encapsulation was originally specified in RFC 1042, which contains recommendations relating to its use. Further recommendations are contained in RFC 1390.
- 2—ISO/IEC TR 11802-5 (IEEE Std 802.1H™-1997) contains recommendations for bridges, addressing the consequences of certain problems that arose from differing interpretations of RFC 1042.

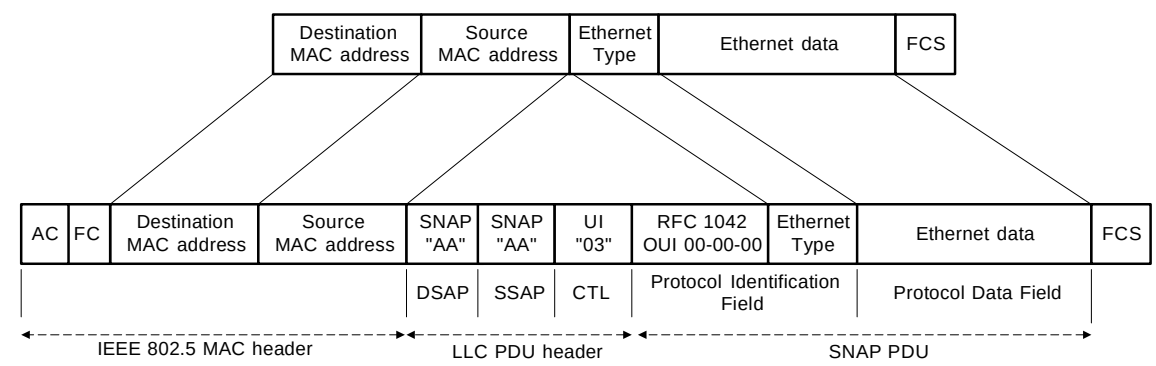


Figure 18—Ethernet frame SNAP-encapsulated in IEEE 802.5 frame

11. ISLAN and MAN support for isochronous bearer services

In addition to the mandatory LAN and MAN services described so far, some IEEE 802 LANs and MANs, notably, ISO/IEC 8802-9 and ISO/IEC 8802-6, also make provision for supporting isochronous bearer services. Isochronous bearer services are distinctive relative to packet services such as the MAC service and LLC, in that they maintain a flow of service data units at constant time intervals from a transmitter to a receiver for the duration of a service. In almost all circumstances, such isochronous bearer services are carried over duplex bidirectional connections thereby providing effective and efficient means of supporting the ubiquitous WAN voice telephony services. IEEE 802 ISLANs and MANs provide isochronous bearer services designed to interwork readily with these WAN services as standardized by ITU-T, in particular, as defined in the I series of ITU-T Recommendations.

NOTE—Use of an end-to-end physical-layer isochronous bearer service, which intrinsically delivers data with timing preserved, needs to be distinguished from use of a packet-based service for conveyance of time-sensitive data such as voice or video. The latter approach can be successful, given adequate bandwidth in the LANs and bridges, and provided that bridges do not introduce the possibility of excessive delay for packets carrying the time-sensitive data.

11.1 Key concepts

Applications requiring sustained periodic use of end-to-end network bandwidth are common. Two of the IEEE 802 standards address this requirement; both the ISO/IEC 8802-9 ISLAN standard and the optional isochronous service on an ISO/IEC 8802-6 Distributed Queue Dual Bus (DQDB) MAN use synchronously cyclic methods to ensure precise timing of the acceptance, transfer, and delivery of continuous streams of information. This applies whether or not asynchronous packet services are also provided concurrently.

The ISLAN employs a time-division multiplex (TDM) bearer mechanism within the Physical layer. The isochronous service on a DQDB MAN uses a Pre-Arbitrated function to ensure precisely timed access to the media, as distinct from the packet-service Queued Arbitrated function. In each case, this permits the support and delivery of a plurality of transparent isochronous service channels, the provision of an octet alignment signal for these channels, and a facility to provide and accept these precise timing signals. It is the provision of the timing signal that principally distinguishes the isochronous services from the asynchronous packet services that are provided from the ISLAN Physical layer.

注释

- 1—这种封装最初在 RFC 1042 中作出了规定，其中包含有关其使用的建议。RFC 1390 中还包含进一步的建议。
- 2—ISO/IEC TR 11802-5（IEEE Std 802.1H™-1997）包含有关网桥的建议，旨在解决因对 RFC 1042 的不同解释而产生的某些问题所带来的影响。

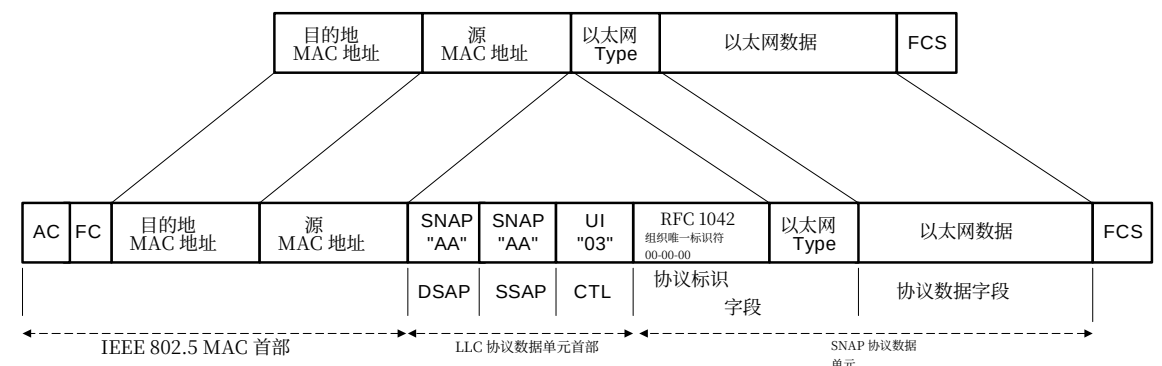


图 18—封装在 IEEE 802.5 帧中的 SNAP 以太网帧

11. ISLAN 和 MAN 对等时承载业务的支持

除上述强制性局域网和城域网服务外，某些 IEEE 802 局域网和城域网（尤其是 ISO/IEC 8802-9 和 ISO/IEC 8802-6）还规定支持等时承载业务。与 MAC 服务和逻辑链路控制等分组服务相比，等时承载业务的特点是：在服务持续期间，以恒定的时间间隔将服务数据单元从发送方传送到接收方。在几乎所有情况下，这类等时承载业务都通过双向双工连接承载，从而为支持普遍存在的广域网语音电话业务提供有效且高效的方式。IEEE 802 综合业务局域网和城域网提供的等时承载业务，旨在与这些由 ITU-T 标准化的广域网服务轻松互通，特别是与 ITU-T 建议书 I 系列中定义的服务互通。

注——应将端到端物理层等时承载业务的使用（该业务本身能够在传送数据时保留定时信息），与使用基于分组的服务来传送语音或视频等对时间敏感的数据区分开来。后一种方式在局域网和网桥具有足够带宽，且网桥不会导致承载时间敏感数据的分组可能出现过度延迟的情况下，可以取得成功。

11.1 关键概念

需要持续、周期性地使用端到端网络带宽的应用十分常见。IEEE 802 标准中有两项标准针对这一需求；ISO/IEC 8802-9 ISLAN 标准和 ISO/IEC 8802-6 分布式队列双总线 (DQDB) 城域网的可选等时服务，均采用同步循环方法，确保连续信息流的接收、传输和交付时序精确。无论是否同时提供异步分组服务，均适用。

ISLAN 在物理层采用时分复用（TDM）承载机制。DQDB 城域网中的等时服务使用预先仲裁功能，以确保对介质的访问时序精准；这与分组服务使用的排队仲裁功能不同。在这两种情况下，都可以支持并传送多个透明的等时服务信道，为这些信道提供八位字节对齐信号，并提供发送和接收这些精准定时信号的功能。定时信号的提供，正是等时服务与 ISLAN 物理层所提供的异步分组服务之间的主要区别。

Both methods for providing isochronous bearer services require the prior establishment of a connection for each isochronous information flow. For the DQDB MAN isochronous services, the mechanisms used for establishing and clearing connections are left outside of the scope of the IEEE 802 Standard. For ISLAN, see 11.4 below for an overview of the mechanisms used.

11.2 Applications

Applications for isochronous bearer services include the following:

- PBX interconnections at DS1 (1.544 Mbit/s) or E1 (2.048 Mbit/s) rates
- Low (384 kbit/s) to medium (44.2097 Mbit/s) bandwidth constant bit-rate compressed video
- Channels with bandwidth in multiples of 56 or 64 kbit/s for conveyance of voice and audio
- Multimedia combinations of these along with data services

Multimedia applications can require simultaneous, integrated use of two or more of these audio/voice, video, text, and graphics information streams. This can require the conveyance over common bearer channels of multiple isochronous and bursty information streams, in distinct channels with specific timing relationships. This provides the main rationale for incorporating isochronous services in LANs and MANs.

The provision of isochronous (TDM bearer) services enables direct interworking with a WAN network such that the access unit or hub (LAN/WAN interworking unit) can forward information in its native form, so that ISLAN terminals and access units do not have to provide gateway functions to transform the user information streams. In addition, the provision of isochronous bearer services greatly reduces the latency as information is queued/dequeued at the Physical service interface. This is of value in meeting established norms of loop delay on end-to-end connections for many human interactive services.

Typical isochronous services include the following:

- Unrestricted 64 kbit/s information transfer
- Restricted 56 kbit/s information transfer
- Synchronous data
- Facsimile data
- Wideband video and image transfer

The ISO/IEC 8802-9 ISLAN is specifically designed to provide concurrent support for LLC conformant packet services and narrowband ITU-T conformant ISDN services, both packet and isochronous, as defined in ITU-T I-series Recommendations.

11.3 Isochronous service access points and PhSAPs

The ISLAN and DQDB MAN standards both support concurrent packet and isochronous services within the Physical layer by means of convergence functions. In both cases, there is therefore a need for explicit identification of the distinct packet and isochronous channels provided over the common media supporting the ISLAN or MAN. In addition, both offer support for multiple isochronous bearer channels, and thus, there is a need to distinguish among multiple Physical service connections.

In the ISLAN standard, multiple PhSAPs are used to provide for the plurality of connections at the Physical service boundary, and for access by the user to the Physical-layer services for packet transfer and for

提供等时承载服务的两种方法，都要求预先为每个等时信息流建立连接。对于 DQDB 城域网的等时服务，建立和清除连接所用的机制不在 IEEE 802 标准的范围内。关于 ISLAN，请参见下文 11.4 节，了解所用机制的概述。

11.2 应用程序

等时承载业务的应用程序包括：

- DS1（1.544 兆比特/秒）或 E1（2.048 兆比特/秒）速率的专用小交换机互连
- 低带宽（384 kbit/s）至中等带宽（44.2097 Mbit/s）的恒定比特率压缩视频
- 带宽为 56 或 64 kbit/s 整数倍的信道，用于传输语音和音频
- 多媒体：上述内容与数据服务的组合

多媒体应用可能需要同时、集成地使用两种或多种音频/语音、视频、文本和图形信息流。这可能要求通过公共承载信道传输多个等时和突发信息流，并将它们置于具有特定定时关系的不同信道中。这在局域网和城域网中引入等时业务提供了主要依据。

提供等时（TDM 承载）服务，可实现与广域网的直接互通，使接入单元或集线器（局域网/广域网互通单元）能够以信息的原生形式转发信息，从而综合业务局域网终端和接入单元无需提供网关功能来转换用户信息流。此外，提供等时承载业务可大幅降低信息在物理服务接口处排队/出队时产生的时延。这对于满足许多人机交互业务端到端连接中既定的环路时延标准具有重要价值。

典型的等时业务包括以下内容：

- 不受限制的 64 kbit/s 信息传送
- 受限制的 56 kbit/s 信息传送
- 同步数据
- 传真数据
- 宽带视频和图像传送

ISO/IEC 8802-9 ISLAN 专门设计用于同时支持符合逻辑链路控制要求的分组服务，以及符合 ITU-T 要求的窄带综合业务数字网服务，包括分组服务和等时业务，定义见 ITU-T I 系列建议。

11.3 等时业务接入点和物理层服务接入点

ISLAN 和 DQDB 城域网标准都通过汇聚功能，在物理层中同时支持分组服务和等时业务。因此，在这两种情况下，都需要明确识别通过支持 ISLAN 或城域网的公共介质提供的不同分组信道和等时信道。此外，两者都支持多个等时承载信道，因此需要区分多个物理层服务连接。

在 ISLAN 标准中，使用多个物理层服务接入点，以便在物理层服务边界提供多个连接，并供用户访问物理层服务，用于分组传送和

D channel signaling (see Figure 19). For the DQDB MAN isochronous services, Connection Endpoint (CEs) identifiers are used for a similar purpose.

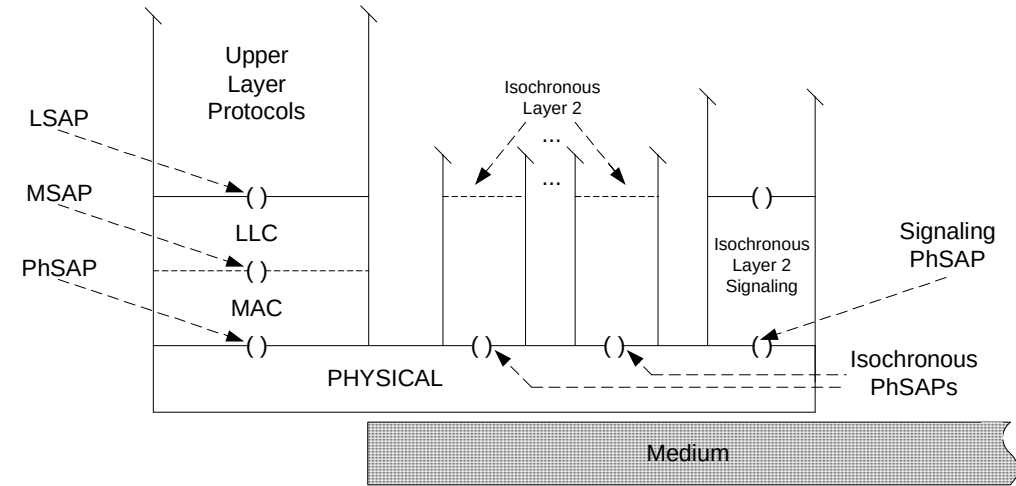


Figure 19—ISLAN RM (end station)

PhSAPs and CEs are the architectural mechanism by which symbol streams are passed to the Physical service provider by the Physical service user, and to the Physical service user by the Physical service provider. The distinction of different PhSAPs at a single ISLAN Physical service boundary is required because that service boundary is used simultaneously to provide both packet-mode and (multiple) circuit-mode Physical services.

It is a function of the layer management of the PHY multiplexing sublayer to provide each Physical service user with both the information stream and a channel identifier that is mapped onto the PhSAP relevant to the service provided to that user.

11.4 ISLAN signaling

The ISLAN standard provides for direct interworking with ITU-T conformant ISDN I.series Recommendations. These require means of establishing, maintaining, and disestablishing end-to-end connections across (in the ISLAN case) both the ISLAN and intervening WAN. To this end, ISO/IEC 8802-9 includes specifications for extensions to the ITU-T I- and Q-series signaling protocols carried in a signaling-specific D channel. This is a packet-based protocol, but to achieve interworking with the ITU-T Recommendations, it is distinct from the LLC packet service provided over ISLAN.

The signaling service provided by the protocols carried in the D channel permits negotiation of end-to-end network resources such that a guaranteed service can be provided to the users of an end-to-end isochronous channel. Thus, the ISLAN management signalling entity is responsible, as a management agent, for performing the following tasks:

- Negotiation of bandwidth (configuration) management, fault management, performance management, and security management of the access link between device and hub
- Negotiation of service provisioning over the local ISLAN interface in order to access the WAN-based ISDN services

The ISLAN signaling message elements are provided in compliance with international (ITU-T) network signaling methods with appropriate protocol extensions.

D 信道信令（参见图19）。对于 DQDB 城域网的等时服务，连接端点（CE）标识符也用于类似目的。

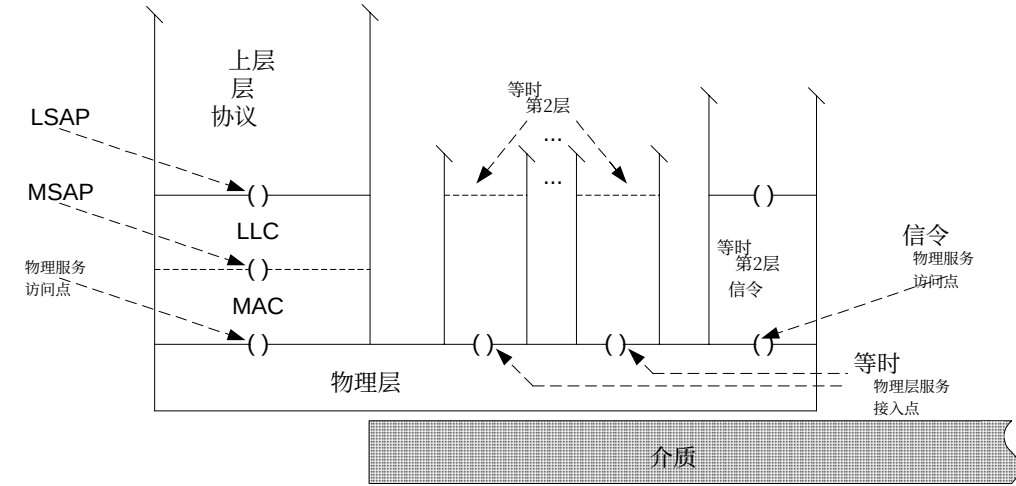


图19—ISLAN RM（终端站）

物理层服务接入点和连接端点是体系结构机制，物理服务用户通过该机制将符号流传递给物理服务提供者，物理服务提供者也通过该机制将符号流传递给物理服务用户。之所以需要在单个 ISLAN 物理服务边界上区分不同的物理层服务接入点，是因为该服务边界同时用于提供分组模式和（多个）电路模式物理层服务。

PHY 复用子层的层管理负责向每个物理服务用户提供信息流和信道标识符；该标识符会映射到与提供给该用户的服务相关的物理服务接入点。

11.4 ISLAN 信令

ISLAN 标准规定了与符合 ITU-T ISDN I 系列建议的系统直接互通。这些建议要求具备在（对于 ISLAN 而言）ISLAN 和中间广域网上建立、维持和断开端到端连接的手段。为此，ISO/IEC 8802-9 纳入了对通过专用信令 D 信道传送的 ITU-T I 系列和 Q 系列信令协议的扩展规范。这是一种基于分组的协议，但为了实现与 ITU-T 建议书的互通，它不同于 ISLAN 提供的逻辑链路控制分组服务。

通过 D 信道承载的协议提供的信令服务，允许协商端到端网络资源，从而为端到端等时信道的用户提供保证服务。因此，ISLAN 管理信令实体作为管理代理，负责执行以下任务：

- 协商 带宽（配置）管理，故障管理，性能接入链路的管理和安全管理（该链路位于设备与集线器之间）
- 为访问基于广域网的综合业务数字网服务，通过本地 ISLAN 接口协商业务配置

ISLAN 信令消息元素符合国际（ITU-T）网络信令方法，并采用适当的协议扩展。

Annex A

(informative)

Bibliography (Additional references for LAN/MAN-related standards)

A.1 General LAN/MAN standards and specifications

[B1] *Ethernet Version 2.0, A Local Area Network—Data Link Layer and Physical Layer Specifications*. Digital Equipment Corp., Intel Corp., and Xerox Corp., November 1982.

[B2] IEEE 100™, *The Authoritative Dictionary of IEEE Standards Terms*, Seventh Edition.⁸

[B3] IEEE Std 802.10-1992, IEEE Standards for Local and Metropolitan Area Networks: Interoperable LAN/MAN Security (SILS) (includes IEEE Std 802.10b™-1992).⁹

[B4] ISO/IEC TR 8802-1:1997, Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Technical reports and guidelines—Part 1: Overview of Local Area Network Standards.

[B5] ISO/IEC 8802-3:1999 (IEEE Std 802.3-1998), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 3: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications.

[B6] ISO/IEC 8802-4:1990 (IEEE Std 802.4-1990), Information processing systems—Local area networks—Part 4: Token-passing bus access method and physical layer specifications.

[B7] ISO/IEC 8802-5:1998 (IEEE Std 802.5-1998), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 5: Token ring access method and physical layer specifications.

[B8] ISO/IEC 8802-6:1994 (IEEE Std 802.6™-1994), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 6: Distributed Queue Dual Bus (DQDB) access method and physical layer specifications.

[B9] ISO/IEC 8802-9:1996 (IEEE Std 802.9™-1996), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 9: Integrated Services (IS) LAN Interface at the Medium Access Control (MAC) and Physical (PHY) Layers.

[B10] ISO/IEC 8802-11:1999 (IEEE Std 802.11™-1999), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) specifications.

⁸The IEEE standards and products referred to in Annex A are trademarks owned by the Institute of Electrical and Electronics Engineers, Incorporated.

⁹IEEE publications are available from the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, USA (<http://standards.ieee.org/>).

附录 A

(资料性)

参考文献（局域网/城域网相关标准的补充参考文献）

A.1 局域网/城域网标准和规范

[B1] 以太网 2.0 版，局域网——数据链路层和物理层规范。数字设备公司、英特尔公司和施乐公司，1982年11月。

[B2] IEEE 100™，《IEEE 标准术语权威词典》第七版。⁸

[B3] IEEE 标准 802.10-1992，局域网和城域网 IEEE 标准：可互操作局域网/城域网安全（SILS）（包括 IEEE Std 802.10b™-1992）。⁹

[B4] ISO/IEC TR 8802-1:1997，信息技术——系统间电信和信息交换——局域网和城域网——技术报告和指南——第1部分：局域网标准概述。

[B5] ISO/IEC 8802-3:1999（IEEE Std 802.3-1998），信息技术——系统间电信和信息交换——局域网和城域网——特殊要求——第3部分：载波侦听多路访问／冲突检测（CSMA/CD）访问方法和物理层规范。

[B6] ISO/IEC 8802-4:1990 (IEEE Std 802.4-1990)，信息处理系统—局域网—第 4 部分：令牌传递总线访问方法和物理层规范。

[B7] ISO/IEC 8802-5:1998 (IEEE Std 802.5-1998)，信息技术—系统间电信和信息交换—局域网和城域网—特殊要求—第 5 部分：令牌环访问方法和物理层规范。

[B8] ISO/IEC 8802-6:1994 (IEEE Std 802.6™-1994)，信息技术—系统间电信和信息交换—局域网和城域网—特殊要求—第 6 部分：分布式队列双总线（DQDB）访问方法和物理层规范。

[B9] ISO/IEC 8802-9:1996 (IEEE Std 802.9™-1996)，信息技术—系统间电信和信息交换—局域网和城域网—特殊要求—第 9 部分：介质访问控制（MAC）层和物理层（PHY）的综合业务（IS）局域网接口。

[B10] ISO/IEC 8802-11:1999（IEEE Std 802.11™-1999），信息技术—系统间电信和信息交换—局域网和城域网—特殊要求—无线局域网介质访问控制（MAC）和物理层（PHY）规范。

⁸附录 A 中提及的 IEEE 标准和产品是电气与电子工程师协会拥有的商标。⁹IEEE 出版物可向电气与电子工程师协会购买，地址：美国新泽西州皮斯卡塔韦 Hoes Lane 445 号，邮政信箱 1331，新泽西州 08855-1331（<http://standards.ieee.org/>）。

[B11] ISO/IEC 8802-12:1998 (IEEE Std 802.12-1998), Information technology — Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 12: Demand-Priority access method, physical layer and repeater specifications.

[B12] ISO 9314-1:1989, Information processing systems—Fibre Distributed Data Interface (FDDI)—Part 1: Token Ring Physical Layer Protocol (PHY).

[B13] ISO 9314-2:1989, Information processing systems—Fibre Distributed Data Interface (FDDI)—Part 2: Token Ring Media Access Control (MAC).

[B14] ISO 9314-3:1990, Information processing systems—Fibre Distributed Data Interface (FDDI)—Part 3: Token Ring Physical Layer Medium Dependent (PMD).

[B15] ISO 9314-6:1989, Information processing systems—Fibre Distributed Data Interface (FDDI)—Part 6: Token Ring Station Management (SMT).

[B16] ISO/IEC TR 11802-5:1997 (IEEE Std 802.1H-1997), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Technical reports and guidelines—Part 5: Media Access Control (MAC) Bridging of Ethernet V2.0 in Local Area Networks.

[B17] ISO/IEC 15802-5:1998 (IEEE Std 802.1G™-1998), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 5: Remote Media Access Control (MAC) Bridging.

[B18] RFC 1042, A Standard for the Transmission of IP Datagrams over IEEE 802 Networks. Postel, J., and Reynolds, J., February 1988.¹⁰

[B19] RFC 1390, Transmission of IP and ARP over FDDI Networks. Katz, D., January 1993.

A.2 Management

NOTE—Many of the standards listed in Clause 2 and A.1 contain specifications of their managed objects.

[B20] IEEE Std 802.1F-1993, IEEE Standards for Local and Metropolitan Area Networks: Common Definitions and Procedures for IEEE 802 Management Information.

NOTE—Some of the content of IEEE Std 802.1F-1993 is also included in ISO/IEC 10742.

[B21] ISO/IEC 7498-4:1989, Information processing systems—Open Systems Interconnection—Basic Reference Model—Part 4: Management framework.

[B22] ISO/IEC 8824-1:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Specification of basic notation.

[B23] ISO/IEC 8824-2:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Information object specification.

¹⁰Internet RFCs are available from the Internet Engineering Task Force on the World Wide Web browser at <http://www.ietf.org/rfc/rfcnnnn.txt> (where nnnn is the four-digit RFC number, padded with leading zeros if necessary). For more information, call the IETF secretariat at +1-703-620-8990.

[B11] ISO/IEC 8802-12:1998（IEEE 标准 802.12-1998）， 信息技术 — 系统间电信和信息交换—局域网和城域网—特殊要求—第 12 部分： 需求优先级访问方法、物理层和中继器规范。

[B12] ISO 9314-1:1989， 信息处理系统—光纤分布式数据接口（FDDI） —第1部分： 令牌环物理层协议（PHY）。

[B13] ISO 9314-2:1989， 信息处理系统—光纤分布式数据接口（FDDI） —第2部分： 令牌环媒体访问控制（MAC）。

[B14] ISO 9314-3:1990， 信息处理系统—光纤分布式数据接口（FDDI） —第3部分： 令牌环物理层介质相关（PMD）。

[B15] ISO 9314-6:1989， 信息处理系统——光纤分布式数据接口（FDDI） ——第 6 部分： 令牌环站管理（SMT）。

[B16] ISO/IEC TR 11802-5:1997（IEEE 标准 802.1H-1997）， 信息技术——系统间的电信和信息交换——局域网和城域网——技术报告和指南——第 5 部分： 局域网中 EthernetV2.0 的媒体访问控制（MAC）网桥。

[B17] ISO/IEC 15802-5:1998（IEEE 标准 802.1G™-1998）， 信息技术——系统间的电信和信息交换——局域网和城域网——通用规范——第 5 部分： 远程媒体访问控制（MAC）网桥。

[B18] RFC 1042， IEEE802 网络上 IP 数据报传输标准。Postel, J. 和 Reynolds, J., 1988年2月。¹⁰

[B19] RFC 1390， FDDI 网络上的 IP 和 ARP 传输。Katz, D., 1993年1月。

A.2 管理

注——第 2 条和 A.1 中列出的许多标准都包含对其受管对象的规范。

[B20] IEEE 标准 802.1F-1993， 局域网和城域网 IEEE 标准： IEEE 802 管理信息的通用定义和规程。

注——IEEE Std 802.1F-1993 的部分内容也收录于 ISO/IEC 10742 中。

[B21] ISO/IEC 7498-4:1989， 信息处理系统—开放系统互连—基本参考模型—第4部分： 管理框架。

[B22] ISO/IEC 8824-1:1995， 信息技术—抽象语法记法一 (ASN.1)—基本记法规范。

[B23] ISO/IEC 8824-2:1995， 信息技术—抽象语法记法一 (ASN.1)—信息对象规范。

¹⁰互联网 RFC 可通过万维网上的互联网工程任务组浏览器在 <http://www.ietf.org/rfc/rfcnnnn.txt> 获取（其中 nnnn 是四位 RFC 编号，必要时在前面补零）。如需了解更多信息，请致电 IETF 秘书处： +1-703-620-8990。

OVERVIEW AND ARCHITECTURE	IEEE Std 802-2001	概述与体系结构	IEEE 802-2001 标准
[B24] ISO/IEC 8824-3:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Constraint specification.		[B24] ISO/IEC 8824-3:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Constraint specification.	
[B25] ISO/IEC 8824-4:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Parameterization of ASN.1 specifications.		[B25] ISO/IEC 8824-4:1995, Information technology—Abstract Syntax Notation One (ASN.1)—Parameterization of ASN.1 specifications.	
[B26] ISO/IEC 8825-1:1995, Information technology—ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER).		[B26] ISO/IEC 8825-1:1995, Information technology—ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER).	
[B27] ISO/IEC 9595:1991, Information technology—Open Systems Interconnection—Common management information service definition.		[B27] ISO/IEC 9595:1991, Information technology—Open Systems Interconnection—Common management information service definition.	
[B28] ISO/IEC 9596-1:1991, Information technology—Open Systems Interconnection—Common management information protocol—Part 1: Specification		[B28] ISO/IEC 9596-1:1991, Information technology—Open Systems Interconnection—Common management information protocol—Part 1: Specification	
[B29] ISO/IEC 10040:1992, Information technology—Open Systems Interconnection—Systems management overview.		[B29] ISO/IEC 10040:1992, Information technology—Open Systems Interconnection—Systems management overview.	
[B30] ISO/IEC 10164-1/8:1993, Information technology—Open Systems Interconnection—Systems management: [various functions].		[B30] ISO/IEC 10164-1/8:1993, Information technology—Open Systems Interconnection—Systems management: [various functions].	
[B31] ISO/IEC 10165-1:1993, Information technology—Open Systems Interconnection—Structure of management information: Management information model.		[B31] ISO/IEC 10165-1:1993, Information technology—Open Systems Interconnection—Structure of management information: Management information model.	
[B32] ISO/IEC 10165-2:1992, Information technology—Open Systems Interconnection—Structure of management information: Definition of management information.		[B32] ISO/IEC 10165-2:1992, Information technology—Open Systems Interconnection—Structure of management information: Definition of management information.	
[B33] ISO/IEC 10165-4:1992, Information technology—Open Systems Interconnection—Structure of management information: Guidelines for the definition of managed objects.		[B33] ISO/IEC 10165-4:1992, Information technology—Open Systems Interconnection—Structure of management information: Guidelines for the definition of managed objects.	
[B34] ISO/IEC 10742:1994, Information technology—Telecommunications and information exchange between systems—Elements of management information related to OSI Data Link Layer standards.		[B34] ISO/IEC 10742:1994, Information technology—Telecommunications and information exchange between systems—Elements of management information related to OSI Data Link Layer standards.	
[B35] ISO/IEC 15802-2:1995 (IEEE Std 802.1B-1995), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 2: LAN/MAN management, service and protocol.		[B35] ISO/IEC 15802-2:1995 (IEEE Std 802.1B-1995), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 2: LAN/MAN management, service and protocol.	
[B36] ISO/IEC 15802-4:1994 (IEEE Std 802.1E™-1994), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 4: System load protocol.		[B36] ISO/IEC 15802-4:1994 (IEEE Std 802.1E™-1994), Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Common specifications—Part 4: System load protocol.	
[B37] RFC 1095, The Common Management Services and Protocol over TCP/IP (CMOT). Warrior, U., and Besaw, L., April 1989.		[B37] RFC 1095, The Common Management Services and Protocol over TCP/IP (CMOT). Warrior, U., and Besaw, L., April 1989.	
[B38] RFC 1155, Structure and Identification of Management Information for TCP/IP-based Internets. Rose, M., and McCloghrie, K., May 1990.		[B38] RFC 1155, Structure and Identification of Management Information for TCP/IP-based Internets. Rose, M., and McCloghrie, K., May 1990.	
[B39] RFC 1157, A Simple Network Management Protocol (SNMP). Case, J., Fedor, M., Schoffstall, M., and Davin, J., May 1990.		[B39] RFC 1157, A Simple Network Management Protocol (SNMP). Case, J., Fedor, M., Schoffstall, M., and Davin, J., May 1990.	
[B40] RFC 1212, Concise MIB Definitions. Rose, M., and McCloghrie, K. (Editors), March 1991.		[B40] RFC 1212, Concise MIB Definitions. Rose, M., and McCloghrie, K. (Editors), March 1991.	
[B41] RFC 1215, A Convention for Defining Traps for use with the SNMP. Rose, M., March 1991.		[B41] RFC 1215, A Convention for Defining Traps for use with the SNMP. Rose, M., March 1991.	

[B42] RFC 1493, Definitions of Managed Objects for Bridges. Decker, E., Langille, P., Rijsinghani, A., and McCloghrie, K., July 1993.

[B43] RFC 1643, Definitions of Managed Objects for the Ethernet-like Interface Types. Kastenholz, F., July 1994.

[B44] RFC 1902, Structure of Management information for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B45] RFC 1903, Textual Conventions for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B46] RFC 1904, Conformance Statements for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B47] RFC 1905, Protocol Operations for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B42] RFC 1493, Definitions of Managed Objects for Bridges. Decker, E., Langille, P., Rijsinghani, A., and McCloghrie, K., July 1993.

[B43] RFC 1643, Definitions of Managed Objects for the Ethernet-like Interface Types. Kastenholz, F., July 1994.

[B44] RFC 1902, Structure of Management information for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B45] RFC 1903, Textual Conventions for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B46] RFC 1904, Conformance Statements for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.

[B47] RFC 1905, Protocol Operations for Version 2 of the Simple Network Management Protocol (SNMPv2). Case, J., McCloghrie, K., Rose, M., and Waldbusser, S., January 1996.